JOHANNES KEPLER
UNIVERSITY LINZ

Author
Christian Wiedner
K52001455

Submission
Institute for Machine Learning

Thesis Supervisor
Assoc. Prof. PD Dipl.-Ing. Dr.
Hrvoje Bogunović

Assistant Thesis Supervisor
Dr. **Dmitrii Lachinov**

October 2026

Neural PDE Solvers for Geographic Atrophy Progression Prediction from Longitudinal OCT Imaging

Master's Thesis

to confer the academic degree of

Master of Science

in the Master's Program

Artificial Intelligence

JOHANNES KEPLER UNIVERSITY LINZ
Altenberger Straße 69
4040 Linz, Austria
jku.at

Abstract

Geographic Atrophy (GA) is the atrophic late form of age-related macular degeneration, manifested as a gradual and irreversible loss of retinal tissue. A forecast of where and how fast the lesion of an individual eye will grow would support patient management, but it is difficult to model: patterns and rates of progression vary widely between patients, visits are irregularly spaced, and cohorts are small. This thesis explores GA progression through computational modelling, investigating the framework required to forecast from longitudinal optical coherence tomography (OCT) data, as well as the optimal operator for mapping the disease state between visits. It contributes a framework for irregular visit schedules in which models of any architecture class are trained and evaluated alike, and a controlled comparison of such models for GA. Among operators with spatial context, the score depends mainly on whether one update reaches as far as the lesion front moves between two visits.

The data comprise 75 eyes of 51 patients from the Medical University of Vienna, with 553 visits over a median follow-up of three years. Each visit is represented on the en-face grid of the OCT scan by a binary GA mask, annotated by graders, and the depth maps of ten segmented retinal layer boundaries. The framework predicts this state at the next visit as the current state plus an update scaled by the time Δt between visits, and trains all models with one loss and one autoregressive curriculum. The operator was filled in turn with graph neural networks, a U-Net, Fourier Neural Operators and Finite Element Networks at similar parameter counts. All models were compared by patient-level five-fold cross-validation on the change-region Dice, the overlap between predicted and true lesion change one year after baseline, reported as mean $\pm$ standard deviation over the folds; a forecast of no change scores zero. Differences were tested against the variation from retraining and repeated with a second seed.

The strongest operator, a Finite Element Network with a learned transport term, reaches 0.54 ± 0.04, followed by a graph network at 0.53 ± 0.05 and a U-Net at 0.50 ± 0.06, so models of three classes reach similar scores. Spatial context and a loss that emphasises the lesion are both required; without either, a model predicts no change. Neither a fourth-order Runge–Kutta step nor larger models help. On the anisotropic grid, the cross-sectional scans (B-scans) lie 121 µm apart, the pixels along a scan only 5.7 µm. In a graph built from pixel indices, the neighbours of a pixel therefore lie within about 11 µm along the B-scan, whereas the lesion front typically advances by about 70 µm between visits. A dilated stencil whose neighbours reach about 120 µm in both directions raises the Dice by 0.064 at an identical parameter count, on all five folds. A learned transport term on a mesh with the same seven-column spacing raises the Finite Element Network by a similar 0.062. In this cohort, the Dice on the newly atrophic region alone is about 0.5 to 0.65 in each year of follow-up, and predicted and measured growth rates correlate at $r = 0.40$.

Kurzfassung

Die Geographische Atrophie (GA) ist eine Spätform der altersabhängigen Makuladegeneration, die durch einen fortschreitenden und irreversiblen Verlust von Netzhautgewebe gekennzeichnet ist. Eine Vorhersage, wo und wie schnell die Atrophie in einem einzelnen Auge fortschreitet, könnte die Betreuung der Betroffenen verbessern. Die Modellierung dieser Progression ist jedoch anspruchsvoll: Verlaufsmuster und Wachstumsgeschwindigkeit unterscheiden sich stark von Patient zu Patient, die Kontrolluntersuchungen erfolgen in unregelmäßigen Abständen, und die verfügbaren Kohorten sind klein. Die vorliegende Arbeit betrachtet die GA-Progression aus der Perspektive des Computational Modelling und untersucht, welches Framework für Vorhersagen auf Basis longitudinaler optischer Kohärenztomographie (OCT) erforderlich ist und welcher Operator sich am besten eignet, um den Krankheitszustand von einer Untersuchung zur nächsten abzubilden. Dazu wird ein Framework für unregelmäßige Untersuchungsintervalle entwickelt, in dem sich Modelle unterschiedlicher Architekturklassen unter identischen Bedingungen trainieren und vergleichen lassen. Unter den Operatoren mit räumlichem Kontext hängt das Ergebnis vor allem davon ab, ob ein einzelner Vorhersageschritt so weit reicht, wie sich der Läsionsrand zwischen zwei Untersuchungen verschiebt.

Datengrundlage sind 75 Augen von 51 Patientinnen und Patienten der Medizinischen Universität Wien mit insgesamt 553 Untersuchungen und einer Nachbeobachtungszeit von im Median drei Jahren. Jede Untersuchung wird auf dem En-face-Raster der OCT durch eine binäre GA-Maske sowie durch Tiefenprofile von zehn segmentierten Netzhautschichtgrenzen beschrieben. Das Framework berechnet den Zustand bei der nächsten Untersuchung, indem es zum aktuellen Zustand eine mit dem zeitlichen Abstand Δt skalierte Änderung addiert. Alle Modelle werden mit derselben Loss-Funktion und demselben autoregressiven Curriculum trainiert. Als Operator kamen Graph Neural Networks, ein U-Net, Fourier Neural Operators und Finite-Elemente-Netzwerke mit vergleichbarer Parameterzahl zum Einsatz. Die Evaluation erfolgte mittels Five-Fold Cross-Validation mit patientenbasierter Aufteilung. Als primäre Metrik dient der Change-Region-Dice, die Überlappung zwischen vorhergesagter und tatsächlicher Läsionsveränderung ein Jahr nach der Ausgangsuntersuchung, angegeben als Mittelwert $\pm$ Standardabweichung über die fünf Folds; eine Vorhersage ohne Veränderung ergibt den Wert null. Ob Unterschiede zwischen den Modellen über die durch erneutes Training bedingte Streuung hinausgehen, wurde statistisch geprüft; die Vergleiche wurden mit einem zweiten Random Seed wiederholt.

Das Finite-Elemente-Netzwerk mit gelerntem Transportterm erzielt mit $0{,}54 \pm 0{,}04$ den höchsten Change-Region-Dice, gefolgt von einem Graph-Netzwerk mit $0{,}53 \pm 0{,}05$ und einem U-Net mit $0{,}50 \pm 0{,}06$; Modelle aus drei Architekturklassen liegen damit nahe beieinander. Unverzichtbar sind räumlicher Kontext und eine Loss-Funktion, die die Läsion stärker gewichtet: Fehlt eines

von beiden, sagt das Modell keine Veränderung voraus. Weder ein Runge–Kutta-Schritt vierter Ordnung noch größere Modelle verbessern die Vorhersage messbar. Das Raster ist stark anisotrop: Die Querschnittsaufnahmen (B-Scans) liegen 121 µm auseinander, die Bildpunkte innerhalb eines B-Scans nur 5,7 µm. In einem Graphen, dessen Nachbarschaften über Pixelindizes definiert sind, liegen die Nachbarn eines Pixels entlang des B-Scans daher höchstens etwa 11 µm entfernt, während sich der Läsionsrand zwischen zwei Untersuchungen typischerweise um etwa 70 µm verschiebt. Ein Dilated Stencil, dessen Nachbarschaft in beiden Richtungen bis etwa 120 µm reicht, steigert den Change-Region-Dice bei gleicher Parameterzahl um 0,064, und zwar in allen fünf Folds. Ein gelernter Transportterm auf einem Gitter mit demselben Abstand von sieben Spalten steigert das Finite-Elemente-Netzwerk um vergleichbare 0,062. In dieser Kohorte liegt der Dice auf der neu atrophierten Fläche allein in jedem Beobachtungsjahr bei etwa 0,5 bis 0,65, und vorhergesagte und gemessene Wachstumsraten korrelieren mit $r = 0{,}40$.

Acknowledgements

I would like to thank the Medical University of Vienna for making this thesis possible and for providing the longitudinal OCT data on which it is built.

I thank Hrvoje Bogunović for supervising this thesis, for the opportunity to work on this topic, and for the guidance throughout the project.

I am especially grateful to Dmitrii Lachinov for working with me throughout this project, for the many discussions, for the support during the experiments, and for the detailed feedback on the drafts of this thesis.

Contents

Abstract ii

Kurzfassung iii

Acknowledgements v

List of Acronyms ix

1 Introduction 1
- 1.1 Clinical Motivation . . . 1
- 1.2 Aim of the Thesis . . . 2
- 1.3 Contributions . . . 3
- 1.4 Thesis Outline . . . 4

2 Background 5
- 2.1 Medical Background . . . 5
  - 2.1.1 Eye Anatomy . . . 5
  - 2.1.2 Retinal Imaging . . . 7
  - 2.1.3 Geographic Atrophy . . . 10
- 2.2 Partial Differential Equations and Numerical Solvers . . . 12
  - 2.2.1 Discretising space: stencils . . . 12
  - 2.2.2 Stepping in time . . . 14
  - 2.2.3 Strengths and limits . . . 14
- 2.3 Neural PDE Solvers and Surrogate Architectures . . . 15
- 2.4 Related Work . . . 18
  - 2.4.1 Predicting the Risk and Rate of Progression . . . 18
  - 2.4.2 Spatial Forecasting of GA Lesions . . . 18
  - 2.4.3 Forecasting from Irregularly Sampled Longitudinal Data . . . 19
  - 2.4.4 PDE-Based Models in Biomedicine . . . 19
- 2.5 Modelling Approach and Problem Statement . . . 20
  - 2.5.1 Choice of the Modelling Approach . . . 20
  - 2.5.2 Problem Statement . . . 22

3 Data and Preprocessing 23
- 3.1 Dataset . . . 23
- 3.2 State Representation . . . 24
- 3.3 Spatial Standardisation and its Measured Cost . . . 26
- 3.4 Normalisation . . . 27
- 3.5 Patient-Level Covariates . . . 27
- 3.6 Temporal Structure . . . 28
- 3.7 Splits and Cross-Validation . . . 29

4 Method 31
- 4.1 Overview . . . 31

4.2 The Fixed Framework . . . 32
4.2.1 Δt-Conditioned Residual Formulation . . . 33
4.2.2 Multi-Channel State Propagation . . . 34
4.2.3 Conditioning: Patient Covariates . . . 34
4.2.4 Objective . . . 34
4.2.5 Time-Budgeted Pushforward Curriculum . . . 35
4.2.6 Optimisation . . . 36
4.3 The Operators . . . 37
4.3.1 The Operator Contract . . . 37
4.3.2 Message-Passing Graph Neural Network . . . 38
4.3.3 Graph Construction on an Anisotropic Grid . . . 40
4.3.4 Dense Operators: U-Net, FNO and a Local-Kernel Hybrid . . . 42
4.3.5 The Locality Floors . . . 44
4.3.6 The Finite Element Network . . . 45
4.3.7 Fixed-Step Runge–Kutta Integration . . . 46
4.3.8 Parameter Matching . . . 48
4.4 Implementation Details . . . 49

5 Experiments **51**
5.1 Evaluation Protocol . . . 51
5.1.1 Evaluation Challenges . . . 51
5.1.2 Metrics . . . 52
5.1.3 Rollout Evaluation . . . 53
5.1.4 Late-Epoch Mean . . . 53
5.1.5 Noise Level and Comparisons . . . 54
5.2 Main Results . . . 55
5.3 Forecasts Beyond One Year . . . 58
5.4 Ablation Studies . . . 59
5.4.1 Spatial Context . . . 59
5.4.2 Physical Reach at Lesion Scale . . . 60
5.4.3 Global and Local Context Together . . . 61
5.4.4 An Explicit Transport Term . . . 62
5.4.5 Summary of the Ablation Studies . . . 63
5.5 Differences Between Folds . . . 63
5.6 Negative Results . . . 66
5.6.1 Time Handling . . . 66
5.6.2 The Objective . . . 67
5.6.3 Capacity . . . 69
5.7 Additional Experiment: Adaptive Meshes . . . 70
5.8 Validity Diagnostics . . . 73
5.8.1 The Step-Size Test . . . 73
5.8.2 Numerical Stability of the T-FEN . . . 76
5.9 Comparison with Prior Work . . . 79
5.10 Computational Cost . . . 82
5.11 Summary . . . 83

6 Discussion **85**
6.1 Interpretation of Results . . . 85
6.1.1 The Framework . . . 85
6.1.2 The Operator . . . 86
6.1.3 Capacity . . . 88

6.1.4 A Missing Counterpart to the Equation Features 88
6.1.5 Predictions Between Visits 88
6.1.6 Implications for Modelling 89
6.2 Clinical Implications 89
6.3 Limitations 91
6.4 Future Work 92

7 Conclusion **94**

Bibliography **96**

Appendix A Dataset Details **102**
A.1 Visit Schedule and Lesion Scale 102
A.2 Native Grids and the Crop 103

Appendix B Additional Rollout Figures **104**

Appendix C Ablation Details **106**
C.1 Border-Touching and Interior Eyes 106
C.2 Growth-Speed Measures for All Arms 107
C.3 Statistical Criteria 107
C.4 Statistics of the Comparisons 108
C.5 The Step-Size Test 112

Appendix D Code Structure **114**

Appendix E The Moving-Mesh Extension (MM-PDE) **116**
E.1 Background: MM-PDE 116
E.2 The Moving-Mesh Extension for GA 117
E.2.1 Data-Free Mesh Mover for GA 118
E.2.2 Dual-Branch Composition 119
E.3 Moving Mesh Quality 121
E.4 Details of the Result 122
E.5 Outlook 123

List of Acronyms

AMD	Age-related Macular Degeneration
CAM	Classification of Atrophy Meetings
CFL	Courant–Friedrichs–Lewy
cRORA	Complete RPE and Outer Retinal Atrophy
DMM	Data-free Mesh Mover
FAF	Fundus Autofluorescence
FEN	Finite Element Network
FET-PET	FET Positron Emission Tomography
FNO	Fourier Neural Operator
GA	Geographic Atrophy
GNN	Graph Neural Network
iRORA	Incomplete RPE and Outer Retinal Atrophy
ItpNet	Interpolation Network
MLP	Multi-Layer Perceptron
MM-PDE	Moving Mesh Partial Differential Equation solver
MP-PDE	Message-Passing Partial Differential Equation solver
MRI	Magnetic Resonance Imaging
MSE	Mean Squared Error
MUW	Medical University of Vienna
OCT	Optical Coherence Tomography
ODE	Ordinary Differential Equation
PDE	Partial Differential Equation
RK4	Classical fourth-order Runge–Kutta method
RMSE	Root Mean Squared Error
RORA	RPE and Outer Retinal Atrophy
RPE	Retinal Pigment Epithelium
SDE	Stochastic Differential Equation
SLO	Scanning Laser Ophthalmoscopy
T-FEN	Finite Element Network with a learned transport term

Chapter 1

Introduction

1.1 Clinical Motivation

Age-related macular degeneration (AMD) is the leading cause of irreversible central vision loss in older adults in industrialised countries (Boopathiraj et al., 2024). Worldwide, it ranks as the third most frequent cause of moderate-to-severe vision impairment and the fourth most frequent cause of blindness (Flaxman et al., 2017). The number of people with some form of AMD was projected to reach 196 million in 2020 and 288 million by 2040 (Wong et al., 2014). AMD progresses from early and intermediate stages to two late forms: neovascular AMD, with abnormal blood vessel growth, and macular atrophy (Vogl et al., 2021). The early stages are marked by drusen, extracellular deposits of cellular debris (Boyer et al., 2017). The atrophic late form without neovascularisation is termed *Geographic Atrophy* (GA). It is an advanced, irreversible and progressive atrophy of the macula, the central region of the retina that provides sharp vision. GA causes a marked loss of visual acuity, the sharpness of vision, once the fovea, the centre of the macula, is involved (Vallino et al., 2024). Approximately one million people in the United States and five million worldwide are affected by GA (Singh et al., 2025), with around 160,000 new diagnoses per year in the United States alone (Vallino et al., 2024). Unlike cataract, glaucoma or diabetic retinopathy, AMD is not counted among the avoidable causes of vision loss (Flaxman et al., 2017). Avoidable causes are those that can be prevented or treated by known, cost-effective means. Clinical care therefore focuses on *managing* the progression of the disease once it has appeared.

The disease burden is concentrated in the older population. Prevalence of GA rises by roughly a factor of four per decade after age fifty (Singh et al., 2025), and a South Korean national survey reports overall AMD prevalence increasing from 3.6 % in the fifth decade of life to 34.0 % at the age of eighty and above (Song et al., 2025). GA often affects both eyes: bilateral GA was present in 39 % of the patients of one imaging cohort (Yehoshua et al., 2011), and in the cohort of the Age-Related Eye Disease Study (AREDS) the cumulative probability of bilateral GA rose from 7 % at first GA development to 62 % at ten years (Boyer et al., 2017). The two eyes of a patient also show a high degree of concordance in the incidence, progression and area of GA (Boyer et al., 2017). Progression is also highly variable between patients, with the lesion area growing by 0.53 to 2.6 mm^2 per year (Mai et al., 2024). The clinically relevant question is therefore how fast and where in the macula the lesion will grow.

Until 2023, no treatment for GA was approved (Trincão-Marques et al., 2024), and numerous phase 1–3 trials of GA and intermediate AMD, addressing mech-

anisms such as neuroprotection, visual cycle modulation, antiamyloid pathways and cell replacement, had failed during the preceding decade (Lad and Chakravarthy, 2023). This changed in 2023, when the U.S. Food and Drug Administration approved two complement inhibitors, pegcetacoplan and avacincaptad pegol (Singh et al., 2025; Trincão-Marques et al., 2024). Both drugs block part of the complement cascade, a component of the immune system. So far they are approved in the United States but not in the European Union (European Medicines Agency, 2024a; European Medicines Agency, 2024b). In the phase 3 trials, both drugs reduced the rate of GA enlargement by roughly 14 % to 27 % over twelve to twenty-four months. The lesions nevertheless continued to grow. Best-corrected visual acuity, measured with the best optical correction, did not differ significantly from a sham (simulated) injection at the trial endpoints (Lad and Chakravarthy, 2023). Hence, two things matter for clinical practice. First, treatment timing: an intervention that only slows progression is most valuable when administered before foveal involvement, because central vision is largely preserved until the atrophy reaches the fovea (Boyer et al., 2017; Vallino et al., 2024). Second, both drugs are given as injections into the eye, which carry patient burden and procedural risk. Identifying which eyes will progress rapidly enough to justify therapy is therefore the critical decision-support task.

Both decisions need a forecast of how fast, and where, the lesion of an individual eye will grow. Optical coherence tomography (OCT) produces three-dimensional images of the retina (§2.1.2), and measurable features in these images, called biomarkers, indicate the course of the disease. As Vallino et al. (2024) note in their conclusions, combining artificial intelligence with structural OCT biomarkers holds significant promise: it could provide predictive probabilities of GA development over time. This thesis pursues that avenue.

1.2 Aim of the Thesis

Predicting GA progression from imaging is framed as a *spatiotemporal forecasting* task on the en-face grid of the OCT scan, the retina seen from the front. For each eye, longitudinal imaging yields a sequence of macula-centred grids that carry the outline of the lesion, annotated by graders, and the depths of the boundaries between the retinal layers. Given the most recent observed state and a target horizon Δt, the task is to predict the extent and shape of the lesion at the later visit. Earlier learning-based work on AMD predicts the conversion of an eye to a later disease stage (Schmidt-Erfurth et al., 2018; Vogl et al., 2021); more recent models forecast the GA lesion itself, among them that of Mai et al. (2024) on the cohort used here (Section 2.4). Like these, this thesis forecasts a complete map of the lesion at the future visit.

Such a forecast is made by a learned update from the state at one visit to the state at a later one. Its learned part, the change it adds per unit time, is called the operator in this thesis. Many architectures can serve as the operator, among them the neural solvers for partial differential equations on which this thesis builds (Chapter 2). Which architecture GA progression requires cannot be decided by testing a single one: if it succeeds, the reason may be the architecture, its size, or the setup around it. This thesis therefore separates the operator from

the setup around it, called the framework. The framework fixes how elapsed time enters the prediction and how the operators are trained and evaluated. It is fixed once, and only the operator changes, so that architectures from different families can be compared on equal terms. The closest forecasts of this kind are discussed in Section 2.4.

Research question. This thesis addresses one question, divided into two parts:

> *What is needed to forecast the progression of Geographic Atrophy from longitudinal OCT?*
>
> 1. *What kind of framework does the forecast need?*
> 2. *What kind of operator does that framework need?*

The first part concerns the framework described above. The second concerns the operator that maps one visit to the next. Chapter 4 develops a framework that answers the first part and leaves the operator as its only free component. Chapter 5 then answers the second by comparing different operators within that framework.

1.3 Contributions

The main contributions of this thesis are the following.

- A framework for irregular visit schedules that is independent of the architecture inside it: a residual update conditioned on the interval Δt between visits, a training curriculum that also feeds the model its own earlier predictions, with its depth measured in elapsed days, and one loss and evaluation for all models. Inside it, every operator with spatial context learned to predict change, and the loss decided whether any change was learned at all.
- A controlled comparison of operators from four architecture families (graph networks, a U-Net, Fourier Neural Operators and Finite Element Networks) for forecasting GA on 75 eyes, at matched parameter counts and with patient-level five-fold cross-validation. The highest operators of three families score within about 0.04 of each other at one year.
- The finding that, among operators with spatial context, the score depends mainly on how far one update reaches compared with how far the lesion front moves between two visits. Placing the neighbours of a graph network at lesion scale and adding a transport term, on a mesh with the same spacing, to a Finite Element Network each raise the score by about 0.06 against their own controls; this is consistent with, but does not prove, the reach explanation.
- Results on what did not help, each within the noise level: larger operators, a higher integration order and a monotonic-growth penalty. An adaptive moving mesh, tried in a single additional experiment on the k-NN graph network, gave no measurable gain either (§5.7); it is not a verdict on mesh adaptation.

1.4 Thesis Outline

The remainder of this thesis is organised as follows. Chapter 2 introduces the anatomy of the eye, retinal imaging and GA. It then covers the numerical methods for partial differential equations (PDEs) and the neural solvers that learn parts of them, reviews related work, and closes with the motivation for the modelling approach and a formal statement of the problem. Chapter 3 describes the Medical University of Vienna GA cohort and how each visit is turned into an eleven-channel state. Chapter 4 addresses the first part of the research question by developing the framework, and describes the operators that fill its one free component. Chapter 5 addresses the second part by comparing these operators, and reports one additional experiment with an adaptive moving mesh. Chapter 6 discusses what the results mean clinically and methodologically, together with their limitations, and Chapter 7 concludes. The appendices give further details on the data, additional rollouts, the statistics of the comparisons and the code; Appendix E gives the background and design of the experiment with the adaptive mesh.

Chapter 2

Background

This chapter provides the background on which the later chapters build. Section 2.1 introduces the anatomy of the eye, the methods used to image the retina and the disease itself. Section 2.2 introduces partial differential equations and the numerical methods that solve them, and Section 2.3 the neural PDE solvers, which replace parts of such a solver, or the whole update, with a neural network trained on data. Section 2.4 reviews related work. Section 2.5 connects the two strands: it explains why the structure of a numerical solver suits the forecasting of GA and states the problem formally.

2.1 Medical Background

This section introduces the medical terms on which the data and the forecast rely. §2.1.1 describes the anatomy of the eye and the layered retina, §2.1.2 the imaging of the retina, from the OCT volume to the en-face maps the model operates on, and §2.1.3 the disease and how it appears on these images. Together they explain the structure of the data: a binary lesion mask with the geometry of the surrounding retinal layers (Chapter 3).

2.1.1 Eye Anatomy

Figure 2.1 (left) shows the eye in cross-section. Light enters at the front through the *cornea* and *lens*, traverses the gel-filled vitreous body, and is focused onto the *retina*. The retina is a thin neural-tissue lining at the back of the eye that converts the incoming light pattern into electrical signals. The conversion takes place in the *photoreceptors*, the light-sensitive cells of the retina. Those signals leave the eye via the *optic nerve*, which exits the retina at a specific anatomical landmark called the *optic-nerve head* (ONH, also called the *optic disc*). Behind the retina lie two further layers that support and supply it from beneath: the *choroid*, a richly vascularised tissue whose innermost capillary bed is the *choriocapillaris*, and the *sclera*, the white, mechanically rigid outer shell. The inside of the back of the eye, as seen through the pupil, is called the *fundus*.

The retina is not uniform across its surface. A small disc-shaped region, the *macula*, is specialised for high-acuity central vision, while the rest of the retina handles peripheral vision at lower resolution. The macula has a diameter of roughly five to six millimetres and is where GA develops (§2.1.3). The OCT scans of the cohort are centred on the macula; how they are brought to one modelling

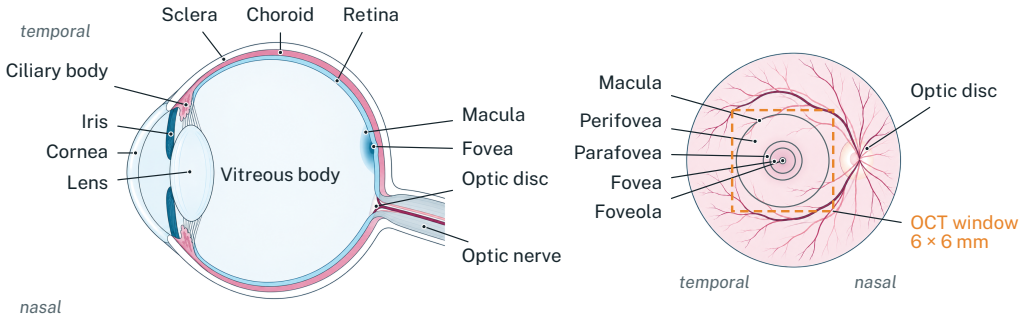


Figure 2.1: Anatomy reference for the terms used throughout this thesis (schematic). *Left:* horizontal cross-section of a right eye, seen from above with the front of the eye on the left, so that the temporal side is at the top and the nasal side at the bottom. Light enters through the cornea and lens, traverses the vitreous body and falls on the retina; the *macula* is a small specialised region of the retina, and the *fovea* is its centre, where photoreceptor density and visual acuity peak. The optic disc (also called the optic-nerve head, ONH) is the point at which the optic nerve leaves the eye. Retina and choroid are drawn thicker than they are. *Right:* fundus of a right eye, i.e. the inside of the back of the eye seen through the pupil, with the optic disc on the nasal side. The circles mark the zones of the macula around the fovea, with approximate diameters of 0.35 mm (foveola), 1.5 mm (fovea), 2.3 mm (parafovea) and 5.5 mm (perifovea, the outermost region of the macula) (Koksaldi et al., 2025). The panel is scaled so that the fovea lies 4.36 mm horizontally from the centre of the optic disc, the mean measured in healthy right eyes (Nair et al., 2021). The dashed square is the modelling window of about 6×6 mm used in this thesis (§3.3). Drawings generated with ChatGPT; labels added and verified by the author.

window of about 6×6 mm is described in §3.3. Within the macula, four concentric anatomical zones are conventionally distinguished, in order of increasing radius from the centre: the *foveola*, the *fovea*, the *parafovea*, and the *perifovea* (Figure 2.1, right). The fovea is the small central pit at which photoreceptor density and visual acuity reach their peak, and the term "foveal" refers to anything in or close to that pit; "parafoveal" and "perifoveal" describe progressively more eccentric annular regions. Two qualifiers used in the GA literature follow directly from this geography: a lesion is called *subfoveal* if it overlaps the fovea (i.e. it directly threatens high-acuity vision) and *extrafoveal* otherwise. As discussed in §1.1, whether a lesion is subfoveal or extrafoveal changes its clinical significance even if its total area is the same.

The retina is also a *layered* tissue, with several distinct strata stacked vertically from the front (closest to the vitreous body) to the back (closest to the choroid). The principal layers, in order from inner to outer, are the inner limiting membrane, the retinal nerve fibre layer, the ganglion-cell and inner-plexiform layers, the inner nuclear and outer plexiform layers, the outer nuclear layer, the photoreceptor inner- and outer-segment bands, the retinal pigment epithelium (RPE), and Bruch's membrane; Figure 2.4(a) gives a labelled schematic. The RPE is a single layer of pigmented cells that supports the photoreceptors,

and Bruch's membrane is the thin sheet that separates it from the choroid beneath. For the purposes of this thesis, it suffices to note that the retina is layered, that GA destroys the outer retina together with the choriocapillaris beneath it (§2.1.3), and that the surrounding layer geometry, given as ten boundary depth maps, is passed to the model together with the lesion mask (§3.2).

2.1.2 Retinal Imaging

Optical Coherence Tomography (OCT), introduced by D. Huang et al. (1991), produces micrometre-resolution cross-sectional images of the retina. The principle is analogous to ultrasound imaging of soft tissue, with low-coherence near-infrared light in place of sound. A weak probe beam is directed into the eye, and the strength of the light reflected back from each *depth* along its path is measured. This depth-versus-reflectivity profile is called an *A-scan*. It is displayed as a single column of pixels, with intensity encoding reflectivity and vertical position encoding depth. Resolving the depth dimension requires an optical technique, because conventional electronics cannot time near-infrared light from a single short pulse precisely enough to separate individual layers. The standard solution is interferometry. The source light is split into two beams (Figure 2.2(a)): the *sample arm* is sent into the eye and reflected back from the various retinal layers, and the *reference arm* is sent to a known mirror and reflected back unchanged. The two returning beams are then recombined and produce an interference pattern, and the spectrum of that pattern (in spectral-domain OCT) encodes the depth at which each reflection in the eye occurred. Standard signal processing then recovers the A-scan as a one-dimensional function of depth.

A single A-scan is one column of one image. Scanning the probe beam laterally along one direction produces many adjacent A-scans, and displaying them side by side yields a two-dimensional cross-section through the retina, called a *B-scan* (Figure 2.2(b)). Scanning in both lateral directions yields a three-dimensional *OCT volume* (Figure 2.2(b)), in which each voxel records the reflectivity of one small block of tissue. The MUW volumes are macular scans whose lateral extent differs between eyes (§3.1); all are brought to a common window of about 6×6 mm (§3.3). How such a volume is summarised as a two-dimensional en-face image is described below, and how the varying native grids are brought to one modelling grid in §3.3.

Two further imaging modalities are used alongside OCT. Scanning laser ophthalmoscopy (SLO) scans a laser point by point across the fundus and records the reflected light, giving a sharp two-dimensional image of the retinal surface. The OCT device used for the cohort records a near-infrared reflectance image of this kind together with each volume, aligned with the OCT scans (Mai et al., 2024). Fundus autofluorescence (FAF) imaging records the natural fluorescence of lipofuscin, a breakdown product of photoreceptor outer segments that accumulates in the RPE. Where the RPE is lost, the signal is lost, so GA appears as a dark area with high-contrast borders (Figure 2.3), and FAF has become the gold standard for detecting and measuring GA area (Boopathiraj et al., 2024).

Although FAF is the standard for measuring GA area, Vallino et al. (2024) give three reasons to prefer OCT: it shows each retinal layer in cross-section and so

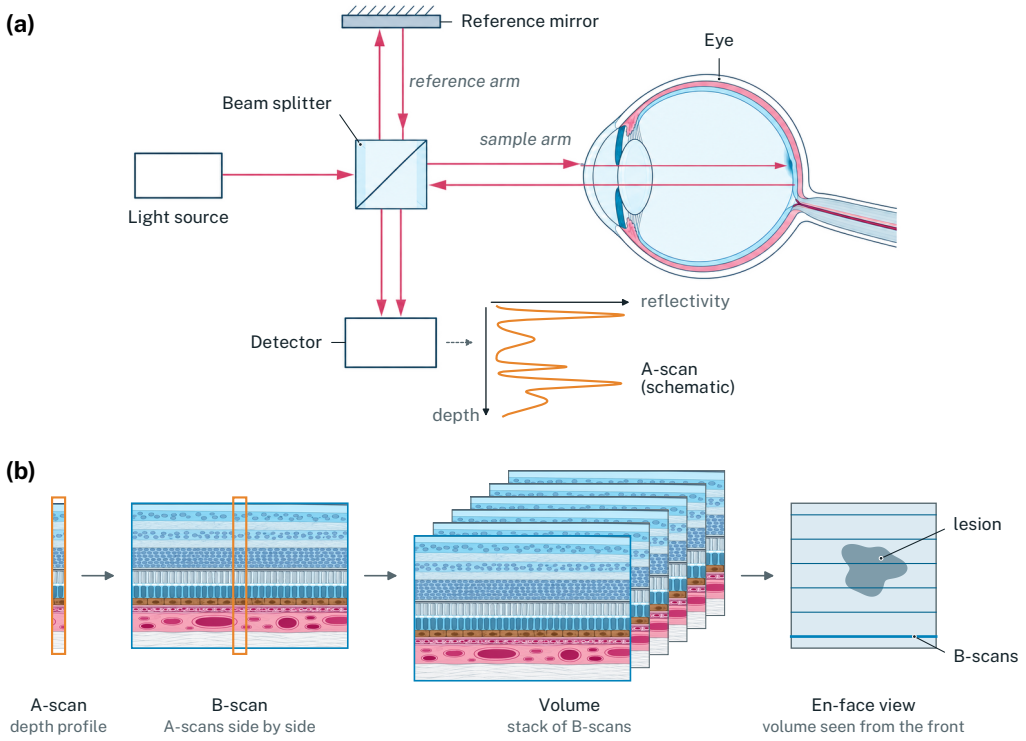


Figure 2.2: Measuring principle and scan geometry of OCT (schematic). *(a)* Light from the source is split by the beam splitter into a reference arm, which is reflected by a mirror, and a sample arm, which enters the eye and is reflected by the retina; both reflections are recombined and recorded by the detector (D. Huang et al., 1991). For one position of the beam this gives a depth profile of reflectivity, an A-scan; the curve shown is schematic. *(b)* From A-scan to en-face view: A-scans recorded side by side along a line form a B-scan, B-scans recorded along parallel lines form the OCT volume, and the volume seen from the front of the eye is the en-face view, in which every line marks the position of one B-scan and a GA lesion appears as a patch. The highlighted A-scan, B-scan and line correspond to each other. The modelling window of this thesis covers about 6×6 mm (§3.3). Drawings generated with ChatGPT; labels added and verified by the author.

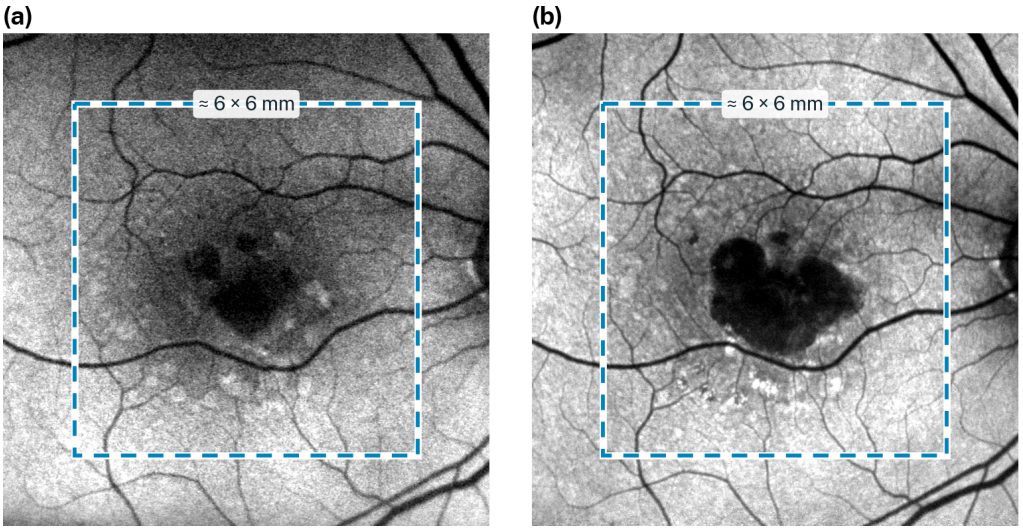


Figure 2.3: Geographic atrophy on fundus autofluorescence in one eye of the cohort. *(a)* First visit. *(b)* The same eye three years later. The lesion is the dark area, where the RPE has been lost. It has grown, and its two largest patches have merged. The dashed square marks the region the models receive (§3.3). Brightness is not comparable between the two panels.

permits layer-wise quantitative analysis, it detects atrophy earlier in its natural history, and it remains reliable near the fovea, where blue-light FAF is less reliable. OCT-based area measurements are also reproducible and consistent with FAF: Pilotto et al. (2015) report inter-reader intraclass correlation coefficients of 0.97–0.98 for en-face OCT at the outer-retinal level and find the resulting GA areas at that level statistically equivalent to blue-light and near-infrared FAF measurements in a small cohort. In this thesis, the lesion mask and the layer maps are both given on the OCT en-face grid; how the mask was annotated is described in §3.1.

A single OCT volume shows both the lesion and the retinal layers around it. §2.1.3 describes how the lesion appears in it.

From OCT volumes to en-face maps. Models of GA progression have also been trained on full OCT volumes (Lachinov et al., 2024; Mai et al., 2024). The clinical literature on GA area and growth, however, works with an *en-face* image: a two-dimensional map of the retina seen from the front, as in a fundus photograph, in which every pixel corresponds to one A-scan of the volume and summarises what lies along it (Chu et al., 2022; Pilotto et al., 2015; Vallino et al., 2024; Yehoshua et al., 2011) (Figure 2.2(b), right). This thesis adopts the same convention.

Several ways of forming the en-face image from the volume are used in the literature, for example summing the full depth of each A-scan (Yehoshua et al., 2011) or taking only a thin coronal section at a fixed depth relative to Bruch's membrane (Pilotto et al., 2015). No projection is computed in this thesis. Instead, the MUW data provide two kinds of en-face maps for every visit, on the same grid: the GA lesion as a binary mask, and, for each of ten retinal layer

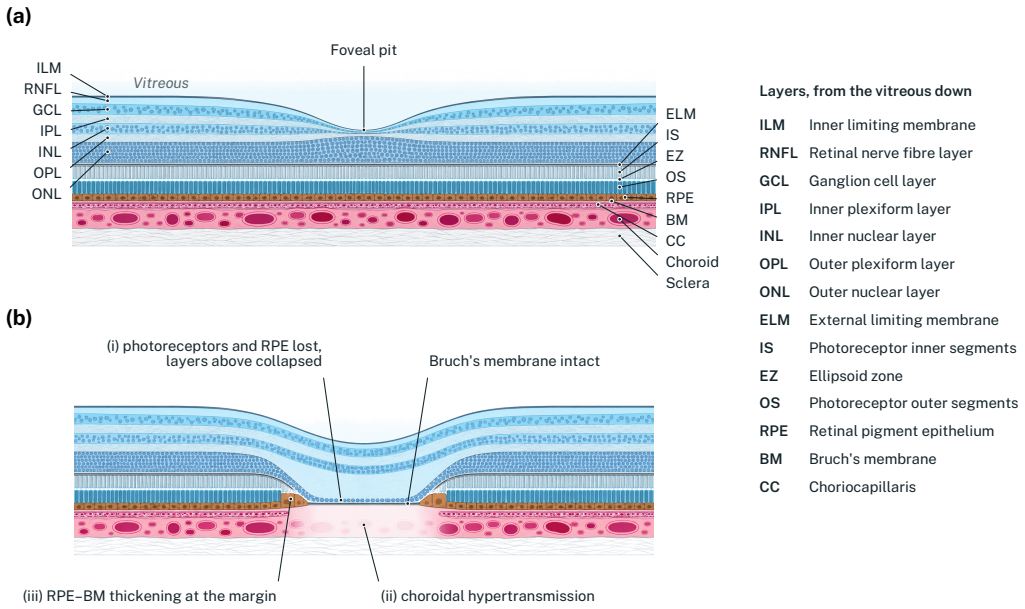


Figure 2.4: Retinal layers and the OCT signs of geographic atrophy (schematic, layer thicknesses not to scale). *(a)* Healthy retina around the foveal pit, with the retinal layers, Bruch's membrane, the choriocapillaris, the choroid and the sclera from the vitreous downwards; the key on the right gives the full names of the abbreviations (Abràmoff et al., 2010; Sadda et al., 2018). In the foveal pit the inner layers thin out, while the layers from the ELM downwards continue underneath. *(b)* The same strip with a GA lesion, marked with the two OCT signatures and the margin feature described in the text: (i) the photoreceptors and the RPE are lost and the ONL is reduced to a remnant, so the layers above it lie closer to Bruch's membrane, which remains; (ii) the choroid beneath the lesion appears brighter (choroidal hypertransmission); (iii) the RPE–BM complex is thickened at the lesion margin (Chu et al., 2022; Sadda et al., 2018). The choriocapillaris is lost under the lesion (Boopathiraj et al., 2024). Drawings generated with ChatGPT; labels added and verified by the author.

boundaries, its depth along every A-scan (§3.1). These maps are the imaging input the models in this thesis receive.

The en-face representation is also the view on which quantitative growth-rate metrics and layer-thickness analyses are reported in the recent literature (Chu et al., 2022; Vallino et al., 2024; Vogl et al., 2021). The en-face grid also reduces the forward problem of each step from three dimensions to two, which matters for the tractability of the models in the following chapters.

2.1.3 Geographic Atrophy

Figure 2.4(a) sketches the principal anatomical strata of the retina in the macula. Geographic atrophy is the disease in which three tightly coupled layers die together in a localised patch of the macula: the retinal pigment epithelium (RPE), the photoreceptors just above it, and the choriocapillaris, the innermost

capillary bed of the choroid below it (Boopathiraj et al., 2024; Boyer et al., 2017). These three layers are anatomically interdependent: the RPE supports the photoreceptors, and the choriocapillaris supplies both. Atrophy of one layer therefore leads to the degeneration of the others, and the resulting lesion appears on imaging as a sharply delineated patch (Figure 2.3) in which the entire outer-retinal complex is missing rather than as the loss of a single isolated layer.

An international consensus group, the Classification of Atrophy Meetings (CAM), has formalised the OCT appearance of this end-stage lesion under the term *complete RPE and outer retinal atrophy* (cRORA) (Sadda et al., 2018; Vallino et al., 2024). cRORA is identified by three concurrent OCT features: *(a)* a region of choroidal hypertransmission of at least 250 µm in diameter; *(b)* co-located attenuation or disruption of the RPE band of at least 250 µm; and *(c)* overlying photoreceptor degeneration, all in the absence of an RPE tear. cRORA is thus the OCT-readable definition of an established atrophic lesion, which can arise with or without macular neovascularisation (the wet form of AMD), and *GA is the subset of cRORA that arises in eyes without macular neovascularisation.* The CAM framework also recognises a precursor stage, *iRORA* (incomplete RPE and outer retinal atrophy), which shows the same OCT pattern below the 250 µm size threshold. In one retrospective study of intermediate-AMD eyes, 93 % of iRORA lesions progressed to cRORA within twenty-four months, with a median conversion time of fourteen months (Vallino et al., 2024). Working on en-face maps rather than on single B-scans is also supported by a caveat documented by Vallino et al. (2024): a lesion classified as iRORA on a single horizontal B-scan can be part of a larger cRORA lesion that is fully visible only in the en-face slab. Channel 0 of the state tensor used in this thesis marks the GA lesion as a binary mask (§3.2); iRORA is not represented as a separate label.

An established GA lesion is identified on a B-scan by two direct OCT signatures (Boopathiraj et al., 2024; Vallino et al., 2024; Yehoshua et al., 2011). *(i)* The outer-retinal layers are missing: the photoreceptor inner-segment / outer-segment junction (the *ellipsoid zone*) and the RPE band drop out of the B-scan, leaving a structurally collapsed column where intact retina used to be. *(ii) Choroidal hypertransmission* appears beneath the lesion: because pigmented RPE no longer absorbs the probe beam, light penetrates deeper into the choroid, and the column-summed signal is higher than in the surrounding intact tissue. This is the same optical effect that makes GA appear bright on summed-voxel en-face projections, discussed in §2.1.2. A third feature lies just outside the lesion: *(iii) RPE–Bruch's-membrane (BM) thickening* is concentrated in a narrow rim just outside the lesion margin and is thought to reflect, at least in part, basal-laminar deposits accumulated in the diseased tissue; the thickness of this rim correlates significantly with the eye's subsequent annual square-root lesion enlargement rate (Chu et al., 2022). The two signatures delineate the lesion; the third feature is a predictive biomarker at its boundary and motivates encoding the surrounding layer geometry in addition to the lesion mask.

2.2 Partial Differential Equations and Numerical Solvers

The models in this thesis are built like numerical solvers for partial differential equations (PDEs): a state is held on a grid and advanced step by step by an update rule. The later chapters therefore use the vocabulary of numerical solvers throughout. The update of the framework is one explicit Euler step in residual form (Chapter 4). The operators placed in it are learned stencils, global spectral methods and finite elements. The experiments ask how far one update reaches, whether a higher-order time integrator helps, and when a step becomes unstable (Chapter 5). This section introduces these mechanisms.

A *temporal* PDE describes a field $u(t, x)$ that changes over time t at every point x of a spatial domain $\mathcal{X} \subset \mathbb{R}^D$. It states how fast the field changes at each point, given the current field around that point:

$$\partial_t u \;=\; F(t, x, u, \partial_x u, \partial_{xx} u, \dots), \qquad (t, x) \in [0, T] \times \mathcal{X}, \tag{2.1}$$

where $u : [0, T] \times \mathcal{X} \to \mathbb{R}^n$ has n components and F is the law of the process. Because F involves only u and its derivatives at x, and derivatives are local quantities, the rate of change at a point depends only on the field in an arbitrarily small neighbourhood of it. The stencils of §2.2.1 exploit this locality. Together with an initial condition $u(0, x) = u^0(x)$ and a condition on the domain boundary, (2.1) determines $u(t, \cdot)$ for all $t > 0$. Closed-form solutions exist only in special cases, so the solution is computed numerically. A numerical solver does this in two steps: it replaces the continuous domain by a grid and the spatial derivatives by operations on grid values (§2.2.1), and it then advances the grid values in discrete time steps (§2.2.2). §2.2.3 then summarises the strengths and limits of this design.

2.2.1 Discretising space: stencils

The domain is partitioned into a grid of N small cells c_i with centres x_i, and the solution is stored as one value per cell and time step, $u_i^k \approx u(t_k, x_i)$; $u^k = (u_1^k, \dots, u_N^k)$ denotes the whole grid. The simplest grid is a uniform one, a regular array of equal rectangular cells with constant spacings Δx and Δy. The two spacings need not be equal. On such an anisotropic grid, a stencil that is symmetric in grid indices reaches different physical distances along the two axes.

Finite differences and finite volumes. The finite difference method (FDM) replaces each spatial derivative by a difference of neighbouring grid values (LeVeque, 2007). On a uniform one-dimensional grid, the central approximation of the first derivative is

$$\left.\frac{\partial u}{\partial x}\right|_{x_i} \;\approx\; \frac{u_{i+1} - u_{i-1}}{2\Delta x}. \tag{2.2}$$

In general, the n-th derivative at cell i is estimated as a weighted sum over a small neighbourhood $\mathcal{N}(i)$,

$$\left[\partial_x^{(n)} u\right]_i \;\approx\; \sum_{j \in \mathcal{N}(i)} \alpha_j^{(n)} u_j, \tag{2.3}$$

with fixed coefficients $\alpha_j^{(n)}$ obtained by fitting a local polynomial to the neighbourhood and differentiating it. The neighbourhood together with its coefficients is called a *stencil*. Inserting such stencils into F gives a function f that maps the grid values u^k to a rate of change for every cell. The finite volume method (FVM) follows the same pattern, but applies its stencils to the flow of a quantity between neighbouring cells rather than to derivatives at cell centres (LeVeque, 2002).

Finite elements. The finite element method (FEM) divides the domain into simple elements, typically triangles, and represents the solution as a sum of basis functions, each of which is non-zero only on the elements around one node (Courant, 1943; Langtangen and Mardal, 2019). With piecewise-linear basis functions, the coefficients are exactly the node values. Instead of requiring (2.1) to hold at every point, the method requires it to hold on average against each basis function, the so-called *weak form*. Because each basis function overlaps only those of its immediate neighbours, every node is coupled only to the nodes it shares an element with. The result is again a local stencil, but one defined by the mesh rather than by a regular grid, which makes the method well suited to irregular domains. The Finite Element Networks of §2.3 learn their dynamics inside this discretisation.

Adaptive stencils. A fixed stencil that straddles a sharp front produces spurious oscillations next to it. Weighted essentially non-oscillatory (WENO) schemes avoid this by computing several candidate estimates on shifted sub-stencils. The estimates are combined with weights that depend on how smooth the data are in each sub-stencil, so that sub-stencils crossing the front receive almost no weight (Jiang and Shu, 1996; X.-D. Liu et al., 1994). The stencil thereby becomes a non-linear function of its input, which makes WENO the classical mechanism closest to what a learned operator does.

Global methods. Pseudospectral methods compute derivatives in Fourier space instead of from a neighbourhood: the field is transformed, each mode is multiplied by $(i\omega)^n$, and the result is transformed back,

$$\partial_x^n u \;=\; \mathcal{F}^{-1}\big[(i\omega)^n\,\hat{u}(\omega)\big], \qquad \hat{u} = \mathcal{F}[u].$$

The derivative at every point then depends on all grid values. The accuracy is excellent for smooth solutions on periodic domains, but the method breaks down for non-smooth solutions (Gottlieb and Orszag, 1977; Trefethen, 2000).

Locality and reach. Stencil methods are *local*: the rate of change of a cell is computed from a small neighbourhood, so one evaluation of f moves information by at most one stencil radius. This distance, the *reach* of the stencil, is set in physical units by the width of the stencil and by the grid spacing. The contrast between local stencils and global methods reappears in the neural architectures of §2.3.

2.2.2 Stepping in time

The stencils of §2.2.1 reduce the spatial part of the problem to the rate function f; the grid then has to be advanced in time with this rate. The two parts are independent: the same f can be combined with any time-stepping scheme, and the choice of scheme does not change how f is computed. The simplest time-stepping scheme is the explicit Euler step (Butcher, 1987; Hairer et al., 1993),

$$u_i^{k+1} \;=\; u_i^k \;+\; \Delta t\, f_i\Big(t_k, u^k\Big), \qquad \Delta t = t_{k+1} - t_k, \tag{2.4}$$

which holds the rate constant over the step. Runge–Kutta methods reach higher accuracy by evaluating f several times within a step and combining the results; the classical fourth-order scheme (RK4) uses four evaluations.

Spatial stencil and time integrator together define one update step

$$u^{k+1} \;=\; \mathcal{A}\Big(\Delta t, u^k\Big), \tag{2.5}$$

and a trajectory is produced by applying $\mathcal{A}$ repeatedly from u^0, each step consuming the output of the previous one. This repeated application is called a *rollout*. Figure 2.5 summarises the path.

Two properties decide whether such an update is usable. The first is *consistency*: as the step shrinks to zero, the update must reproduce the exact solution, $\lim_{\Delta t \to 0} \| \mathcal{A}(\Delta t, u^0) - u(\Delta t) \| = 0$ (Brandstetter et al., 2022). An update written as the current state plus Δt times a rate, as in (2.4), is said to be in *residual* form; it returns the current state at $\Delta t = 0$ whatever the rate. Consistency in the stricter sense used for convergence also requires the rate to approach the right-hand side F. The second is *stability*: errors made in one step must not grow from step to step. For explicit schemes this bounds the step size. The Courant–Friedrichs–Lewy (CFL) condition (Courant et al., 1928) states that, for a process that moves information at speed c, one step may not carry information further than the stencil reaches,

$$\frac{c\,\Delta t}{\Delta x} \;\leqslant\; C_{\max}, \tag{2.6}$$

with a constant $C_{\max}$ of order one that depends on the scheme. Equivalently, a stencil used with a given step must be wide enough to cover the distance the solution moves during that step.

2.2.3 Strengths and limits

Classical solvers are well understood. Each component has a known error behaviour, convergence to the true solution can be proven for broad classes of problems as the grid is refined (LeVeque, 2002), and no training data are needed.

Their limits follow from the same design. The equation F, including its coefficients, must be known, because the solver can only compute what F prescribes. Good schemes are hand-crafted for each class of equation. Following Bartels (2016), Brandstetter et al. (2022) describe the field of numerical methods as a "splitter field" that builds a dedicated solver for each sub-problem, rather than a "lumper field" that seeks one method for all of them. Accuracy is also expensive. Fine features need fine grids, the CFL condition then forces small time steps, and the cost grows faster than the resolution.

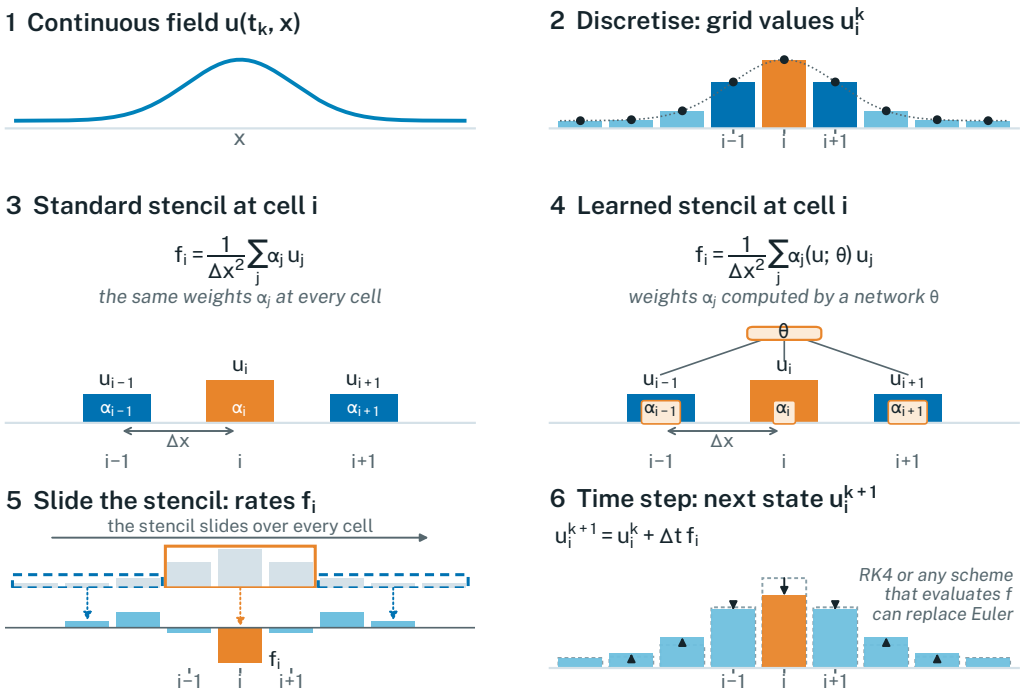


Figure 2.5: The computational path of a PDE solver, in one dimension. Each panel title names the step that leads to it; cell i is orange throughout. *(1)* Continuous field. *(2)* Discretisation: the field is sampled at the cell centres. *(3)* On a uniform grid with spacing Δx, a standard stencil is a weighted sum of the values of cell i and its two neighbours (blue) (Eq. 2.3, with the factor $1/\Delta x^2$ written separately). It uses the same weights α_j at every cell, chosen depending on the derivative it approximates; from this derivative the PDE gives the rate f_i. *(4)* A learned stencil, for example that of Bar-Sinai et al. (2019), keeps the same sum, but a network with parameters θ computes the weights from the local values, so they differ from cell to cell. Message-passing networks are another example: they learn the whole combination of the neighbours' values (§2.3). *(5)* The stencil is applied at every cell (three positions shown); each position gives the rate of its centre cell. *(6)* One explicit Euler step (Eq. 2.4) moves each cell by Δt times its rate; dashed: the values before the step, arrows: the change $\Delta t\, f_i$. Any scheme that only evaluates f, such as Runge–Kutta, can replace Euler without changing the stencil. The bar heights are illustrative.

2.3 Neural PDE Solvers and Surrogate Architectures

This section introduces the learned counterparts of these mechanisms: how learned solvers keep or replace parts of the classical computation, the two ways of using them in time, the error accumulation of autoregressive rollouts and the training remedies against it, and the architecture families that Chapter 5 compares.

Every mechanism of §2.2 is a function from a neighbourhood of grid values to a number: the stencil coefficients, the WENO weights, the rate that the time

step applies. Any of them can be learned from data instead of derived from an equation, and much of the neural-solver literature does exactly that while keeping the rest of the computation. *Hybrid* solvers keep the classical scheme and learn selected parts of it. Bar-Sinai et al. (2019) learn the stencil coefficients of (2.3) and obtain accurate solutions on considerably coarser grids than hand-designed stencils allow; Kochkov et al. (2021) apply the same idea inside a solver for turbulent flow. *End-to-end* solvers learn the whole update $\mathcal{A}$ but keep its shape: a state on a grid or mesh, a local operator, a step in time. Graph neural networks built this way simulate fluids, deformable materials and cloth (Pfaff et al., 2021; Sanchez-Gonzalez et al., 2020). A graph-network weather forecaster trained on decades of reanalysis data outperforms a leading operational numerical model on most of its verification targets (Lam et al., 2023).

The message-passing solver of Brandstetter et al. (2022) (MP-PDE) makes the correspondence between message passing and a classical stencil explicit. The grid becomes a graph whose nodes are the cells and whose edges connect each cell to its stencil neighbourhood. One round of message passing computes, at every node, a learned function of its neighbours' values and of their differences to its own value; it thus acts as a learned, non-linear stencil (Figure 2.5, panel 4). The authors show that one, two and three rounds can represent finite differences, finite volumes and WENO respectively, so the classical schemes are special cases of the learned one. The output is applied in the residual form of (2.4), so the update reduces to the current state as the step shrinks. The graph operator of this thesis is built on this solver and is described in §4.3.2.

A learned solver does not require f to be written down, because f is fitted to data. The solver can therefore be used where the equation is expensive to solve, known only approximately, or unknown, and it can often run on coarser grids with larger steps. In exchange, the classical guarantees are lost. A learned solver needs training data, has no convergence proof, and in a rollout it consumes its own predictions, so its errors can accumulate from step to step. This problem and its remedies are discussed later in this section.

Learned solvers also differ in how they use time. A direct approach learns a continuous map $M : [0, T] \times \mathcal{F} \to \mathcal{F}$ such that $M(t, u_0) = u(t)$, and so predicts the solution at any time directly from the initial condition. By contrast, autoregressive solvers learn the update $\mathcal{A}$ of (2.5) and apply it repeatedly to unroll a trajectory. The two framings concern the use of a network, not its architecture: a neural operator such as the FNO can serve as a direct map or, applied step by step, as an update (Z. Li et al., 2021), which is how it is used in this thesis (§4.3.4). When the step size Δt is fixed by the dataset, the dependence on it is usually omitted from the notation. Here it cannot be omitted, because the intervals between clinical visits are irregular.

Finite-dimensional neural operators are bound to the grid geometry they were trained on, whereas infinite-dimensional ones are defined independently of the spatial discretisation. The Fourier Neural Operator (Z. Li et al., 2021) parameterises a kernel integral operator in the Fourier domain, which gives the network global spatial support within a single layer. DeepONet (Lu et al., 2021) instead factorises the operator into a branch network that encodes the input function and a trunk network that encodes the continuous query coordinate. Used as direct maps, both share a well-known limitation: a trained operator is tied to the equation family it was trained on, and its extrapolation beyond the temporal horizon seen during training is consistently poor.

The autoregressive framing is the natural fit for clinical longitudinal data. Each visit-to-visit transition constitutes one step. Since there is no shared temporal origin ($t = 0$) across patients, time can only be measured from each eye's own visits, not on a time axis shared by all eyes; an autoregressive update needs only the interval to the next visit. In addition, the prediction horizon is set by the clinic rather than by the model, so the model has to support as many steps as a patient's visit schedule requires.

The principal cost of this framing is the error accumulation noted at the start of this section: after the first step, the solver consumes its own output instead of numerical ground truth, so the inputs it receives during a rollout differ systematically from those seen in training. This distribution shift is the central training difficulty for autoregressive rollouts.

Together with their message-passing solver, introduced above, Brandstetter et al. (2022) propose two mechanisms against this distribution shift. The first is temporal bundling: the model predicts K future steps in one forward pass. A rollout of a given length then needs K times fewer solver calls, and the model is exposed to fewer shifts in its input distribution. The second is the pushforward trick. During training, the input state is replaced by the model's own one-step prediction from the preceding state, so the model is trained on the kind of input it will meet during a rollout. Two steps are unrolled, and gradients are propagated only through the last. Backpropagating through the first step as well would teach the model to make its errors small rather than to correct them.

The authors relate the pushforward trick to zero stability. An update is zero-stable if a small perturbation of its input changes the output only by a bounded multiple, $\|\mathcal{A}(u^0 + \epsilon) - u^1\| \leqslant \kappa\|\epsilon\|$, and the pushforward loss reduces the constant κ. Together with consistency (§2.2.2), zero stability is the classical basis of a convergence argument. Chapter 4 describes how the framework of this thesis treats temporal bundling and the pushforward trick.

Beyond graph networks, several distinct architecture families are used as PDE surrogates. The U-Net (Ronneberger et al., 2015) is an encoder-decoder architecture with skip connections and a multi-scale receptive field. It serves as a standard strong surrogate baseline in the neural-PDE literature (Gupta and Brandstetter, 2023). The architectures also differ in their spatial context: global spectral operators such as the Fourier Neural Operator couple all grid points within one layer, whereas convolutional and message-passing operators are local. Hybrid architectures add a small local kernel bypass to a global operator to capture high-frequency details more effectively (Liu-Schiaffini et al., 2024). Finite Element Networks (Lienen and Günnemann, 2022) build the neural update directly from a finite-element discretisation with learned dynamics, and their transport variant carries an explicit transport (advection) term. Finally, because a residual step resembles one explicit Euler step, the network is sometimes read as the right-hand side of an ordinary differential equation, the Neural-ODE reading (Chen et al., 2018). A higher-order integrator such as a Runge–Kutta scheme can then in principle replace the residual step. This reading is only legitimate if the trained model is nearly invariant to the solver used at inference. The property has to be tested empirically, because classical solver guarantees do not automatically transfer to neural parameterisations (Krishnapriyan et al., 2023; Ott et al., 2021).

The architecture families differ mainly in four structural properties: whether

the operator has spatial context at all, how far that context physically reaches, whether it combines global and local information, and whether it carries an explicit transport term. Such built-in assumptions of a model are called its inductive biases. Chapter 5 measures the four properties one at a time.

2.4 Related Work

This section places the thesis among four groups of prior work and identifies what they leave open. Section 2.4.1 covers models that predict the risk or the rate of AMD and GA progression, §2.4.2 models that forecast where a GA lesion will grow, §2.4.3 forecasts from irregularly sampled longitudinal data in general, and §2.4.4 PDE-based models in biomedicine.

2.4.1 Predicting the Risk and Rate of Progression

Schmidt-Erfurth et al. (2018) predicted from OCT, demographic and genetic features whether an eye with intermediate AMD converts to neovascular AMD or GA within two years (495 eyes; area under the curve 0.80 for conversion to GA). Bogunović et al. (2017) predicted the regression of individual drusen, a sign of disease activity, in intermediate AMD from OCT (61 eyes; area under the curve 0.75 within two years). Vogl et al. (2021) provided a topographic analysis of the retinal and choroidal layer changes that precede the conversion of intermediate AMD to macular atrophy or neovascularisation.

Deep-learning models have also been trained to predict the growth rate of GA in an individual eye. From OCT volumes, four different approaches reached a squared correlation of 0.33 to 0.35 between predicted and observed growth rate on a holdout set (Yoshida et al., 2025). For growth-rate models trained on FAF images, occlusion experiments showed that a rim of tissue around the lesion contributes more to the prediction than the lesion itself or the background (Julia Cluceru et al., 2024). Most of this literature formulates prediction at cohort level or per eye, as a categorical time to conversion or a single rate. The formulation used here instead requires a per-eye, per-time-step prediction of the full spatial next state.

2.4.2 Spatial Forecasting of GA Lesions

A single rate per eye does not describe where a lesion grows, because the border of a GA lesion advances at different local speeds. The lateral spread of GA was slowest at regular borders, at about 63 µm per year, faster at irregular borders, at about 91 µm per year, and fastest at splitting borders, at about 183 µm per year (Lindner et al., 2018). Closer to the task of this thesis are therefore models that forecast the lesion itself. Gigon et al. (2021) predicted the progression of RPE and outer retinal atrophy (RORA; cf. cRORA, §2.1.3) from OCT en-face maps of layer thicknesses, mean reflectivity and drusen height in 129 eyes of 119 patients. Their model gives each en-face pixel a probability of atrophy over time, from which a map of the time to conversion follows; the Dice score ranged from 0.73 to 0.80 for the total RORA area and from 0.46 to 0.72 for the growth region.

Lachinov et al. (2024) encode each pixel of a baseline scan as a feature vector and evolve these vectors with a Neural ODE (Chen et al., 2018) to any requested time point, using a fourth-order Runge–Kutta scheme. Their model was developed on 967 OCT volumes of 100 patients with GA and also applied to brain magnetic resonance imaging (MRI) in Alzheimer's disease. The Runge–Kutta-integrated operators of this thesis (§4.3.7) are related to this continuous-time formulation.

The closest prior work on the specific prediction task addressed here is by Mai et al. (2024), who predicted individual GA progression from a single baseline OCT with this Neural ODE model, on the MUW GA cohort (184 eyes of 100 patients). The data of this thesis are drawn from the same cohort, with tighter registration bounds and cleaner segmentations (75 eyes of 51 patients, §3.1). The input regime, however, is different. That work operates directly on raw OCT volumes, whereas the present thesis uses pre-segmented lesion masks and layer surfaces. As a precedent for dense convolutional architectures on this task, Salvi et al. (2025) predicted future GA growth with a two-dimensional U-Net on fundus autofluorescence images.

2.4.3 Forecasting from Irregularly Sampled Longitudinal Data

Irregular intervals between observations are a general property of longitudinal clinical data. Latent ODE and ODE-RNN models let a hidden state evolve in continuous time between observations and so handle arbitrary time gaps (Rubanova et al., 2019). For images, ImageFlowNet learns a multiscale latent representation and a flow field in it with a neural ODE or a stochastic differential equation (SDE), and forecasts image-level trajectories of disease progression. It was evaluated on FAF images of GA, on multiple sclerosis and on glioblastoma, an aggressive brain tumour (C. Liu et al., 2025). In brain imaging, 4D-DANI-Net generates subject-specific longitudinal MRI scans that mimic neurodegeneration in ageing and dementia (Ravì et al., 2022), and BrLP combines a latent diffusion model with an auxiliary model of regional volume changes to predict the evolution of individual 3-D brain MRIs (Puglisi et al., 2024). These models treat time continuously or condition on it. The default update of this thesis takes the interval as an input to a single residual step (§2.5); the Runge–Kutta wrapper of §4.3.7 instead integrates an operator without a Δt input over the interval, which is closer to these continuous-time models.

2.4.4 PDE-Based Models in Biomedicine

Where the governing equation of a disease process is known, PDE models are fitted to individual patients. For glioblastoma, tumour growth models based on the Fisher–Kolmogorov equation are personalised with Bayesian inference from MRI and FET positron emission tomography (FET-PET) scans to design radiotherapy plans (Lipková et al., 2019). Learned components speed this up: a neural surrogate that replaces the numerical solver of such a tumour model is about 50 times faster (Ezhov, Mot, et al., 2022), and a network that solves the inverse problem infers the patient-specific tumour cell distribution within minutes (Ezhov, Scibilia, et al., 2023). Physics-informed machine learning in medical image analysis more broadly has been reviewed in a survey of over 80

papers (Banerjee et al., 2025). In all of these, the equation is known and the network learns to solve or invert it. Without a known equation, GA progression has been modelled with learned continuous-time dynamics (Lachinov et al., 2024; C. Liu et al., 2025).

Across the four groups, a controlled comparison of operators from different architecture classes inside one fixed framework has, to the author's knowledge, not been reported for GA. It is the comparison this thesis makes.

2.5 Modelling Approach and Problem Statement

This section brings the clinical and the numerical background together. It first explains why the structure of a numerical solver suits the forecasting of GA (§2.5.1) and then states the problem formally, in the notation used in the remaining chapters (§2.5.2).

2.5.1 Choice of the Modelling Approach

A clinical OCT follow-up gives the data three properties that a model must handle. First, follow-up visits are *irregularly spaced* in time and the spacing varies substantially between patients and between consecutive visits of the same patient. Irregular spacing rules out methods that assume a fixed sampling rate, and the lack of a shared temporal origin across eyes rules out methods that assume a common $t = 0$. Second, the disease state is *multi-channel*: a binary mask marks the GA lesion, and ten retinal layer boundaries describe the retinal structure in which the atrophy lies. The layer channels are relevant because prognostic biomarkers documented in the clinical literature, such as RPE–Bruch's-membrane thickening (Chu et al., 2022) and thinning of the outer retinal layers (Vallino et al., 2024), are not contained in the lesion mask alone. Third, every patient carries *covariates*, in particular age and sex. Age is strongly associated with the risk of developing the disease (§1.1), and the evidence on sex is mixed; whether either also helps to forecast where and how fast a lesion grows is not established.

Handling these properties depends on the model and on how it is used. A typical image-to-image model maps one image to another; on its own, it has no notion of elapsed time and no built-in way to advance a state from one visit to the next. Both abilities have to be supplied by a framework around the model that conditions the prediction on Δt and advances the state from one visit to the next. Such a framework can borrow several of its techniques from numerical and neural PDE solvers, as the remainder of this subsection explains. Which model should sit inside it is a separate question.

GA expansion is, to a first approximation, a process of *local boundary dynamics*. Yehoshua et al. (2011) found that the enlargement rate of the square root of the lesion area does not depend on the baseline lesion size. This is what is expected if the radius of a GA lesion grows at a roughly constant rate while its enclosed area scales quadratically. With $A(t)$ the lesion area and $r(t)$ an effective radius,

$$\frac{dr}{dt} \approx \text{const.} \quad \Longleftrightarrow \quad \sqrt{A(t)/\pi} \;=\; r_0 + \nu\, t, \tag{2.7}$$

which is consistent with the data within a clinical follow-up horizon of six months to slightly over two years (Yehoshua et al., 2011); the square-root enlargement rate has been re-used in subsequent biomarker work (Chu et al., 2022). The interior of the lesion is already atrophic, and the retina further away changes little. Most of the change between visits occurs at the lesion boundary, although new atrophy can also appear at some distance from it (§5.5). Such behaviour is characteristic of a system governed by a temporal PDE with strong spatial localisation.

The same boundary, however, does not advance uniformly in all directions. GA growth is *anisotropic*: lesions progress faster towards the retinal periphery than towards the fovea, and growth towards the fovea peaks at approximately one millimetre from the fovea (Singh et al., 2025). In eyes treated monthly with pegcetacoplan, progression towards the fovea was slowed more than progression towards the periphery (Singh et al., 2025). The map from the present state to the future state is therefore expected to act mainly locally, and its reach would have to cover the distance the front moves between two visits, which differs between directions. In a classical solver, this reach is set by the size and shape of the stencil (§2.2.1).

GA progression has no known governing equation, and its data are a few observed states per eye (§3.1) instead of a densely simulated trajectory. A classical solver therefore cannot be used. The structure such a solver is built from, however, matches the problem in four respects, and the thesis adopts it for these reasons.

- **A state on a fixed grid.** Every visit is represented on the same uniform en-face grid (§3.2), so the state is exactly the vector of grid values u^k that a solver advances.
- **Change driven by the neighbourhood.** As described above, GA grows at the margin of the lesion, while its atrophic interior and the surrounding retina change little (Figure 2.3). Whether a pixel becomes atrophic depends on what surrounds it, which is the situation a stencil is designed for.
- **Slow change.** Consecutive visits differ in a small part of the grid. The residual form predicts only the change and returns the current state as $\Delta t \to 0$; its default, the unchanged state, suits data that change slowly.
- **Irregular steps.** The intervals between visits differ from eye to eye and visit to visit. A solver step is parameterised by Δt, so each step can use the actual interval between two visits instead of a fixed one.

Because the governing equation is unknown, the update has to be learned from data, as in the neural PDE solvers of §2.3. These solvers differ in the built-in assumptions named at the end of §2.3. For GA, where the governing equation is unknown and only a few dozen patients are available for training, these assumptions are likely to matter more than model size.

These four points motivate the structure but are not evidence that it fits. They also leave open what kind of spatial operator should compute the update. The CFL argument suggests that an operator applied once per visit has to see at least as far as the lesion front moves between visits. On the anisotropic en-face grid, this distance corresponds to very different numbers of pixels along the two axes: the spacing between B-scans is about 21 times larger than the spacing between A-scans (§3.3), so a neighbourhood that is symmetric in pixels is

strongly lopsided in millimetres. Whether a local stencil, a global operator or a combination of both is needed is therefore not assumed here. §2.3 introduced the operator families that are candidates for this role, and Chapter 5 compares them.

2.5.2 Problem Statement

Each eye of the cohort is observed at a sequence of visits at times $t_0 < t_1 < \cdots < t_N$, with between five and thirteen visits per eye and irregular intervals $\Delta t_k = t_{k+1} - t_k$, measured in years. At visit k, the state of the eye is a tensor

$$u_k \in \mathbb{R}^{C \times H \times W}, \qquad C = 11,\ H = 49,\ W = 1024,$$

defined on the en-face lattice $\Omega = \{1, \dots, H\} \times \{1, \dots, W\}$ (§3.2). Channel 0 is the binary GA mask, $m_k \in \{0, 1\}^{H \times W}$, and channels 1–10 are the depth maps of ten retinal layer boundaries. The lattice is anisotropic: neighbouring rows are $h_y = 121.18\,\mu\text{m}$ apart and neighbouring columns $h_x = 5.68\,\mu\text{m}$, a ratio of about 21 (§3.3). Each eye also carries patient covariates $\mathbf{z}$, its age and sex (§3.5).

The task is to forecast the state at a later visit from the state at an earlier one. The forecast for the next visit is written as one residual update,

$$\hat{u}_{k+1} = u_k + \Delta t_k\, f_\theta(u_k, \Delta t_k, \mathbf{z}), \tag{2.8}$$

where f_θ is a learned map with parameters θ. Throughout this thesis, f_θ is called the *operator*. The term is used in a general sense: through Equation (2.8) it maps the state at one time to the state at a later time. It is not used in the narrower sense of the neural-operator literature (§2.3), where it denotes models such as the FNO or DeepONet that learn maps between function spaces. A U-Net and a graph network are operators in the first sense but not in the second. A forecast over several visits is obtained by applying (2.8) repeatedly to its own output,

$$\hat{u}_{k+n} = \mathcal{A}_\theta\big(\Delta t_{k+n-1}, \dots \mathcal{A}_\theta(\Delta t_k, u_k)\big), \qquad \mathcal{A}_\theta(\Delta t, u) = u + \Delta t\, f_\theta(u, \Delta t, \mathbf{z}),$$

which is the rollout of §2.2.2 with a learned update. The parameters θ are fitted to the observed visits of the training eyes. The forecast is judged mainly on the mask: the predicted mask about one year after a visit is compared with the observed one in the region where the lesion changed (§5.1).

In this notation, the research question of §1.2 reads as follows. The first part asks what form the update and its training need: how Δt enters (2.8), how the rollout is trained, and with which objective. These choices are the *framework*. The second part asks which operator f_θ, with which built-in assumptions, gives the best forecasts on the anisotropic lattice Ω, once the framework is fixed.

Chapter 3

Data and Preprocessing

This chapter describes the data on which every operator in this thesis is trained and evaluated. Section 3.1 introduces the MUW cohort: its eyes and patients, the images and maps recorded at each visit and the origin of the lesion mask. Section 3.2 defines the eleven-channel state tensor built from these maps, and Section 3.3 explains how native grids of different sizes are brought to one common grid and what this costs. Sections 3.4 and 3.5 describe the normalisation of the channels and the patient covariates. Section 3.6 turns the visits of an eye into training windows with irregular intervals, and Section 3.7 describes the patient-level cross-validation. Together, these steps produce the input that the framework of Chapter 4 receives.

3.1 Dataset

The data used in this thesis comes from a longitudinal cohort of eyes with Geographic Atrophy and was provided by the Medical University of Vienna (MUW). It is a subset of 75 of the 184 eyes of the GA cohort described by Mai et al. (2024), selected by MUW with tighter registration bounds and cleaner segmentations, and was acquired in routine clinical practice at a single site in Vienna. The sample unit throughout is the individual eye. The cohort contains 75 eyes, which together contribute 553 visits, where a visit is one OCT acquisition. Ordering the visits of each eye in time yields 478 visit-to-next-visit intervals ($553 - 75$). The number of visits per eye ranges from five to thirteen, with a median of seven. The follow-up span, from the baseline visit to the last recorded visit of an eye, ranges from 720 to 2160 days, with a median of 1080 days, or roughly three years.

The acquisition setting recorded in the data is the same for all 553 visits. The vendor is logged as Heidelberg Engineering and the protocol as a volume scan in ART mode. For this cohort, Mai et al. (2024) report acquisition on a Spectralis device.

The 75 eyes come from 51 patients: 24 patients contribute both eyes and 27 contribute a single eye. Of the 75 eyes, 47 belong to female and 28 to male patients. Age at baseline ranges from 60.7 to 88.0 years, with a mean of 76.4 years over the eyes. Ages are decimal values, computed from the date of birth to the baseline visit.

The data does not contain the raw three-dimensional OCT volumes. Instead, each visit provides a set of registered two-dimensional en-face maps derived from them. The first is a binary GA mask on the native OCT en-face grid, stored

as an image with the values 0 and 255. The second is an array of layer-boundary depth maps of shape (rows, columns, 10), stored as 16-bit integers in units of axial pixels. Each visit further provides a scanning laser ophthalmoscopy (SLO) fundus image, a fundus-autofluorescence image and a second GA mask, all three in the SLO reference frame, together with two files of coordinate transforms between the OCT and the SLO frame. Only the OCT-grid mask and the layer array are required by the pipeline. An eye is included if at least two of its visits carry both, which guarantees at least one transition. All 75 eyes meet this condition.

Although it is stored on the OCT grid, the GA mask was not segmented on the OCT. For this cohort, Mai et al. (2024) report that GA was outlined manually on the fundus-autofluorescence image by certified graders of the Vienna Reading Center, as well-demarcated areas of markedly decreased or absent autofluorescence (Figure 2.3 shows such an image). The outline was then registered automatically to the near-infrared image that is aligned with the OCT, which yields the en-face annotation on the OCT grid. Figure 3.1 shows one visit on the FAF image and on the near-infrared image, together with the GA mask on the modelling grid and a layer-boundary depth map.

The native en-face grids do not all have the same size. The 553 visits have 67 distinct grid shapes, with 38 to 66 rows and 961 to 1719 columns. The shape is constant across the visits of one eye but differs between eyes. The variation stems from the raw acquisitions, whose B-scan spacing ranged from 115 to 258 µm. Upstream of this work, the stored en-face arrays were resampled onto one common pixel spacing: 121.18 µm between B-scans, 5.68 µm between A-scans, and 3.867 µm per axial pixel. Pixel spacing is therefore treated as constant across the cohort. Pixel counts differ between eyes, and so does the physical extent of each native grid.

Lesion growth is summarised as the yearly change in the square root of the lesion area, a transform used for GA because it removes the dependence of growth rates on baseline lesion size (Feuer et al., 2013; Yehoshua et al., 2011). Over the 75 eyes, the median square-root-area growth rate is 0.234 mm per year. The interquartile range spans 0.153 to 0.340 mm per year, and the 90th percentile is 0.473 mm per year. The baseline lesion area on the modelling grid ranges from 0.11 to 20.8 mm^2, with a median of 6.7 mm^2 and an interquartile range of 2.37 to 9.45 mm^2. Both distributions are shown in Appendix A (Figure A.1).

3.2 State Representation

The state of an eye at a visit is a tensor with $C = 11$ channels. Channel 0 holds the binary GA mask, which is both the prediction target and the clinical deliverable. Channels 1–10 hold the ten layer-boundary depth maps, ordered from the most superficial boundary to the deepest. They supply structural context and serve as auxiliary prediction targets. The layer boundaries were segmented with the Iowa Reference Algorithms (Retinal Image Analysis Lab, Iowa Institute for Biomedical Imaging, Iowa City, IA) (Abràmoff et al., 2010; Garvin et al., 2009; K. Li et al., 2006). Of the eleven surfaces the software segments, the data hold ten. From the most superficial to the deepest, they are the inner limiting membrane (ILM), the boundaries between the retinal nerve fibre and ganglion

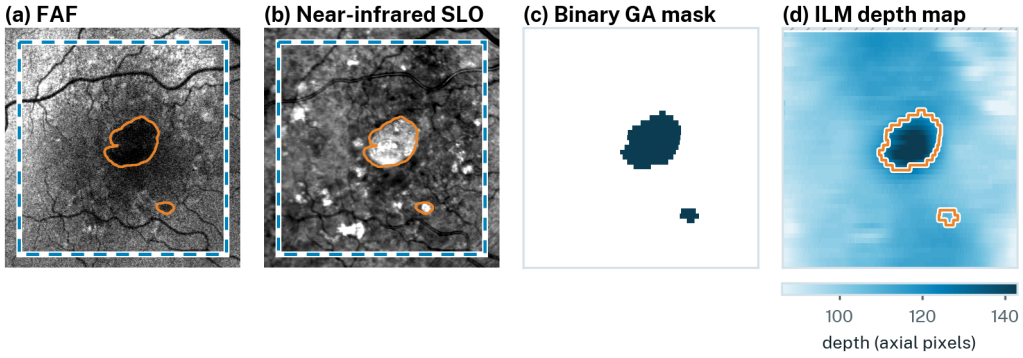


Figure 3.1: One visit of the MUW cohort. In (a) and (b), the dashed square marks the 49×1024 modelling window and the orange line the outline of the GA annotation. *(a)* FAF image, on which the lesion was annotated. The lesion appears dark. *(b)* Near-infrared SLO image of the same visit. The lesion appears bright. The other bright spots are not dark on FAF and are not part of the annotation. In (a) and (b) the annotation is drawn at the resolution of the fundus image, about 12 µm per pixel. *(c)* The binary GA mask (channel 0) on the modelling grid, drawn at its physical aspect ratio of about 5.94×5.82 mm. The rows are 121 µm apart (one B-scan each), the columns 5.7 µm, so each pixel is about 21 times taller than it is wide, which shows as steps along the lesion outline. *(d)* The depth map of the inner limiting membrane (channel 1) on the same grid, in axial pixels, with the lesion outline from (c). The lesion lies on the fovea (subfoveal, §2.1.1), so the membrane lies deeper inside it for two reasons: the foveal pit, where the inner layers thin out (Figure 2.4a), and the loss of the outer retina in GA (§2.1.3). Hatched: positions without a depth value. Panels (c) and (d) show two of the eleven channels of the state tensor before normalisation.

cell layers, between the ganglion cell and inner plexiform layers, between the inner plexiform and inner nuclear layers, between the inner nuclear and outer plexiform layers and between the outer plexiform and Henle's fibre layers, the junction of the photoreceptor inner and outer segments, the inner boundary of the outer photoreceptor layer, and the inner and outer boundaries of the retinal pigment epithelium (RPE).

The tensor is assembled per visit from the two required files. The stored mask, which has the values 0 and 255, is thresholded above 127. This yields a binary field with the value 1 inside the lesion and 0 outside it. The 16-bit layer array is cast to floating point. Mask and layers are then concatenated along the channel axis into a tensor of shape $11 \times$ rows $\times$ columns.

Each layer channel is a depth map. Fixing a pixel of the en-face grid selects a single A-scan of the underlying volume. The channel value at that pixel is the axial pixel index at which the boundary surface lies along that A-scan. Together, the ten channels describe the vertical layer structure of the retina at every en-face position. Across the cohort, the raw depth values range from -14 to 499 axial pixels, where 0 is the top of the scan volume and 496 its nominal depth.

Retinal thickness, the distance between the deepest and the shallowest bound-

ary, is not computed anywhere in this pipeline. It could be recovered from the layer channels, but it is neither stored nor used as an input.

All ten layer channels are kept next to the mask because documented prognostic biomarkers for GA progression are found throughout the vertical retinal column, not only inside the lesion itself. At the 0–300 µm rim around the GA boundary, the RPE–Bruch's-membrane thickening described in §2.1.3 correlates with the annual square-root enlargement rate at $r = 0.595$ (Chu et al., 2022). Outer retinal layer thinning, in particular of the outer nuclear layer and of the RPE-photoreceptor complex, accelerates in the months preceding macular-atrophy conversion (Vogl et al., 2021). The thickness of the inner nuclear layer in the retina immediately adjacent to a GA lesion correlates with the yearly progression rate at Spearman $r = 0.48$ (Ebneter et al., 2016). Vallino et al. (2024) further report that RPE-to-Bruch's-membrane distance combined with outer-retinal-layer thickness jointly predicts approximately 62 % of the annual GA progression, while outer-retinal-layer thickness alone contributes only 6 %. The layer channels therefore carry information on the growth rate that the mask alone cannot provide, and the layer geometry is kept in full rather than reduced to the lesion mask. How the channels are processed by the models is described in §4.2.2.

3.3 Spatial Standardisation and its Measured Cost

Because the native en-face grids differ in size, every state tensor is brought to one common grid of 49 rows by 1024 columns. Per axis, a symmetric centre crop is applied when the native size is at least the target size, and a symmetric zero-pad is applied when it is smaller. Both cases are common: 149 of the 553 visits are padded along the rows, because they have fewer than 49 rows, and 115 are padded along the columns. Cropping and padding are used instead of resampling because they introduce no interpolation artefacts and keep the stored pixel spacing identical across the cohort.

The resulting window spans 49×121.18 µm by 1024×5.68 µm, or about 5.938×5.816 mm, and holds 50 176 grid positions. Figure 2.3 marks this window on the fundus of one eye. The spacing between B-scans is **21.335 times larger** than the spacing between A-scans. The grid is therefore strongly anisotropic: moving by one row covers about the same physical distance as moving by 21 columns. This property matters for every spatial operator applied to the grid and is taken up again in Chapter 4.

The 6 mm window is a standard in the literature. Pilotto et al. (2015) explicitly exclude eyes whose GA extends beyond a 6×6 mm fovea-centred square. The same window is used by Yehoshua et al. (2011) (200×200 raster at 30 µm / pixel lateral) and by Chu et al. (2022) (Zeiss PLEX Elite 9000 swept-source OCT, 6×6 mm scan window with 1536 axial pixels and 500 A-lines per B-scan), and it serves as the de-facto analysis area for GA OCT studies.

Zero-padding raises a less obvious problem. Physically, a zero means "no signal": a mask value of 0 means not atrophic, and a layer depth of 0 is the top of the volume, above the retina. Zero also cannot serve as a sentinel that marks padding, because real layer values range from -14 to 499 and so include zero. The models, however, do not see raw values. Cropping and padding run before

the normalisation statistics are computed (§3.4), so the padded zeros enter the per-channel mean and standard deviation. After normalisation, the zero fill of the layer channels sits between -2.35 and -2.94 standard deviations, which is an extreme value. The loss treats these positions like any other, so the models are trained to reproduce this value at full weight.

The crop also has a measured cost for the pathology itself. A census over all 553 visits shows that the 49×1024 window cuts off real lesion area in 27.3 % of visits. It removes more than 5 % of the lesion in 6.0 % of visits, and 25.7 % of the lesion in the worst case. In addition, in 31.1 % of visits the lesion touches the edge of the imaged field inside the window: the border of the window or, for a padded visit, the edge of the scan itself. Growth across that edge is not observed (it is censored), and no flag in the data marks it. The models are therefore trained on censored targets. At the one-year anchor (§3.6), the lesion of 21 of the 75 eyes touches this edge (§5.1.2); for these eyes, the change-region metrics score only the change inside the window, and areas and growth rates measured on the window are biased low. The censoring also limits comparability with literature results computed over the whole field of view.

The 49×1024 window was nevertheless kept deliberately. Larger windows reduce the truncation, but they require up to 35 % zero-padding and about 1.9 times as many grid positions. They also cannot remove the problem: for about 19 % of lesions, border contact is unavoidable with any window, because those lesions reach the edge of the scan itself. The censoring therefore remains in the data and in the reported metrics, and it is discussed as a limitation (§6.3).

3.4 Normalisation

After spatial standardisation, each channel is normalised. The statistics are computed per channel over the training visits of the current fold only, and are then applied to every visit, including the validation visits. Because they are computed on the cropped and padded states, the padded zeros contribute to each channel's mean and standard deviation (§3.3).

The normalisation is a z-score,

$$x' = \frac{x - \mu_c}{\sigma_c},$$

where μ_c and σ_c are the training mean and standard deviation of channel c. A vanishing standard deviation is replaced by one to avoid division by zero.

The binary mask is z-scored in the same way as the layer channels. On fold 2, the background value 0 maps to -0.602, the foreground value 1 maps to $+1.660$, and the binarisation threshold of 0.5 maps to $+0.529$. The normalised mask therefore has a hard step at the lesion edge. Evaluation maps every threshold into normalised space rather than mapping the prediction back to the original scale.

3.5 Patient-Level Covariates

Age and sex are included as patient-level covariates because they are plausible modulators of how fast GA progresses. Each transition window, a pair of

consecutive visits of one eye (§3.6), is accompanied by two patient-level covariates, age and sex. They come from a clinical covariates table with 187 rows, which covers the wider study cohort and all 75 eyes used here. The table has one row per eye, so a patient who contributes two eyes appears in two rows with the same age and sex. Sex is encoded as 0 for female and 1 for male. A missing value is imputed with the male fraction of the training split.

Age is z-scored with training statistics. Two encodings are implemented. In the baseline mode, each eye keeps its baseline age for all of its windows. In the per-visit mode, which every reported run uses, each window carries the age at its own start visit, that is, the baseline age plus the elapsed years. The z-score statistics for this mode are computed over the start-visit ages of all training windows. Because each window contributes only the age at its start visit, the final visit of each training eye does not enter them. On fold 2 this gives 378 ages, with a mean of 77.653 years and a standard deviation of 7.220 years.

In the per-visit mode, the age covariate advances along a rollout. This leaks no information about the future, because the age at the next step follows from the current age and the input Δt alone.[1] How the covariates enter each operator is described in §4.2.3.

3.6 Temporal Structure

The longitudinal sequences are split into transition windows. For each eye, the visits are sorted by day in study, and every pair of consecutive visits forms one window. Window i consists of the state at visit i, the state at visit $i + 1$, and the interval Δt between them. Each window holds exactly one transition. This is fixed by the framework and cannot be changed by configuration. Δt is stored in years, computed as the gap in days divided by 365, so a gap of 180 days gives $\Delta t = 0.4932$. Each window also stores the raw gap in days, the two covariates and the position of the window within its eye. Over the cohort this gives 478 windows.

Table 3.1: Distribution of the intervals between consecutive visits over the 478 windows of the cohort, with the number of eyes that contain at least one interval of each length.

Interval	Windows	Share	Eyes
90 days	67	14.0 %	10
180 days	364	76.2 %	72
270 days	2	0.4 %	2
360 days	38	7.9 %	28
540 days	6	1.3 %	5
720 days	1	0.2 %	1

[1] The conversion of elapsed days into years uses 365.25 days per year for the age covariate but 365 days per year for Δt. The resulting difference is at most 0.07 %. It is deliberately left unchanged, because aligning the two would invalidate every precomputed dataset.

Table 3.1 gives the distribution of the intervals between consecutive visits. Three properties of this schedule shape the rest of the pipeline. First, every interval is a multiple of 90 days, and the most common interval, 180 days, is the base of the time budget used in training (§4.2.5). Second, the short 90-day intervals are concentrated in a small subgroup: all 67 of them belong to 10 eyes from 7 patients. A training scheme restricted to the 90-day intervals would therefore learn from this densely imaged subgroup alone.

Third, the schedule limits how far an eye can be rolled forward during training, because the training curriculum chains visits within a fixed time budget (§4.2.5). With a budget of 90 days, none of the 75 eyes can chain two visits, so no step is unrolled. With 360 days, 74 of the 75 eyes reach at least one unrolled step. With 540 days, 68 eyes reach at least two, and with 720 days, 73 eyes do.

Beyond training, the schedule also affects the evaluation. The change-region metrics of Chapter 5 are taken at a one-year anchor, the first rollout step whose cumulative time reaches about 347 days. This one-year anchor has no upper bound. For 68 of the 75 eyes, it falls at exactly 360 days. The remaining 7 eyes (9.3 %) are scored at 450 to 720 days, where the true change region is 1.3 to 1.7 times larger. This is accepted so that every eye contributes one score at the anchor. It departs from the yearly bins of Mai et al. (2024), which would place these visits in the second year. The evaluation protocol is defined in §5.1.

3.7 Splits and Cross-Validation

The data is partitioned into five folds for a patient-level cross-validation. The folds come from split files supplied with the data, in which patients were assigned at random, stratified by quartiles of their mean GA area over all visits. The five validation sets do not overlap, and together they cover all patients. The folds are defined on patient identifiers, so both eyes of a patient always fall into the same partition. This prevents leakage between the eyes of one patient, which matters because GA is frequently bilateral (39 % of the patients in one OCT cohort; Yehoshua et al., 2011) and the two eyes of a patient show a high degree of concordance (Boyer et al., 2017).

Table 3.2: Number of training and validation patients, eyes and windows (pairs of consecutive visits) in each of the five cross-validation folds. The split is made at the patient level, so both eyes of a patient always fall into the same partition.

	Training			Validation		
Fold	Patients	Eyes	Windows	Patients	Eyes	Windows
0	38	55	343	13	20	135
1	42	63	413	9	12	65
2	40	59	378	11	16	100
3	41	60	386	10	15	92
4	43	63	392	8	12	86

The per-fold counts are uneven because patients contribute one or two eyes; Table 3.2 lists them. The folds also differ in baseline lesion size. The median baseline lesion area is about 4.1 mm^2 on fold 4, against 6.0 to 7.9 mm^2 on the other four folds. This heterogeneity is one reason why arms are compared fold by fold (§5.1). The table also gives the number of windows, pairs of consecutive visits, in each partition.

No separate test set is held out. With 75 eyes from 51 patients, a held-out test set would leave too few eyes both for training and for a reliable test. Instead, every eye serves as a validation eye in exactly one of the five folds. The protocol is therefore 5-fold train/validation cross-validation, and every result reported in this thesis is a validation result.

In summary, each eye provides a sequence of eleven-channel states on a 49×1024 grid whose rows are about 21 times farther apart than its columns, the intervals between its visits and two covariates. Growth is censored at the window edge in almost a third of the visits. The 478 transitions are split into five patient-level folds. Chapter 4 describes the framework that is trained on them.

Chapter 4

Method

4.1 Overview

This chapter answers the first part of the research question: what kind of framework the forecast needs. The framework defines everything except the operator: how the state is represented, how the elapsed time between visits enters the prediction, and how every model is trained and evaluated. Holding these parts fixed removes them as confounding factors when operators are compared. If every candidate architecture came with its own training pipeline, differences in data handling, time representation or optimisation would be confounded with the architectural differences under test. A single pipeline therefore provides one swappable slot for the operator f_θ, the only component that changes between experiments. Each setting of the slot is referred to below as an *arm* of the comparison.

The framework developed here consists of a residual update conditioned on the elapsed time Δt (§4.2.1), a zero-initialised output so that every operator starts from persistence, one objective that weights the lesion mask (§4.2.4) and a training curriculum measured in elapsed days (§4.2.5). Which of these parts the results depend on is tested in §5.6.

Everything outside the slot is held fixed. The input is the eleven-channel state on the 49×1024 grid of Chapter 3. Time is handled by a residual update conditioned on the interval between visits (§4.2.1). Every operator takes one Euler step, except the FEN, which is integrated with RK4 in 45-day sub-steps without Δt as an input (§4.3.6, §4.3.7). The loss, the training curriculum, the optimiser, the epoch budget, the cross-validation folds, the random seeds, the evaluation and the model selection rule are shared by every run. An operator placed in the slot contains only its architecture. It receives the current state and its conditioning inputs in a common batch format, and it returns the full predicted next state. It may not change the optimisation loop or carry training settings of its own.

The persistence baseline, which predicts no change, is computed in every run by the same evaluation code, and its one-year Dice and intersection over union are identical across all arms in every fold. This indicates that every arm was evaluated on the same validation eyes with the same evaluation routine.

The operators run through this pipeline are a message-passing graph neural network (GNN) on two different neighbourhood graphs (§4.3.3), a U-Net, a Fourier Neural Operator (FNO) and a hybrid Finite Element Network (FEN). Several of these are run in more than one variant (§4.3). The GNN on one of the

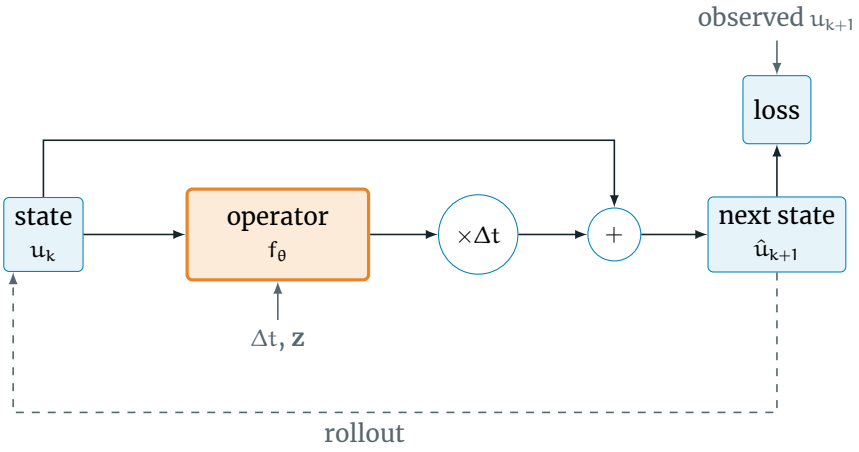


Figure 4.1: The fixed framework and its swappable operator slot. The operator f_θ (orange) receives the current state u_k together with the interval Δt to the next visit and the patient covariates $\mathbf{z}$. Its output is multiplied by Δt and added to the unchanged current state, which gives the predicted next state (Equation 4.1). During training, the prediction is compared with the observed next state in the shared loss (§4.2.4). In the rollout, the prediction is fed back as the input of the next step. Everything except the operator is the same in every experiment; the slot is filled in turn by the graph network, the U-Net, the Fourier Neural Operator and the finite-element network (§4.3).

two graphs is the *canonical* model of this thesis. It is the reference configuration: the ablations of the framework in Chapter 5 are run on it, and the other arms are matched to its 67 147 parameters within a narrow band; the matching and its exceptions are described in §4.3.8. A fixed-step Runge–Kutta wrapper can be placed around any of these operators. A per-pixel model with no spatial context serves as the lowest reference point of the comparison (§4.3.5).

§4.2 defines the framework, and §4.3 describes all operators on the same terms: each was taken from the literature, and for each the original is summarised before its adaptation to the framework. §4.4 closes with the implementation. Figure 4.1 summarises the pipeline. A moving-mesh extension of the graph operator, following MM-PDE (Hu et al., 2024), was tested once as an additional experiment. It is not part of the framework and is reported in §5.7, with its design in Appendix E.

4.2 The Fixed Framework

This section defines what every arm shares: the input, the time handling (§4.2.1), the tensor layout of the state (§4.2.2), the conditioning inputs (§4.2.3), the objective (§4.2.4), the training curriculum (§4.2.5) and the optimisation (§4.2.6). These parts are the same for every operator, except that the FEN, like any operator run in the Runge–Kutta wrapper, replaces the single step by an RK4 integration (§4.3.7); the shared evaluation follows in §5.1. With these parts, any operator can be trained on visits with irregular intervals, starts from exact persistence and is optimised under the same loss and curriculum.

Every operator receives exactly one visit, without earlier visits as input, and predicts the state at the next visit. With five to thirteen visits per eye (§3.1), using several past visits as input would leave fewer visits as prediction targets; an operator with memory of the trajectory is left for future work (§6.4). The framework also predicts one step at a time. Some neural PDE solvers, such as MP-PDE, predict several future steps in one forward pass (temporal bundling, §2.3), but not all of the operators compared here support this. Bundling is therefore removed, so that every operator makes the same kind of prediction.

4.2.1 Δt-Conditioned Residual Formulation

Neural PDE solvers such as MP-PDE (§4.3.2) use a fixed time step and receive the absolute time as an input feature. The clinical data offers neither. Visits occur at irregular intervals, and patients share no common time origin. The absolute time is therefore replaced by the interval between two consecutive visits, Δt, measured in years (§3.6).

The update of the problem statement (§2.5.2, Equation (2.8)) writes the transition from the current state u_k to the predicted next state $\hat{u}_{k+1}$ as a residual update scaled by the interval:

$$\hat{u}_{k+1} \;=\; u_k \;+\; \Delta t_k \cdot f_\theta(u_k,\, \Delta t_k,\, \mathbf{z})\,, \tag{4.1}$$

where f_θ is the operator in the swappable slot and $\mathbf{z}$ denotes the patient covariates. Δt therefore enters twice. It is an input feature of the network, so the operator can adapt its output to the length of the interval. It is also the multiplier of the network output.

If f_θ stays bounded as Δt approaches zero, the predicted change $\Delta t \cdot f_\theta$ vanishes and the prediction collapses to $\hat{u}_{k+1} = u_k$. Because Δt is also an input of f_θ, this is not automatic. It is guaranteed if f_θ is Lipschitz-continuous in Δt with a constant L, since then

$$\begin{aligned} &\|f_\theta(u, \Delta t, \mathbf{z}) - f_\theta(u, 0, \mathbf{z})\| \leqslant L\,|\Delta t|, \\ &\qquad \|\Delta t \cdot f_\theta(u, \Delta t, \mathbf{z})\| \leqslant |\Delta t|\big(\|f_\theta(u, 0, \mathbf{z})\| + L\,|\Delta t|\big) \xrightarrow{\Delta t \to 0} 0. \end{aligned}$$

The operators used here consist of linear layers, Lipschitz-continuous activations and normalisation layers with a positive stabilising constant, so the assumption holds. The constant L is not controlled during training and can be large; on the mask channel the operator in fact learns to scale its output with $1/\Delta t$ within the observed intervals (§5.6.1). The statement therefore concerns only the limit and says nothing about how quickly it is approached. The leading u_k term is also a deliberate prior: consecutive visits are almost identical, because GA progresses little over a typical interval between visits. Passing the current state through unchanged relieves the network of reproducing the whole state. It only has to model the change over the interval.

The output of f_θ is signed and unbounded, so the model can in principle also predict a shrinking lesion. Atrophic tissue does not regenerate, and a shrinking lesion is not clinically plausible. A constraint that prevents the lesion from shrinking is nevertheless not imposed, because the reference segmentations themselves shrink locally between most pairs of visits (§6.3). The persistence prior therefore rests on slow change alone.

The final layer of every operator's output head is initialised to zero, weights and bias alike. Before training, f_θ therefore outputs zero for every input, and Equation (4.1) reduces to exact persistence. Every arm starts from the same function.

Equation (4.1) has the form of one explicit Euler step (§2.2.2) whose step size differs from window to window. The Euler form does not, however, make f_θ a time derivative: a network trained with this update need not be a consistent discretisation of any continuous-time system (Ott et al., 2021). Whether such a reading holds is tested in §5.8.1.

4.2.2 Multi-Channel State Propagation

The full eleven-channel state of §3.2 is carried through every component of the framework; this subsection describes only how the channels are laid out as tensors. The state $u_k \in \mathbb{R}^{C\times H\times W}$ of §2.5.2 is stored per window as a matrix of $N = H \cdot W = 50\,176$ grid nodes by $C = 11$ channels, with the nodes ordered row-major over the 49×1024 grid. At run time, a batch of B windows is stacked into a node-feature matrix of shape $(B \cdot N,\ C)$. Operators that work on the image grid (§4.3.4) use the layout (B, C, H, W) instead.

All eleven channels are both input and output of the operator. The mask channel is the clinical target, and the ten layer channels are auxiliary targets. Each output channel receives its own independent prediction. The loss weights the mask channel more heavily than the layer channels, with the weights set in §4.2.4.

4.2.3 Conditioning: Patient Covariates

Every window carries two patient-level covariates, the age of the patient at the start visit and the sex. Their extraction and encoding are described in §3.5. Both covariates are global quantities: a visit has one value of each, and that value is shared by all $50\,176$ grid positions. Along the trajectory of one eye, the sex is constant and the age advances only with the elapsed time. Because the covariates carry no spatial information, they act on the prediction of an operator only as a whole and cannot point to a location on the grid or shape the lesion boundary locally.

The covariates are included because age and sex are plausible modulators of how fast GA progresses. They are enabled in the canonical configuration of every operator. Each operator receives them together with Δt; how they enter its forward pass is described with the operator in §4.3.

4.2.4 Objective

The loss is computed on the normalised state (§3.4) and has two terms, a channel-weighted mean squared error (MSE) and a soft-Dice term on the mask channel:

$$\mathcal{L} \;=\; \frac{1}{\sum_c w_c} \sum_{c=0}^{10} w_c\, \mathcal{L}_{\mathrm{MSE}}^{(c)} \;+\; \lambda_{\mathrm{Dice}}\, \mathcal{L}_{\mathrm{Dice}}, \tag{4.2}$$

where $\mathcal{L}_{\mathrm{MSE}}^{(c)}$ is the mean squared error of channel c and w_c its weight. The mask channel ($c = 0$), which is the clinical target, has the weight $w_0 = 5$, and each of

the ten layer channels has the weight $w_c = 1$. Dividing by the sum of the weights keeps the term on the scale of an unweighted MSE; with equal weights it equals the MSE averaged over the eleven channels. The layer channels are auxiliary targets, but the operator must still predict them, because they are part of the next state that is fed back during a rollout. The same loss is used in training and in validation.

The second term is a soft-Dice loss on the mask channel (Milletari et al., 2016), with the weight $\lambda_{\text{Dice}} = 5$. It counters an imbalance in the data: between two consecutive visits, almost all mask pixels are unchanged, so a plain MSE is dominated by static pixels and gives little gradient at the moving lesion boundary. A model trained on it can learn to copy its input. The soft-Dice term restores gradient in the band around the binarisation threshold, which is where the boundary lies. Unlike the original soft-Dice loss, which scores a segmentation probability map, the term is applied here to a regression output. The prediction p of the mask channel is first turned into a soft mask,

$$p_{\text{soft}} = \sigma(s\,(p - \tau)), \tag{4.3}$$

where σ is the logistic sigmoid, $s = 2$ the sharpness and τ the physical threshold of 0.5 mapped into normalised space. The evaluation uses the same mapping (§3.4), so the loss and the metric agree on where the threshold lies. With the hard-thresholded target mask t, the term is $\mathcal{L}_{\text{Dice}} = 1 - 2\sum p_{\text{soft}}\, t \,/\, (\sum p_{\text{soft}} + \sum t)$.

The soft-Dice term is computed per eye and then averaged over the batch. A Dice score is a ratio of sums. If the sums were pooled over the whole batch, the gradient for one eye would depend on which other eyes it was batched with. Scoring per eye also matches the evaluation, which scores each eye separately before averaging.

The soft-Dice term has one accepted side effect. Because the sigmoid of Equation (4.3) does not saturate at the physical mask values, the term stretches the raw mask predictions away from the threshold. This cannot change a thresholded prediction, but it changes the raw error of the mask channel. It is one reason why the evaluation does not compare operators on raw errors (Chapter 5).

A monotonic-growth penalty can be added to the objective, which penalises a decrease of the mask where the previous state is atrophic. It is set to zero for every operator of the comparison (§4.2.1); §5.6.2 tests it on the canonical operator.

4.2.5 Time-Budgeted Pushforward Curriculum

The operators are trained with the pushforward trick of Brandstetter et al. (2022) (§2.3): the model is rolled forward on its own predictions without gradients, and only the final step is supervised. The original counts the depth of this rollout in steps. With irregular visit intervals (§3.6), a step count is not a comparable quantity: two steps can mean 180 days for one eye and 720 days for another. The framework therefore counts the depth in elapsed days.

Training proceeds in passes; in each pass, every training eye contributes one randomly chosen start window. Each training pass draws a time budget B uniformly from $\{0, 1, \ldots, \min(e, U)\} \times 180$ days, where e is the current epoch and U the

largest number of 180-day units in a budget. The base of 180 days is the modal visit interval of the cohort (§3.6). Every reported run uses $U = 2$, so the largest budget is 360 days, which matches the one-year evaluation anchor. Because the upper limit is $\min(e, U)$, the largest budget grows by 180 days per epoch over the first U epochs; after that, budgets of 0, 180 and 360 days are drawn with equal probability. A budget of zero gives a plain one-step pass on observed inputs.

For a given budget, only start windows that can realise it are sampled. The chain of consecutive visits from such a window must reach at least B days in total, and its first interval must not exceed B. If an eye has no such window, sampling for that eye falls back to all of its windows. The number of unrolled steps for each eye then follows an undershoot rule: it is the largest n for which the n rolled intervals and the final, supervised interval together stay within B. The chain never overshoots the budget, so eyes on different visit schedules are trained to the same elapsed horizon. The undershoot rule is also the reason for the 180-day base: a budget of 90 days, the shortest interval, can never chain two visits (§3.6), so the curriculum would not unroll.

Within a batch, the eyes are grouped by their number of unrolled steps. Each group is rolled forward n steps without gradients: at every step, the next window of the eye is fetched and its input state is replaced by the model's own prediction. All channels are rolled, including the predicted mask. An eye whose sequence ends is dropped from the chain. The group then receives one forward pass with gradients, and the loss is taken against the observed next state. The group losses are combined, each weighted by the share of the batch its group holds.

The curriculum is used in every reported run; no run trains without unrolling or with a curriculum counted in steps, so its effect on the forecast is not measured.

4.2.6 Optimisation

Table 4.1 lists the training settings that every operator shares; this subsection explains the optimiser and the schedule. All operators are trained with AdamW (Loshchilov and Hutter, 2019), a learning rate of 2×10^{-3} and a weight decay of 1×10^{-2} on all trainable parameters. The learning rate is multiplied by 0.4 at the start of epoch 5 and again at the start of epoch 20, counting from 0, so the first five epochs run at the initial rate. These settings were not tuned per operator. A shared recipe prevents differences in training from being confounded with the architectural differences, but it may suit some architectures better than others.

Training runs for 30 epochs of 20 passes each, with a batch size of four eyes. Because the number of training eyes differs between the folds (§3.7), an epoch has between 280 and 320 optimiser steps, depending on the fold.

Gradients are clipped to a norm of 1.0 separately for each trainable module, so that a large gradient in one module cannot shrink the updates of another. For the single-network operators of the main comparison, this amounts to one clip over the whole model. The separation matters for the moving-mesh extension of Appendix E.

Unless stated otherwise, every reported run uses seed 42; replication runs with seed 7 are reported in Chapter 5 and Appendix C. The model is validated on the

Table 4.1: Training settings, identical for every operator of the survey.

Setting	Value
Optimiser	AdamW
Learning rate	2×10^{-3}, multiplied by 0.4 at the start of epochs 5 and 20 (counted from 0)
Weight decay	1×10^{-2}
Epochs $\times$ passes per epoch	30×20
Batch size	4 eyes
Gradient clipping	Norm 1.0, per module
Loss weights	Mask MSE 5, layer MSE 1 each, soft-Dice 5 (sharpness 2), monotonic penalty 0
Pushforward budget	Base 180 days, maximum 360 days ($\mathrm{U} = 2$)
Seeds	42 (replication: 7)

held-out fold after every epoch (§3.7); §5.1.4 sets out which epochs enter the reported numbers and why.

In summary, the framework fixes one input visit, the residual update of Equation (4.1) with a zero-initialised output, the loss of Equation (4.2), the time-budgeted curriculum and the settings of Table 4.1; §4.3 describes the operators that fill the slot.

4.3 The Operators

This section describes the operators placed in the slot of §4.2. Each was chosen to isolate one property: the reach of the spatial context of one step (the message-passing graph neural network on two different graphs, §4.3.2–§4.3.3), the amount of spatial context needed (two truncated versions of it, §4.3.5), multiscale local detail (the U-Net, §4.3.4), global spectral support without and with a local path (the Fourier Neural Operator and a hybrid, §4.3.4), an explicit transport term (the Finite Element Network, §4.3.6) and the order of time integration (a Runge–Kutta wrapper around any of these operators, §4.3.7). The interface shared by all operators is stated first (§4.3.1), and the parameter matching last (§4.3.8).

4.3.1 The Operator Contract

Every operator satisfies the same interface: it is built from the same metadata (one time step, eleven state channels and two covariates) and consumes the same batch. It returns the full next state in the residual form of Equation (4.1), and its final layer is initialised to zero (§4.2.1), so that every operator starts training at exact persistence. The residual and the next state are computed from one shared internal computation, so the two cannot become inconsistent; the integration wrapper of §4.3.7 integrates this same residual.

No operator uses a normalisation that keeps batch statistics. The training curriculum of §4.2.5 rolls the model forward on its own predictions, without gradients but in training mode, so the states it produces drift away from the observed ones. Batch normalisation (Ioffe and Szegedy, 2015) would normalise these drifted states with the statistics of the current batch and so remove the perturbation the curriculum is meant to expose the model to. It would also write the drifted statistics into the running estimates that evaluation later uses, so that training and evaluation would behave as two different operators.

The operators that work on the image grid (the U-Net, the Fourier Neural Operator and its local-kernel hybrid) receive identical inputs, so that a difference between them cannot come from the information they receive. Each takes sixteen input channels: the eleven state channels, Δt broadcast as a constant plane, the two covariates broadcast as constant planes, and two coordinate ramps running from zero to one along each axis, following the CoordConv construction of R. Liu et al. (2018).

4.3.2 Message-Passing Graph Neural Network

The graph operator is the message-passing neural PDE solver (MP-PDE) of Brandstetter et al. (2022), the learned-stencil solver introduced in §2.3. It builds on the graph neural network model of Scarselli et al. (2009) and the message-passing formulation of Gilmer et al. (2017). Three choices made in adapting it became part of the shared framework: the Δt-scaled residual update, the zero-initialised output layer and the one-step prediction. The original is summarised first, followed by the modifications proposed in this work and a summary of them.

The original MP-PDE. MP-PDE is an autoregressive neural solver for time-dependent PDEs. It operates on a spatial graph, built on the analogy that grid points or finite-volume cells become nodes and numerical stencils become edges. The graph is built from a cut-off radius on regular grids, or by connecting the k nearest neighbours on irregular meshes.

The architecture follows the encode–process–decode pattern (Battaglia et al., 2018; Sanchez-Gonzalez et al., 2020). An encoder, a per-node two-layer multilayer perceptron (MLP), lifts the node features into a latent space. It computes $f_i^0 = \epsilon([u_i^{k-K:k}, x_i, t_k, \theta_{PDE}])$ from the last K states of the node, its position, the current time and the equation features.

The processor applies several rounds of message passing to the latent node states (six in the original paper). The message function is $m_{ij}^m = \phi(f_i^m, f_j^m, u_i^{k-K:k} - u_j^{k-K:k}, x_i - x_j, \theta_{PDE})$, and the node update is $f_i^{m+1} = \psi(f_i^m, \sum_j m_{ij}^m, \theta_{PDE})$. Both ϕ and ψ are two-layer MLPs with skip connections and normalisation layers. The relative position $x_i - x_j$ builds translational symmetry into the network. The solution difference $u_i - u_j$ lets each message act as a learned local difference operator, that is, a learned stencil.

The equation feature vector θ_{PDE} holds the PDE coefficients and the boundary-condition identifiers. It is injected at the encoder and at every message-passing layer. Injecting it at every layer allows a single trained solver to generalise

across a family of equations, and removing it degrades performance most when the equation parameters vary across the dataset.

The decoder is a shallow one-dimensional convolutional network with weights shared across space. It treats the final hidden vector of each node as a temporal signal and convolves along it, producing one update per bundled step, $d_i = (d_i^1, \dots, d_i^K)$. The residual update $u_i^{k+l} = u_i^k + (t_{k+l} - t_k)\, d_i^l$ gives the prediction for each substep $1 \leqslant l \leqslant K$. As the step size approaches zero, the prediction approaches the current state, which keeps the update consistent (§2.2.2). The model is trained with temporal bundling and the pushforward trick (§2.3). The original models have roughly one million parameters and a hidden size of the order of 128, and they are trained with AdamW.

The original paper names two limitations: it needs high-quality numerical ground truth for training, and it has no formal accuracy guarantees. A further one, not named by its authors, is that a uniform mesh is wasteful in smooth regions and under-resolved at sharp gradients; it motivates the moving-mesh extension of Hu et al. (2024), which was tested once as an additional experiment (§5.7, Appendix E).

Proposed modifications. The forward pass consists of an encoder, a number of message-passing rounds and a decoder. The notation follows the summary above. Each node i carries the normalised eleven-channel state u_i and its physical position (x_i, y_i) in millimetres. The positions enter the network divided by the domain size, as $(x_i/L_x,\ y_i/L_y)$.

Each node also receives a conditioning vector $\mathbf{c}_i$. In the canonical configuration it has three entries: the interval Δt in years, the z-scored age and the sex of the patient. All three are graph-level quantities, broadcast to every node of the graph. The conditioning vector takes the place of the equation feature vector θ_{PDE} of the original: no governing equation or coefficients are known for GA, so the interval and the covariates are used instead. As in the original, the vector is injected at three points: the encoder input, every message and every node update.

The encoder maps the concatenated node input to a hidden vector,

$$f_i^0 \;=\; \epsilon\big(\big[u_i,\ x_i/L_x,\ y_i/L_y,\ \mathbf{c}_i\big]\big), \tag{4.4}$$

where ϵ consists of a linear layer, layer normalisation, a ReLU activation, a second linear layer and a final layer normalisation, with no activation after it.

Each message-passing round m computes a message along every edge from node j to node i and then updates node i:

$$m_{ij}^m \;=\; \phi\big(f_i^m,\ f_j^m,\ u_i - u_j,\ \Delta\mathbf{p}_{ij},\ \mathbf{c}_i\big), \tag{4.5}$$

$$f_i^{m+1} \;=\; \mathrm{LN}\left(f_i^m + \psi\left(f_i^m,\ \frac{1}{|\mathcal{N}(i)|}\sum_{j\in\mathcal{N}(i)} m_{ij}^m,\ \mathbf{c}_i\right)\right). \tag{4.6}$$

Here $\Delta\mathbf{p}_{ij}$ is the offset between the normalised positions of the two nodes, $\mathcal{N}(i)$ is the set of neighbours of node i, and LN denotes layer normalisation. The message function ϕ and the update function ψ each consist of two linear layers, each followed by a ReLU activation. As in the original, the state difference $u_i - u_j$ lets the message act as a learned local difference operator. Every round receives the

state u_i, the positions and the conditioning vector again, so this information is available at every depth. Each update ends in a residual connection followed by a normalisation layer; the original also uses skip connections in its message-passing layers, and only the type of normalisation differs. The canonical configuration aggregates messages by their mean, as written in Equation (4.6), and uses two message-passing rounds and a hidden width of 64; larger and smaller settings are compared in §5.6.3.

All normalisation layers are per-node layer normalisations (Ba et al., 2016). Each node is normalised over its own feature vector, so the operator keeps no batch statistics, as the contract of §4.3.1 requires.

The one-dimensional convolutional head of the original, which exists to emit several bundled steps, was replaced by an MLP decoder, because the framework predicts one step per pass (§4.2). It maps the final hidden vector f_i^M to the eleven channels: a linear layer, layer normalisation and a ReLU, applied twice, followed by a final linear layer whose full-rank output gives every channel its own independent prediction. The final layer has no normalisation or activation, because the output is a signed and unbounded residual, and it is initialised to zero, which yields the persistence start of §4.2.1. The predicted next state of node i is then $u_i + \Delta t \cdot \delta(f_i^M)$, where δ denotes the decoder; this is Equation (4.1) at node level.

Summary of the modifications. Table 4.2 summarises the modifications to MP-PDE (Brandstetter et al., 2022). The entries for MP-PDE refer to the published paper, which evaluates the method mainly on one-dimensional equations, with additional two-dimensional smoke-inflow experiments, and uses different settings for the two.

4.3.3 Graph Construction on an Anisotropic Grid

The graph determines which nodes exchange messages, and therefore how far information travels in one round. The modelling grid has 49 rows and 1024 columns, or 50 176 nodes. Rows are 121.18 µm apart, while columns are only 5.68 µm apart (§3.3). Along a row, the nodes are therefore about 21 times denser than across rows. The node positions are stored in physical millimetres.

The usual construction on irregular meshes, a k-nearest-neighbour graph, connects each node to the k nodes at the smallest Euclidean distance between their physical positions (Figure 4.2a). On this grid it does not connect the rows: the twelve physically nearest neighbours of every node all lie in its own row, and no edge crosses rows. Message passing on such a graph runs along rows only, no information can cross from one row to the next at any network depth, and the grid decomposes into 49 independent one-dimensional rows. This construction is therefore not used in any reported experiment; it is described because it is the natural first choice and shows that the anisotropy of the grid has to be taken into account.

Measuring the Euclidean distance between integer grid indices (row, column) instead restores two-dimensional connectivity (Figure 4.2b). With $k = 12$ and without self-loops, the neighbourhood of an interior node is exactly the set of

Table 4.2: Differences between the original MP-PDE (Brandstetter et al., 2022) and the graph solver of this thesis. *1-D* and *2-D* refer to the one-dimensional experiments and the two-dimensional smoke-inflow experiments of the original. For MP-PDE, the table gives the sum written in its node-update equation; the paper does not describe the aggregation used in its reference implementation.

Aspect	Original MP-PDE	This thesis
State	PDE solution (mostly scalar-valued)	Eleven-channel clinical state on a 2-D grid
Temporal bundling	K future steps per forward pass	Removed; one step per pass ($K = 1$)
Time	Fixed time step; current time as an input	Irregular per-window Δt as input and as multiplier; no shared time origin
Equation identity	Known coefficients and boundary conditions (θ_{PDE})	None known; Δt and covariates in its place (§4.2.3)
Graph	Cut-off radius (1-D, 2-D) or k-nearest neighbours (wave equations), on positions	Index-space k-nearest neighbours, then the dilated stencil (§4.3.3)
Aggregation	Sum over neighbours	Mean
Activation	Swish (1-D); ReLU (2-D)	ReLU
Normalisation	Instance normalisation (1-D); batch normalisation (2-D)	Per-node layer normalisation
Decoder	1-D convolution along the node's hidden vector, read as a temporal signal, emitting K steps	Full-rank three-layer MLP; last layer zero-initialised

nodes at squared index distance at most four. On the integer grid this set coincides with the ball of radius two in geodesic (hop) distance on the 4-connected lattice, a diamond: the four direct neighbours, the four diagonal neighbours and the four neighbours two steps away along either axis. The resulting graph has 602 112 directed edges; about 33 % connect nodes in the same row, 50 % nodes one row apart and 17 % nodes two rows apart. About 8.5 % of the nodes (4 276), on and next to the border of the grid, do not have the full interior neighbourhood.

This index-space graph is the graph of the k-nearest-neighbour arm in Chapter 5. It restores two-dimensional connectivity, but its physical reach differs strongly between the two axes. A neighbourhood that is symmetric in grid steps inherits the 21:1 anisotropy of the grid. One hop reaches about ± 242 µm across rows but only about ± 11 µm along a row. After the two message-passing rounds of the canonical configuration, a node has received information from at most four columns away, about ± 23 µm along the row.

For comparison, the GA front advances a median of about 12 columns (68 µm) between two visits 180 days apart; the upper quartile is 17 columns (97 µm) and the 90th percentile 23 columns (131 µm). These values are the per-eye growth rate of the effective lesion radius, $\sqrt{A/\pi}$, over 180 days, for the 75 eyes. A single hop along a row therefore covers only a small fraction of the distance the

front typically moves between two visits, and the four columns reached after two rounds are about a third of the median advance. The index-space graph restores connectivity but not reach, which motivates a second graph construction whose reach along a row matches the scale of the lesion change.

The dilated stencil is designed to close this gap (Figure 4.2c). Each interior node is connected to the nodes at row offsets $\{-1, 0, +1\}$ and column offsets $\{0, \pm 7, \pm 14, \pm 21\}$, excluding the node itself: 20 neighbours in total. Because every column offset is a multiple of seven, a node exchanges messages, at any depth, only with columns whose index differs from its own by a multiple of seven; the grid splits into seven interleaved sub-lattices 40 µm apart, which share the network weights. It follows the idea of dilated convolutions (Yu and Koltun, 2016), which apply the same kernel at a larger range by spacing its taps apart, so that the receptive field grows while the number of parameters stays the same; here the dilation is seven along rows and one across rows. Dilated neighbourhoods have also been used in graph networks (G. Li et al., 2019), where the k neighbours are taken from the $k \cdot d$ nearest ones by skipping every d-th, measured in the feature space of the current layer. Here the dilation is instead set in physical distance, because what one hop has to cover is the physical advance of the front between two visits; an isotropic dilation in grid steps would keep the 21:1 imbalance of the index-space graph. One hop reaches ± 121 µm across rows and ± 119 µm along a row, nearly equal on both axes at about 120 µm per hop. Along a row, one hop spans 21 columns, close to the 90th percentile of the per-visit front advance. The design thus derives the reach from the physical scale of the lesion. The dilated stencil changes only the edge set; the network architecture and every other component are the same as on the index-space graph, and the stencil adds no parameters. Chapter 5 tests this choice against smaller and larger reaches. The offsets were fixed from the front-advance statistic before this comparison was run.

4.3.4 Dense Operators: U-Net, FNO and a Local-Kernel Hybrid

The three dense operators provide properties that the graph network does not have: multiscale local detail (U-Net), global spectral support (FNO) and global support with a local path (the local-kernel hybrid). They are dense grid operators: they act on the state as an image on the full 49×1024 grid and receive sixteen input channels: the eleven state channels, Δt, the two covariates and two coordinate ramps (§4.3.1). The first dense operator is a U-Net (Ronneberger et al., 2015). The original was designed for biomedical image segmentation. It consists of a contracting and an expansive path. Each contracting step applies two unpadded 3×3 convolutions, each followed by a ReLU, and then a 2×2 max pooling with stride two; the number of feature channels doubles at each downsampling step. Each expansive step upsamples, halves the channels with an up-convolution, concatenates the correspondingly cropped feature map from the contracting path, and applies two further 3×3 convolutions. A final 1×1 convolution maps to the output classes. Gupta and Brandstetter (2023) study modernised U-Nets as PDE surrogates and describe them as handling local and global information through the down- and up-sampling paths. In the GA domain, Salvi et al. (2025) predict future lesion growth with a two-dimensional U-Net on fundus autofluorescence images.

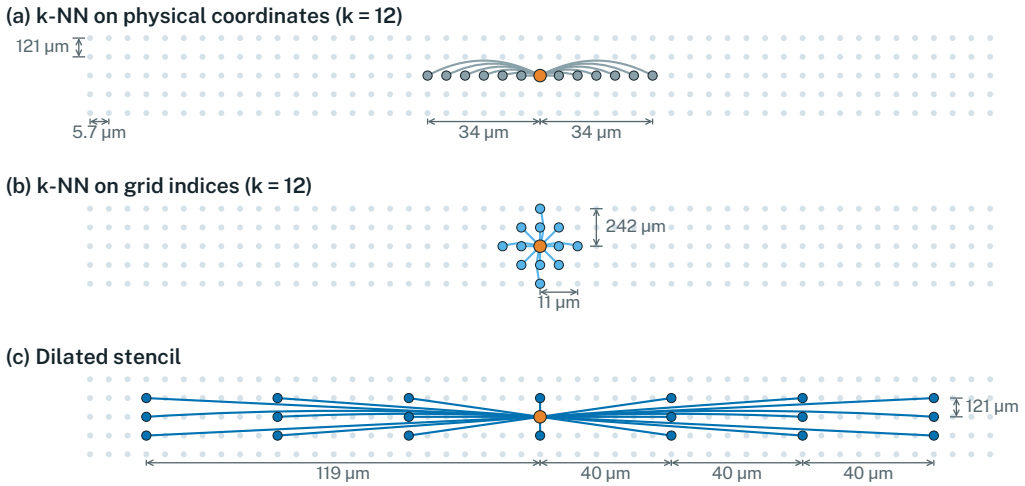


Figure 4.2: The three graph constructions of the graph network on the same lattice, drawn in grid indices around one interior node (orange); lines connect it to the nodes it exchanges messages with in one round. Rows are B-scans 121 µm apart, columns A-scans 5.7 µm apart, so on the retina the vertical spacing is about 21 times the horizontal one. Distances are physical. Edges that would run through another neighbour on the same line are drawn bent. (a) k nearest neighbours ($k = 12$) on physical coordinates: all twelve lie in the node's own row, at most 34 µm away, so no message crosses rows. (b) k nearest neighbours ($k = 12$) on grid indices: a diamond of two rows and two columns, reaching 242 µm across rows but only 11 µm along a row. (c) The dilated stencil: 20 neighbours at rows $\{-1, 0, +1\}$ and columns $\{0, \pm 7, \pm 14, \pm 21\}$, reaching 121 µm across rows and 119 µm along a row. For scale, the GA front advances along a row by a median of 12 columns (68 µm) between two visits 180 days apart, and by 23 columns (131 µm) at the 90th percentile. One hop of (a) or (b) covers only part of the median advance; one hop of (c) covers about the 90th percentile.

The U-Net used here departs from the original in four ways. First, each convolution is followed by group normalisation with a single group before the ReLU; the original uses no normalisation. With one group, group normalisation normalises each sample over all its channels and positions and keeps no batch statistics (Wu and He, 2018). Second, pooling is anisotropic. The factors (rows, columns) are $(1, 2)$, $(1, 2)$, $(2, 2)$ and $(2, 2)$, so the grid shrinks from 49×1024 through 49×512, 49×256 and 24×128 to 12×64. Rows are downsampled by a factor of four in total and columns by a factor of sixteen, which reduces the 21:1 ratio of row to column spacing to about 5:1 without removing it. The schedule was fixed before training and not tuned, because a setting chosen on the validation folds would carry a selection bias, as there is no test split (§3.7). Third, upsampling is nearest-neighbour instead of an up-convolution, and the output has the grid size of the input. Fourth, the final 1×1 layer is zero-initialised and returns a residual rather than class scores. A base width of five, that is, five channels at the input stage, gives 83 081 parameters.

The second dense operator is the Fourier Neural Operator (FNO) of Z. Li et al. (2021). The input is first lifted to a wider representation by a local map. Each

layer then combines a local linear transform with a non-local integral operator and applies a nonlinearity. The integral operator transforms its input to Fourier space, multiplies the retained low modes by learned complex weights, and transforms back. Modes above a threshold are truncated; both the low positive and the low negative frequencies are kept. A local projection maps the result to the output. The paper shows that the Fourier layers are invariant to the discretisation, and that truncation does not limit the output to the retained modes, because the nonlinearities and the projection recover higher frequencies. For its two-dimensional problems, the original uses four Fourier layers, ReLU activations with batch normalisation, 12 modes per dimension and a width of 32. It notes that the local linear term helps the model handle non-periodic boundaries.

The FNO used here keeps four blocks, each a spectral convolution plus a pointwise 1×1 path, and departs from the original in the following respects. It uses GELU activations between blocks instead of ReLU, and no normalisation. It retains 14 modes per axis instead of 12, and its width is 5 instead of 32, which gives 79 160 parameters. The output passes through a 1×1 convolution to 32 channels, a GELU and a zero-initialised 1×1 projection. Because the domain here is a crop and not a periodic physical domain (§3.3), the field is zero-padded by about one eighth per axis before the transform, from 49×1024 to 56×1152, and cropped back after the spectral blocks; the original does not pad and relies on the local linear term instead. The equal mode count on both axes follows from the geometry: the physical domain is nearly square (5.94×5.82 mm), so equal mode counts give nearly equal physical bandwidth on both axes, a property the index-space graph of §4.3.3 does not have. Dividing the padded extent by the highest retained mode index, 13, gives a finest retained wavelength of about 520 µm across rows and 500 µm along columns. The mode count was not ablated.

The third dense operator is a local-kernel hybrid. Liu-Schiaffini et al. (2024) observe that global operations in Fourier space are prone to over-smoothing and may miss local details. They add localised operators in parallel to the Fourier layers and the pointwise path. Their differential kernels subtract the kernel mean and rescale by the reciprocal grid width, so that the convolution converges to a differential operator as the grid is refined; their localised integral kernels use discrete-continuous convolutions. The hybrid used here is the simplest instance of this idea. It widens the pointwise 1×1 path of each FNO block to an unconstrained 3×3 convolution, so a local path runs in parallel with the spectral one. Unlike the original, it operates at a single training resolution and uses no mean subtraction, rescaling or discrete-continuous convolutions, so it is not the resolution-consistent operator of that paper. The spectral path keeps global support and the local path gains one ring of neighbours, at a cost of 800 parameters (79 960 in total). With a kernel size of one, the model reduces exactly to the plain FNO. All three dense operators lie within a factor of 1.25 of the canonical parameter count (§4.3.8).

4.3.5 The Locality Floors

Two arms are used to measure how much spatial context the prediction needs. Both are the graph solver on the index-space k-nearest-neighbour graph

of §4.3.3 with fewer message-passing rounds, so neither needs a separate module. The one-hop floor uses a single round at width 64, so each node sees exactly one ring of neighbours; it has 40 971 parameters. The per-pixel floor uses no message passing, so each node's next state depends only on its own state and its conditioning inputs and the model has no spatial context; it has 14 795 parameters. Together with the two-round index-space arm, they form a locality ladder of zero, one and two rounds.

4.3.6 The Finite Element Network

The Finite Element Network (FEN) of Lienen and Günnemann (2022) builds the update from a finite-element discretisation of an unknown PDE. Its architecture defines the output as a time derivative instead of a next state; it is the only operator in the survey that does so. Whether the trained model behaves as a valid continuous-time system is a separate question, tested in Chapter 5.

The original uses a Galerkin method with the P1 basis of piecewise-linear hat functions, by default on a Delaunay triangulation of the observation points. The resulting system involves the mass matrix of inner products between basis functions, which is lumped by row-wise summation, a standard approximation for time-dependent problems that makes its inversion trivial. The inner products are factored so that a network predicts one coefficient per cell, vertex and feature from the time, the cell centre, the local vertex coordinates and the vertex features; this network is the free-form term. The transport variant, T-FEN, adds a convection term. It starts from the conservative form $-\nabla \cdot (v u)$, applies the product rule and assumes a divergence-free velocity, which leaves the advective form $-v \cdot \nabla u$. A second network predicts one velocity per cell and feature; since the velocity is constant within each cell, it is divergence-free there, but not across cells. Both networks are tanh MLPs, and the resulting ODE is solved with the adaptive Dormand–Prince solver (dopri5).

The FEN used here triangulates the regular grid by splitting every grid cell into two right triangles, which gives 98 208 triangles over the 50 176 nodes; each node belongs to one, two, three or six triangles. The mass matrix is lumped as in the original. On this equal-area lattice, the lumped assembly is exactly a mean over the triangles adjacent to a node. The FEN is therefore a message-passing network with mean aggregation over triangles, each linking three nodes, and its finite-element formulation differs from such a network only at the boundary. The free-form term is a tanh MLP of depth four and width 96, without normalisation and with a zero-initialised head. It receives the three vertex states of a triangle, a triangle-type indicator for the two triangle orientations, the covariates and optionally the normalised triangle centre, and it outputs one coefficient per vertex and channel.

The transport term is a second MLP that predicts one velocity per triangle and channel, in millimetres per year. The assembled operator is the advective form $-v \cdot \nabla u$ with a velocity that is constant within each triangle, as in the original. As in the original, the velocity is not divergence-free across triangles, so reading it as a physical advection speed rests on an assumption that the model does not enforce globally. The gradient coefficients are assembled in grid units, so the 21:1 axis imbalance never reaches the MLPs; only the velocity conversion uses millimetres. The transport term runs on its own coarser mesh, made of

seven interleaved triangulations, each of which connects every seventh column (97 632 triangles in total), while the free-form term uses the full-resolution mesh. This coarser mesh is the finite-element counterpart of the dilated stencil of §4.3.3.

The implementation departs from the original in the following points. A regular lattice replaces the Delaunay triangulation. A triangle-type indicator replaces the local vertex coordinates, which on this lattice take only two forms. The covariates are added as inputs. The time is not an input, so the learned time derivative has no explicit dependence on time (it is autonomous), a variant the original paper explicitly allows. Integration uses a fixed-step scheme instead of the adaptive dopri5 solver. The advective form and the per-cell velocity are the same as in the original; the energy-conserving form of the transport operator described below is a further departure.

One evaluation of the time derivative only couples nodes that share a triangle, so spatial reach has to come from sub-steps. The FEN is therefore always run inside the Runge–Kutta wrapper of §4.3.7, with classical RK4 and a fixed physical step of 45 days; Δt sets only the integration horizon. RK4 is used because, in the interior of the lattice, the assembled transport operator has a purely imaginary spectrum: an explicit Euler step would amplify every mode for any nonzero velocity, whereas RK4 has a stability interval on the imaginary axis. It remains stable only while the velocity times the step, the Courant number in the sense of the stability condition of Courant et al. (1928), stays below a limit of about 2.4 for this stencil.

The assembled transport operator was trained in two forms. The first is the Galerkin form described above, as in the original. At a constant velocity and away from the edge of the domain, it conserves a discrete energy of the state, the mass-weighted sum of squares $u^\top M u$, as expected of pure transport, which moves content without creating it. At the edge of the crop, or where the velocity varies from triangle to triangle, it does not conserve this energy. No flux term is assembled at the boundary, so transported content accumulates there, and the operator has modes that grow instead of travelling. The second form removes this defect. It keeps only the part of the Galerkin operator L that is skew-adjoint with respect to the mass matrix M, which is its energy-conserving part,

$$L_{\text{skew}} = \tfrac{1}{2}\left(L - M^{-1}L^\top M\right), \tag{4.7}$$

which conserves $u^\top M u$ for any per-triangle velocity, including at the edge, where it acts as a closed wall. In the interior and at a constant velocity, the two forms coincide. The energy-conserving form adds no parameters and no measurable cost. Figure 4.3 shows the difference between the two forms on a small grid.

Both forms were trained; Chapter 5 reports both (§5.8.2), and the main comparison uses the energy-conserving form.

The T-FEN has 68 503 parameters; its free-form control is described in §4.3.8.

4.3.7 Fixed-Step Runge–Kutta Integration

The Runge–Kutta wrapper is placed around an operator and changes only how the operator's output is turned into the next state; it adds no parameters. It

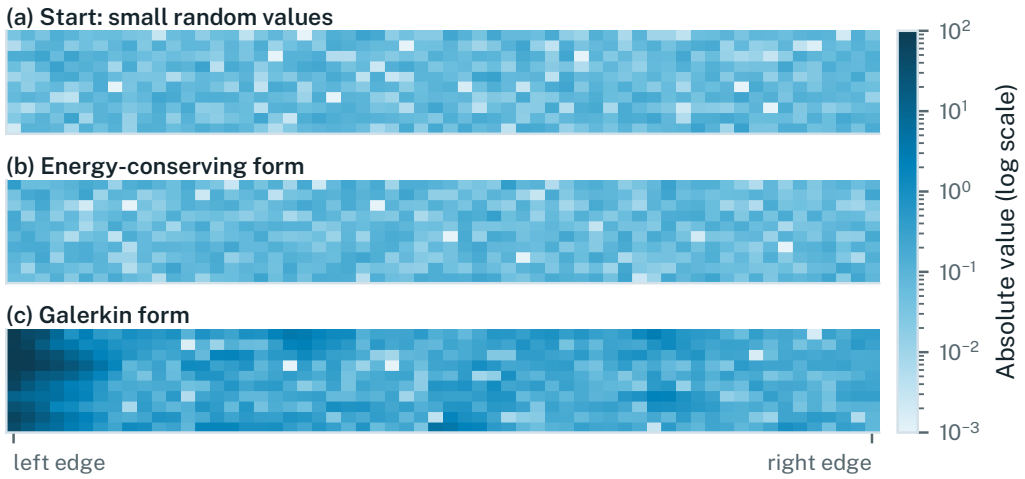


Figure 4.3: The transport operator of the T-FEN in its two forms, on a toy grid of 10×60 nodes with a velocity that points mainly from left to right and varies from cell to cell. No learned part is involved: both forms are applied to the same small random start (a) and integrated over the same time. (b) The energy-conserving form (Equation 4.7) only moves the values; none grows. (c) The Galerkin form lets values grow from about 0.1 to about 145 at the left edge, where content enters the grid and no flux term is assembled; in the interior they stay below 12. The trained Galerkin T-FEN shows the same behaviour on the 49×1024 grid: its free-running rollout diverges at the left or right edge of the crop after about 1.6 to 2 years (§5.8.2), while the energy-conserving form, which the main comparison uses, stays stable.

has three uses. First, the finite-element network outputs a time derivative and needs an integrator (§4.3.6). Second, it allows the integration order to be varied on its own: the same operator is trained with one Euler step and with RK4 (§5.6.1). Third, it makes it possible to test whether a trained model describes a continuous process (§5.8.1).

The canonical update of Equation (4.1) is one explicit Euler step: the operator is evaluated once, at the current state, and its output is applied over the whole interval Δt. A Runge–Kutta scheme evaluates the operator several times within the interval, at intermediate states, and combines the results. The classical fourth-order scheme (RK4) (Kutta, 1901; Runge, 1895) evaluates the operator four times per step and takes a weighted average (Butcher, 1987; Hairer et al., 1993); every reported model trained with the wrapper uses it. The interval can also be split into several sub-steps, each of which is one RK4 step. The sub-step length is set separately for each eye in a batch, either as $h = \Delta t/n$ or from a physical step length in days, with the number of sub-steps rounded up from the ratio of the interval to that length. An eye is therefore integrated with the same steps whether it is batched with others or evaluated alone; eyes with shorter intervals take zero-length steps until the longest interval in the batch is complete. As a check of the implementation, the observed convergence orders on the test problem $du/dt = -\lambda u$ are 0.99, 2.04, 2.04 and 4.35 for the Euler, midpoint, Heun and RK4 schemes, close to their nominal orders of one, two, two and four.

Because RK4 evaluates the operator at intermediate states within the interval, the operator should describe the rate of change at a state, not the change over the full interval. Inside the wrapper, the Δt entry of the operator's conditioning vector is therefore set to zero, so the operator receives no information about the length of the interval. Its output is then a rate of change at the current state with no explicit dependence on time, an *autonomous* field, and the integrator alone determines over which interval this rate is applied; the state still changes by the integrated rate. In this reading the wrapped model is a Neural ODE in the sense of Chen et al. (2018). Positions, graph edges and covariates stay fixed across the stages of one step. The wrapper requires the Δt-multiplied residual form of Equation (4.1), because without the multiplication by Δt the operator's output is not a rate and cannot be integrated.

The canonical model is trained as a single jump: it learns which state follows after an interval of length Δt, in one step. Training does not require two jumps of length $\Delta t/2$ to lead to the same state as one jump of length Δt. In a continuous-time model, by contrast, splitting the interval into more or finer steps must not change the prediction. Ott et al. (2021) show that a model trained with an overly coarse step can reach high accuracy and still lose this property. They test it by evaluating the trained model with solvers of equal or smaller discretisation error and checking whether its predictions stay the same; Krishnapriyan et al. (2023) make the same point for learned models of physical systems. Because the wrapper can integrate any trained operator with other schemes and step counts, it makes this step-size test possible. Its outcome is reported in §5.8.1, and the accuracy of the models trained with the wrapper in §5.6.1.

4.3.8 Parameter Matching

Every architecture arm is matched in size to the canonical graph solver of §4.3.3, which has 67 147 parameters: its width is chosen to come as close to this count as the width granularity allows, so that differences between arms cannot be attributed to extra parameters. Table 4.3 lists every configuration. A separate experiment tests whether the size at which the arms are matched affects the comparison, by scaling every architecture class from about a sixth to three to four times its size (§5.6.3).

All architecture arms lie within a factor of 1.25 of the reference. The two floors lie below this band because they are not architecture arms: the one-hop floor is a rung of the locality ladder of §4.3.5, and the per-pixel floor, with about a fifth of the parameters, is a control that marks what a model without spatial context can do.

The free-form FEN control at width 96 has 34 785 parameters, half the reference and a factor of 1.97 below the T-FEN it controls, because it is the T-FEN with the transport head removed (§4.3.6): of the T-FEN's 68 503 parameters, 34 785 belong to the free-form term and 33 718 to the transport head. Why this gap does not explain the effect of the transport term is discussed in §5.4.4.

Table 4.3: Parameter counts of all arms, with their ratio to the canonical configuration (67 147 parameters) and the property each arm isolates. The architecture arms lie within a factor of 1.25 of the canonical configuration. Three arms lie outside this band: the per-pixel floor, a control without spatial context; the one-hop floor, a rung of the locality ladder and a truncation of the index-space arm; and the free-form FEN at width 96, which is smaller than the T-FEN it is compared with by a factor of 1.97.

Arm	Parameters	Ratio	Isolates
Graph solver, dilated stencil (canonical)	67 147	1.00	reference
Graph solver, index-space k-NN	67 147	1.00	graph geometry
T-FEN	68 503	1.02	finite-element operator with transport
FEN, free-form only, width 96	34 785	0.52	transport term
FNO	79 160	1.18	global support, no local path
FNO with 3×3 local path	79 960	1.19	global support plus local detail
U-Net, width 5	83 081	1.24	multiscale local detail
One-hop floor	40 971	0.61	one ring of spatial context
Per-pixel floor	14 795	0.22	no spatial context (control)
Any operator with RK4 wrapper	unchanged	—	integration order

4.4 Implementation Details

The framework is implemented in Python with PyTorch 2.4.1 and CUDA 12.1 (Paszke et al., 2019). The graph operators use PyTorch Geometric 2.6.1 (Fey and Lenssen, 2019). All dependencies are pinned in a fixed container image. Every reported run was trained on a single NVIDIA RTX A6000 GPU on a cluster managed by Slurm, and the five folds of an experiment were submitted together as a job array. The structure of the code and its availability are described in Appendix D.

The windows of each fold are precomputed once before training. This step performs the crop and pad, the normalisation and the extraction of Δt and the covariates, and it builds the index-space neighbour graph. Because the grid is the same for every visit, one graph serves all windows. The dilated stencil of the canonical graph network (§4.3.3) is built at run time from the grid indices rather than stored. All precomputed data are held in memory during training.

The data loader runs in the main process, because the curriculum changes the sampling restriction of the dataset at every pass and a change made in the main process would not reach separate worker processes. All random number generators (Python, NumPy, PyTorch and CUDA) are seeded, and deterministic

cuDNN algorithms are enforced. Even so, repeated runs of the same configuration do not give identical results on the GPU. Chapter 5 quantifies this run-to-run variation and defines the noise level against which every comparison is read.

Chapter 5

Experiments

This chapter answers the second part of the research question: what kind of operator the framework of Chapter 4 needs. Inside the fixed framework, operators of three different classes reach similar scores at the one-year anchor, and what separates them is how far they reach in one step (§5.4). The operators of §4.3 are run through the same slot under identical conditions. Section 5.1 defines the evaluation, §5.2 compares all operators at one year, §5.3 follows them over the whole follow-up, and §5.4 isolates the properties that separate them. Section 5.5 examines how the folds differ. Section 5.6 tests settings of the framework itself, namely the time handling, the objective and the size of the operators, and so also adds evidence on the first part of the question. Section 5.7 reports a single additional experiment with an adaptive moving mesh. Section 5.8 checks which statements about the operators the results support, §5.9 compares the results with prior work, and §5.10 reports the cost. Section 5.11 summarises the chapter.

5.1 Evaluation Protocol

This section defines how the operators are scored and how two scores are compared. Every operator is scored by the late-epoch mean of the change-region Dice at the one-year anchor, and two operators are compared fold by fold against the variation caused by retraining.

5.1.1 Evaluation Challenges

GA progresses slowly. Between two consecutive visits, almost all pixels of the lesion mask stay unchanged. Persistence, the prediction that nothing changes, is therefore a strong per-pixel baseline. A per-pixel mean squared error or root mean squared error (RMSE) is dominated by the static pixels, so a model can reach a low per-pixel error without predicting any progression.

The same holds for a Dice score over the whole lesion. The Dice score measures the overlap of two regions: it is 1 when they coincide and 0 when they share no pixel. Over the whole lesion it is dominated by the static interior, so persistence already reaches a full-mask Dice of about 0.87, and the score barely responds to progression.

Raw errors are also not compared across architectures. After normalisation, the mask channel takes only two values (§3.4). The clinical metrics only ask

on which side of the threshold between them a predicted value lies, whereas the RMSE measures how far each predicted value lies from its target. A prediction can therefore be correct for the Dice and still carry a large RMSE. Under autoregressive rollout, the dense operators, such as the U-Net and the Fourier Neural Operator, produce more mask values far from the two mask values than the graph networks: the late-epoch mask RMSE of the rollout is roughly twice as large for the U-Net as for either graph network, while its thresholded metrics are higher than those of the k-NN graph network and close to those of the dilated-stencil network. A raw-error comparison across architecture classes would therefore measure how far the raw values stray, not whether the lesion is predicted correctly. In addition, the soft-Dice term of the loss stretches the raw mask values away from the threshold (§4.2.4). RMSE values are computed and logged, but no RMSE value is used as evidence in this chapter.

5.1.2 Metrics

The headline metric is the change-region Dice at the one-year anchor: the first visit of the rollout whose time since baseline reaches about one year (§3.6, §5.1.3). All predicted and ground-truth masks are binarised at the physical threshold of 0.5, which is mapped into normalised space in the same way as in the loss. Let b be the thresholded mask at the baseline visit (the first visit of the rollout), p the thresholded prediction at the one-year anchor and y the thresholded ground truth at the one-year anchor. The predicted and the true change regions are

$$\hat{C} = \{ i : p_i \neq b_i \}, \qquad C = \{ i : y_i \neq b_i \}, \tag{5.1}$$

and the change-region Dice is $2|\hat{C} \cap C| / (|\hat{C}| + |C|)$. Applied to the change regions, the Dice score measures how well the predicted change overlaps the change that occurred; the large part of the lesion that stays the same does not enter it.

Two consequences follow. First, persistence predicts no change, so its score is 0 by construction, and every positive value is an improvement over predicting no progression. Second, the metric is symmetric. A predicted shrinkage counts as predicted change and is scored against the truth, which is consistent with the absence of a monotonic-growth constraint (§4.2.1).

The change-region Dice differs from the growth-region Dice of Mai et al. (2024), which this chapter also uses. The growth-region Dice scores only new atrophy, the pixels that are lesion at the one-year anchor but not at baseline. The change-region Dice scores every changed pixel, growth and shrinkage alike. Both score 0 for persistence and are easy to confuse: Table 5.1 and all differences between arms use the change-region Dice, while the eye counts of §5.1.5 use the growth-region Dice.

Formally, the growth-region Dice compares the true new-atrophy region $\{ i : y_i = 1 \text{ and } b_i = 0 \}$ with its predicted counterpart. It is used for the eye counts of §5.1.5. It is also computed at every rollout step and grouped by the time since baseline into $(0, 1]$, $(1, 2]$, $(2, 3]$ and more than three years. These horizon bins show how the growth-region Dice changes beyond the one-year anchor. Mai et al. (2024) report the same metric in the same bins.

Because growth across the edge of the imaged field is censored (§3.3), the eyes are also split by whether their lesion touches that edge; the split is reported

in §5.2. An eye counts as border-touching when its true lesion at the one-year anchor touches the edge of the imaged field. This edge is the outermost ring of the crop or, for padded visits, the edge of the zero padding, since for those visits the true edge of the field lies inside the crop. Of the 75 eyes, 21 touch it.

Finally, a collapse counter records how many eyes have a thresholded prediction at the one-year anchor that is identical, pixel for pixel, to their baseline mask. It detects a model that has learned to copy its input, which would score exactly as persistence does. The eye-level growth-rate measures used for the clinical comparison are defined in §5.9.

5.1.3 Rollout Evaluation

All experiments use the patient-level five-fold cross-validation of §3.7, which has no test split; every result is a validation result. Two evaluation procedures run after every epoch on the validation eyes of the fold. The one-step evaluation predicts every validation window once from its observed input. It isolates the accuracy of a single step from the drift of a rollout. Every reported number comes from the rollout evaluation. For each validation eye, it starts at the baseline visit and predicts the next visit. The prediction is then fed back as the input of the next window, and the rollout continues over the eye's real visit schedule, with the actual Δt of each window.

The one-year anchor is taken at the first step whose cumulative time reaches it (§3.6). For a few eyes this step lies beyond one year, where the true change region is larger. This is accepted so that every eye contributes one score at the anchor (§3.6). Persistence is computed inside every rollout by the same code; for the one-year anchor metrics it is the baseline mask held constant. Metrics at the one-year anchor are computed once per eye and averaged over the eyes, so every eye counts once. The horizon bins of §5.3 instead pool all follow-up visits in a bin, so an eye with several visits in a bin contributes several values.

5.1.4 Late-Epoch Mean

Each run is trained for 30 epochs and evaluated on the validation eyes of its fold after every epoch. A comparison of operators needs one value per run, and the way this value is formed affects every number in this chapter. Every number reported in this chapter is a late-epoch mean: the mean of the metric over epochs 10 to 29 of one run.

A mean over several epochs is used instead of a single epoch for two reasons: noise and selection bias. The validation score of a run is subject to noise. After an operator has converged, its score still fluctuates between consecutive epochs without a systematic trend, because each fold contains only 12 to 20 validation eyes (§3.7), so small weight updates change the fold average noticeably. Over epochs 10 to 29, this fluctuation has a standard deviation of 0.007 to 0.025 for the arms of Table 5.1, the same order of magnitude as many of the differences between arms. Choosing the epoch with the highest score introduces, in addition, a selection bias. The maximum of thirty noisy scores systematically exceeds the expected score of the operator, because the selection favours epochs whose noise happens to be positive. For the arms of Table 5.1, this bias

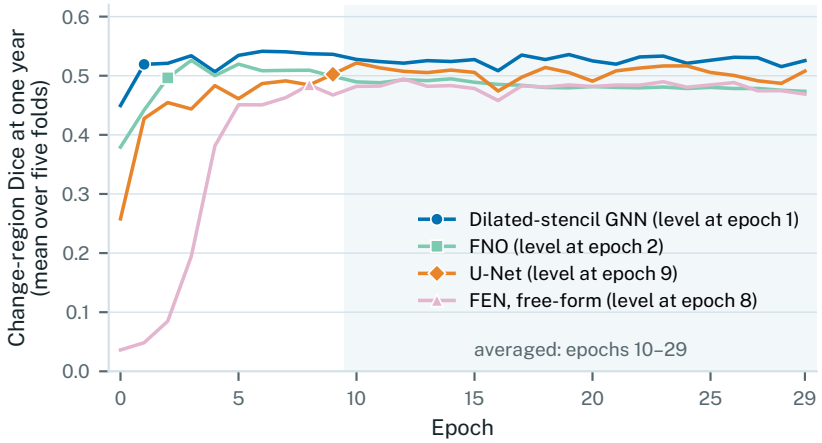


Figure 5.1: Change-region Dice at the one-year anchor over training, mean over the five folds (seed 42), for two operators that learn fast (dilated-stencil GNN, FNO) and two that learn slowly (U-Net, free-form FEN). The marker on each line shows the epoch at which it reaches its level, defined as the first epoch whose fold mean lies within 0.01 of the late-epoch mean. The shaded band marks epochs 10 to 29, over which every reported value is averaged.

amounts to 0.025 to 0.050, and it differs between arms: it is larger both for arms whose scores fluctuate more and for arms whose score peaks early and then settles slightly lower, such as the FNO. It could only be removed by evaluating the selected epoch on independent eyes, which a cross-validation without a test split (§3.7) does not provide.

The window starts at epoch 10 because the operators learn at different speeds (Figure 5.1), and the mean begins only once the slowest of them has reached its level. Most of them reach the level they keep for the rest of training within the first few epochs, and some, such as the FNO, peak shortly afterwards and settle slightly lower. The free-form finite-element network and the U-Net rise more slowly and reach their level only at epochs 8 and 9. A mean that started earlier would include epochs in which these two operators are still learning, which would lower their reported scores below the level they reach after training. Over epochs 10 to 29, every arm has settled: the means over epochs 10 to 19 and 20 to 29 differ by at most 0.014 for every arm.

The late-epoch mean therefore measures every operator after it has reached its level and over the same epochs for every arm. Because no epoch is chosen on the validation scores, it carries no bias from selecting an epoch, and averaging twenty epochs reduces the noise of a single epoch. It does not remove the variation between repeated training runs of the same configuration; this variation sets the noise level against which differences between arms are read (§5.1.5).

5.1.5 Noise Level and Comparisons

Two sources of variation enter every comparison. The first is training noise. Repeated runs of the same configuration do not give identical results on the GPU (§4.4): retraining the same model on the same fold already changes its

late-epoch score by up to about 0.02. The second is the fold. The five folds hold different eyes, and the spread between them is larger than most differences between arms (§5.2). Two arms are therefore always compared fold by fold, on the same five folds, and the five differences are averaged.

A difference between two arms is tested against the variation caused by retraining. Repeated runs of the same configuration on the same fold give a standard deviation of $\sigma \approx 0.0127$ for the late-epoch change-region Dice of one graph network (Appendix C.3). This value is assumed for every arm except the U-Net, whose spread is measured separately below. If two arms did not differ, the mean of their five fold-wise differences would vary with a standard deviation of $\sigma_\Delta = \sqrt{2}\,\sigma/\sqrt{5} \approx 0.008$. The ratio $z = \bar{\Delta}/\sigma_\Delta$ is compared with a standard normal distribution: a difference with $|z| > 1.96$, that is, a mean difference above about 0.016, is unlikely to come from retraining alone (two-sided $p < 0.05$). The runs of the U-Net vary more: its two seeds on the five folds give $\sigma \approx 0.019$, so comparisons with a U-Net are read against a mean difference of about 0.020. A difference is called established only when, in addition, it holds on most folds, the comparison over the 75 validation eyes points the same way clearly (a paired t of about 2.5 or larger, Appendix C.3), it does not change sign when any one fold is left out, and, where the comparison was run at both seeds, it holds at both. Results are stated in these terms, for example "higher by 0.064 on average ($z = 7.9$), on all five folds and on 70 of the 75 eyes". The eye counts and the per-eye test use the growth-region Dice at the one-year anchor, which is recorded per eye. In tables, "mean $\pm$ sd" is the mean and the standard deviation over the five fold values, each the late-epoch mean of one run.

The comparisons listed in Appendix C.4 were repeated with a second random seed, and their conclusions hold at both; the other experiments use seed 42 only. All numbers in this chapter are for seed 42 unless marked as seed 7. Appendix C.3 gives the exact criteria, and Appendix C.4 the standard errors and test statistics of every comparison together with the values at seed 7.

5.2 Main Results

This section compares all arms on the headline metric, the change-region Dice at the one-year anchor (Table 5.1, seed 42). The highest arms of three different architecture classes, the T-FEN, the dilated-stencil graph network and the U-Net, lie within about 0.04 of each other.

The per-pixel floor reads exactly 0.0000 on all five folds. On every validation eye, its thresholded prediction at the one-year anchor is identical to the baseline mask (§5.1.2). Every point of change-region Dice in the table therefore comes from spatial context. The standard deviations in the table are between-fold deviations and are much larger than the noise level of §5.1.5. For the k-nearest-neighbour graph network, the fold values range from about 0.43 to 0.52, and for every arm except the two FNO arms without a local path the best fold is fold 2. Arms are therefore compared fold by fold; how the folds differ is examined in §5.5. Adjacent rows mostly lie within the noise level of each other, so the order of the table alone does not rank them.

The highest arms of the three leading classes are a finite-element network, a graph network and a U-Net. The Runge–Kutta variant of the graph network,

Table 5.1: All arms on the change-region Dice at the one-year anchor, seed 42. Each fold value is the late-epoch mean (epochs 10 to 29) of one run; the table gives the mean $\pm$ sd over the five folds and the highest of the five fold values (best fold), with the index of that fold in brackets. Persistence scores 0 by construction. The ratio is the parameter count divided by that of the canonical dilated-stencil graph network (67 147). Cost is the minimum epoch time in seconds over the five folds; it supports order-of-magnitude statements only (§5.10). The arms are ordered by mean, with the two locality floors and persistence at the bottom; adjacent arms mostly differ by less than the noise level (§5.1.5). The T-FEN uses the energy-conserving form of its transport operator (§5.8.2).

Arm	Params	Ratio	Mean $\pm$ sd	Best fold	s/epoch
T-FEN (§4.3.6)	68 503	1.02	0.5428 ± 0.0401	0.6047 [2]	367
Graph network, dilated stencil (§4.3.3)	67 147	1.00	0.5258 ± 0.0497	0.6060 [2]	91
Graph network, dilated stencil, RK4 (§4.3.7)	67 147	1.00	0.5139 ± 0.0497	0.5907 [2]	439
U-Net, width 5 (§4.3.4)	83 081	1.24	0.5046 ± 0.0632	0.5918 [2]	21
FNO with 3×3 local path (§4.3.4)	79 960	1.19	0.4984 ± 0.0398	0.5509 [2]	29
FNO, RK4 (§4.3.7)	79 160	1.18	0.4893 ± 0.0398	0.5276 [4]	45
FNO (§4.3.4)	79 160	1.18	0.4831 ± 0.0477	0.5391 [4]	23
FEN, free-form only (§4.3.6)	34 785	0.52	0.4810 ± 0.0348	0.5323 [2]	180
Graph network, k-NN, $k = 12$ (§4.3.3)	67 147	1.00	0.4623 ± 0.0361	0.5153 [2]	76
Graph network, k-NN, $k = 20$ (§4.3.3)	67 147	1.00	0.4519 ± 0.0436	0.5223 [2]	108
One-hop floor (§4.3.5)	40 971	0.61	0.4515 ± 0.0243	0.4856 [2]	37
Per-pixel floor (§4.3.5)	14 795	0.22	0.0000 ± 0.0000	0.0000	16
Persistence	0	—	0	0	—

third in the table, is compared with its single-step version in §5.6.1. The T-FEN reported here uses the energy-conserving form of its transport operator (§4.3.6). Its first version, with the Galerkin form, is not stable beyond the training horizon; the two forms are compared in §5.8.2. The T-FEN is higher than the dilated-stencil graph network by 0.017 on average, on four of the five folds and on 46 of the 75 eyes. This lead is small and, as discussed below, not established. The dilated-stencil network is higher than the U-Net by 0.021 on average, but on only three of the five folds. Because the scores of the U-Net vary more between runs than those of the graph networks (Appendix C.3), this lies just beyond the variation of retraining at seed 42 ($z = 2.1$) and within it at seed 7 ($z = 0.8$), and the two cannot be told apart. The T-FEN is higher than the U-Net by 0.038 on average, on all five folds and on 51 of the 75 eyes, and this lead is real.

Growth across the edge of the imaged field is censored (§3.3). On the 54 eyes

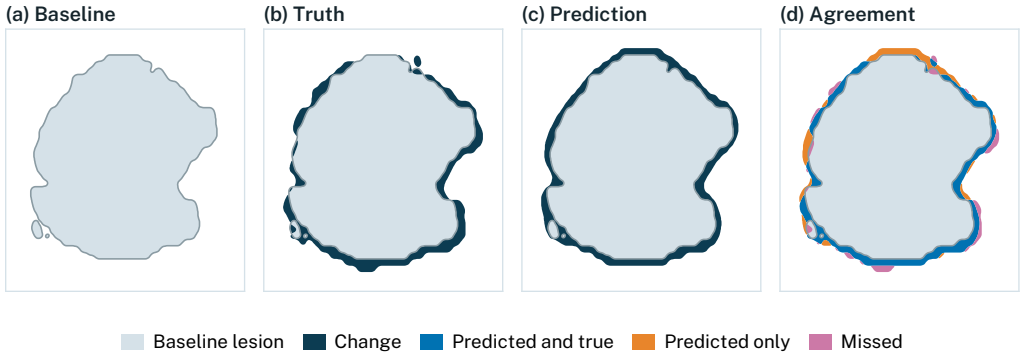


Figure 5.2: Rollout of the canonical model for one validation eye: change at the one-year visit (360 days), cropped to the lesion (each panel 4.4×4.8 mm, physical aspect ratio). (a) The baseline lesion, which the model receives as input. (b) The ground truth: the baseline lesion that remains (light grey) and the true change against it (dark). (c) The prediction of the dilated-stencil graph network, computed with the weights after the last training epoch, in the same encoding. (d) The agreement of (b) and (c): change that is predicted and true (blue), predicted only (orange) and missed (purple). For display, the lesion edges are smoothed; this changes the drawn areas by less than 3 % and removes isolated single pixels, while the Dice is computed on the unsmoothed masks. Pixel-exact rollouts of four further eyes over several years are shown in Appendix B.

whose lesion does not touch that edge (§5.1.2), the arms keep the order of Table 5.1; the values for both groups are given in Table C.1.

Although the T-FEN scores highest at one year, the dilated-stencil graph network serves as the canonical model of this thesis. The lead of the T-FEN, 0.017, only just passes the fold-paired test at both seeds ($z = 2.1$ against a bar of 1.96). In the per-eye test it passes at the seed 7 ($t = 3.1$) but not at seed 42 ($t = 2.2$ against a bar of about 2.5), so by the criteria of §5.1.5 it is not established. The graph network also trains about four times faster, at 91 against 367 s per epoch (§5.10), which matters for the many runs of this chapter that use it as their reference. The canonical model is therefore the reference configuration of the framework.

Figure 5.2 shows the rollout of the canonical model for one eye. Most of the lesion stays as it is, and the metric scores only the band of change along its border. Four further eyes with lesions of different size, number and growth rate are shown in Appendix B. Across the 48 validation eyes of folds 0 to 2, the canonical model predicts less change than occurs in 35, at a median of 0.73 of the true change.

In summary, operators of different classes reach similar scores: the highest arms of three classes lie within about 0.04 of each other, and of the comparisons between them only the lead of the T-FEN over the U-Net is established. The two highest arms share one property. Both extend the reach of a local operator on this grid (§4.3.3): the T-FEN through its transport term, the dilated-stencil network through neighbours placed at lesion scale. Section 5.4 tests such properties one at a time. The Runge–Kutta arms are compared with their single-

step counterparts in §5.6.1, and the cost column is discussed in §5.10.

5.3 Forecasts Beyond One Year

The one-year anchor scores every eye once. The rollout, however, continues over the whole follow-up of an eye, up to about six years. This section follows the forecast over that time. The dilated-stencil network and the T-FEN are the two highest operators in each of the first three years, and every operator scores higher after the first year than within it. For every rollout step, the growth-region Dice of the predicted against the observed visit is assigned to a bin by the time since baseline; the values are averaged over the late epochs and pooled over the five folds (Table 5.2); Figure 5.3 shows the profile of four of them. The four bins hold 149, 149, 109 and 71 follow-up visits. The values are descriptive: no difference between bins or operators is tested, and the later bins hold only the eyes with the longest follow-up.

Table 5.2: Growth-region Dice by time since baseline for the operators of Table 5.1 without the Runge–Kutta variants and the per-pixel floor, seed 42. Each value is the mean over every rollout step in the bin and over the late epochs, pooled over the five folds; the bins hold 149, 149, 109 and 71 follow-up visits. The T-FEN uses the energy-conserving form of its transport operator.

Operator	0–1 years	1–2 years	2–3 years	> 3 years
T-FEN	0.519	0.602	0.633	0.611
Graph network, dilated stencil	0.503	0.600	0.634	0.655
U-Net, width 5	0.458	0.563	0.581	0.555
FNO with 3×3 local path	0.464	0.553	0.571	0.553
FNO	0.446	0.546	0.583	0.594
FEN, free-form only	0.388	0.524	0.536	0.520
Graph network, k-NN, $k = 12$	0.397	0.531	0.557	0.560
One-hop floor	0.395	0.524	0.556	0.571

Every operator scores higher after the first year than within it. Within the first year, the operators fall into three groups. The three operators without a construction for long reach, the k-NN graph network, the one-hop floor and the free-form FEN, score 0.388 to 0.397. The dense operators score 0.446 to 0.464, and the two operators that extend the reach of a local update, the dilated stencil and the T-FEN, score 0.503 and 0.519.

In the second and third year, the dilated stencil and the T-FEN stay level with each other and above all other operators. Beyond three years, the dilated stencil scores highest (0.655), followed by the T-FEN (0.611) and the FNO (0.594); the FNO now lies above the U-Net, which it trailed in the first two years (§5.4.3). This last bin holds the fewest visits.

The dilated stencil and the T-FEN score highest in the first three years; beyond three years the T-FEN lies only 0.017 above the FNO. Among the other operators, the order changes with the horizon: beyond three years the FNO (0.594)

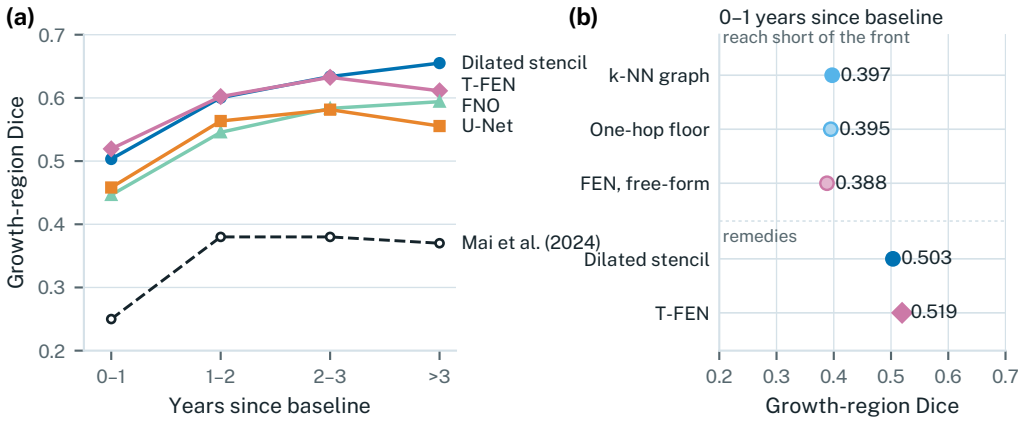


Figure 5.3: Growth-region Dice by time since baseline, averaged over the late epochs and pooled over every rollout step in each bin and over the five folds, seed 42. *(a)* Profile over the four horizon bins for the dilated-stencil graph network, the T-FEN, the U-Net (width 5) and the FNO; the bins hold 149, 149, 109 and 71 follow-up visits. The dashed line gives the values reported by Mai et al. (2024) on the full cohort from which the data here are drawn; it is a reference, not a head-to-head comparison (§5.9). *(b)* The first-year bin alone, on the same scale: the three operators without a construction for long reach (the k-NN graph network, the one-hop floor and the free-form FEN) against the two constructions that extend the reach, the dilated stencil and the transport term of the T-FEN.

lies above the U-Net (0.555) and the FNO with local path (0.553), which it trailed in the first year.

5.4 Ablation Studies

The operators of Table 5.1 differ in more than one respect at once, so their scores alone do not show which property of an operator matters. This section therefore compares operators that differ in one property only, with the rest of the framework held fixed. Four properties are tested, the ones named in §2.3: whether the operator has spatial context at all (§5.4.1), how far that context reaches physically (§5.4.2), whether global context needs a local path (§5.4.3), and whether an explicit transport term helps (§5.4.4). Section 5.4.5 brings the results together. All comparisons are taken fold by fold (§5.1.5).

5.4.1 Spatial Context

The first question is whether an operator needs to see its neighbours at all. It is tested on the graph network on the index-space k-nearest-neighbour graph with zero, one and two message-passing rounds (§4.3.5). With zero rounds, each pixel sees only itself.

Without message passing, the network predicts no change on any eye and scores 0.0000, the same as persistence. One round raises the score to $0.4515 \pm$

0.0243, and a second round to 0.4623 ± 0.0361. The second round adds 0.011 on average. This is within the noise level: the one-hop floor is higher on one of the five folds and on 33 of the 75 eyes.

Spatial context is therefore necessary, and on this graph a single ring of neighbours provides almost all of it. Along a row, one hop of the index-space graph reaches only about 11 µm (§4.3.3). The dilated stencil, tested next, was built to remove this limit.

5.4.2 Physical Reach at Lesion Scale

The second question is whether the physical reach of the neighbourhood matters. The graph network is kept with its loss, its data and its 67 147 parameters, and only its edge set is changed. The index-space k-nearest-neighbour graph, whose hop reaches about 11 µm along a row, is replaced by the dilated stencil, whose hop reaches about 119 µm (§4.3.3, Figure 4.2). The dilated stencil is higher by 0.064 on average, on all five folds and on 70 of the 75 eyes. This result is the reason the dilated stencil is the canonical configuration.

Table 5.3 varies the stencil around this default. A reach of ± 12 columns is lower by 0.019 and a reach of ± 42 columns by 0.026, each higher than the default on at most one fold. The ± 42 geometry keeps the 20 neighbours of the default by doubling the column spacing from 7 to 14, so reach and spacing change together. To separate the two, the ± 35 geometry keeps the default spacing and adds neighbours (32 in total). It is also lower, by 0.010, within the noise level, so a longer reach does not help even at the default spacing. Widening the stencil across rows, to ± 2 rows, lowers the score most of all geometries, by 0.039.

The number of neighbours was also varied at a fixed reach. With $k = 20$ neighbours, the index-space graph reaches as far as with $k = 12$. It is lower by 0.010 on average, higher on only one of the five folds and on 20 of the 75 eyes. The fold average is within the noise level while the eyes lean consistently one way, so the difference is not established; neither view shows a gain from the extra neighbours.

The placement of the neighbours also accounts for the gap between the graph network and the U-Net. On the index-space graph, the parameter-matched U-Net is higher by 0.042 on average, on four of the five folds and on 53 of the 75 eyes; fold 3 goes the other way (-0.042). On the dilated stencil, the two cannot be told apart (§5.2).

Reach along the rows therefore decides how well the graph network does. No tested geometry scores higher than the default hop of ± 21 columns (about 119 µm), which was set from the 90th percentile of the per-visit front advance, 23 columns (§4.3.3). With two rounds, one update of the default network reaches ± 42 columns (about 238 µm), about twice that advance; the ± 12 variant, whose update reaches ± 24 columns, scores lower. Shorter and longer reaches score lower, and more neighbours at the same reach add nothing. More reach through more message-passing rounds does not help either: four rounds instead of two, which double the reach of the stencil, lower the canonical network by 0.009 on average (Table 5.3, §5.6.3). The gap to the U-Net closed when only the placement of the neighbours changed, so it lay in the graph.

Table 5.3: The reach ladder. Each variant of the dilated stencil is compared with the default stencil (rows ± 1, columns $\{0, \pm 7, \pm 14, \pm 21\}$, first row), and the index-space graph with $k = 20$ with the one with $k = 12$. Mean is the change-region Dice at the one-year anchor over the five folds, seed 42; the difference is the mean over the folds of the per-fold difference to the reference; the counts give the folds and the eyes on which the variant is higher than the reference. Standard errors are given in Appendix C.4.

Geometry	Mean	Difference	Folds	Eyes
Dilated stencil				
Columns up to ± 21 (spacing 7, 20 neighbours; default)	0.526	—	—	—
Columns up to ± 12 (spacing 4, 20 neighbours)	0.507	-0.019	1/5	20/75
Columns up to ± 35 (spacing 7, 32 neighbours)	0.516	-0.010	1/5	19/75
Columns up to ± 42 (spacing 14, 20 neighbours)	0.500	-0.026	0/5	16/75
Rows ± 2 (34 neighbours)	0.487	-0.039	0/5	9/75
Four message-passing rounds instead of two	0.517	-0.009	0/5	20/75
Index-space k-NN graph				
$k = 12$ (reference)	0.462	—	—	—
$k = 20$ (same reach)	0.452	-0.010	1/5	20/75

5.4.3 Global and Local Context Together

The third question is whether an operator with global context also needs a local path. The Fourier Neural Operator sees the whole grid in every layer through its Fourier modes but has no local path (§4.3.4). This comparison pairs the plain FNO with the same FNO plus a 3×3 local path, which adds 800 parameters.

The local path raises the FNO by 0.015 on average, on four of the five folds and on 48 of the 75 eyes. The fold average is within the noise level while the eyes lean consistently one way, so the effect is not established.

Against the other operators, the plain FNO lies below the U-Net at the one-year anchor at seed 42: lower by 0.022 on average, higher on only one of the five folds and on 30 of the 75 eyes. At seed 7 it is lower as well, but within the noise level, so the difference is not established (Appendix C.4). It is higher than the k-NN graph network by 0.021 on average, but fold 4 carries all of it (without it, -0.001), so this is not established. With the local path, the FNO cannot be told apart from the U-Net (-0.006 on average). It is higher than the k-NN graph network by 0.036 on average, on four of the five folds and on 49 of the 75 eyes, and this does not depend on any single fold.

Beyond one year, the order between the FNO and the U-Net changes (Figure 5.3a). In the growth-region Dice, the FNO reads 0.546 against 0.563 for the U-Net in the second year, 0.583 against 0.581 in the third and 0.594 against

0.555 beyond three years, which suggests that the FNO gains on the U-Net at longer horizons.

Global context alone scores below the U-Net at the one-year anchor at both seeds, beyond the noise level at one of them. With a local path of 800 parameters, the FNO can no longer be told apart from the U-Net, but the effect of the local path itself is not established.

5.4.4 An Explicit Transport Term

The fourth question is whether an explicit transport term helps. The finite-element network computes the change of the state with a free-form term, a small network evaluated on each triangle. The T-FEN adds a transport term, which moves the state along a velocity predicted by a second network (§4.3.6). The comparison pairs the T-FEN with the same network without the transport term, at the same width of 96. Both arms start at persistence, because their output heads are zero-initialised (§4.2.1). The T-FEN adds the transport term, the velocity network that drives it (33 718 parameters) and the coarser mesh on which it runs (§4.3.6).

The T-FEN is higher by 0.062 on average, on all five folds and on 68 of the 75 eyes. With the earlier Galerkin form of the transport operator, the difference was of the same order, 0.053, again on all five folds. That form becomes unstable in long free-running rollouts, but only well after the one-year anchor (§5.8.2), so its score at the one-year anchor is valid, and the effect does not depend on the form of the transport operator.

The control without the transport term is not parameter-matched, however. It has 34 785 parameters against 68 503 for the T-FEN, so part of the gain could come from the extra parameters. Two results argue against this. The T-FEN at width 48 has 20 455 parameters, fewer than the control, and scores the same as at width 96. In addition, a wider operator does not score higher in any class of the capacity study (§5.6.3). The gain of the transport term is therefore not a matter of parameter count.

The first-year values show where the transport term acts. In the growth-region Dice up to one year, the free-form FEN reads 0.388, close to the one-hop floor (0.395) and the k-NN graph network (0.397). All three operators reach less far in one step than the lesion front advances between two visits (§4.3.3). With the transport term, the T-FEN reads 0.519, level with the dilated stencil (0.503), whose single hop was designed to match that advance (Figure 5.3b). Over the later years the two stay close: 0.602 against 0.600 in the second year and 0.633 against 0.634 in the third. Only beyond three years does the stencil lie higher (0.655 against 0.611, Figure 5.3a). The velocity field learned by the T-FEN is not interpreted here; its stability is examined in §5.8.2.

The transport term therefore improves the finite-element network, also when the T-FEN has fewer parameters than its control (width 48), and it brings the first-year values to those of the dilated stencil.

5.4.5 Summary of the Ablation Studies

Two of the tested properties improve an operator against its own control, and both concern how far an operator reaches in one step. The GA front advances a median of 12 columns (68 µm) between two visits 180 days apart, with a 90th percentile of 23 columns (131 µm, §4.3.3). One hop on the index-space graph reaches only ± 2 columns, about 11 µm along a row, so a single step of such an operator covers only a small part of that advance.

Two different constructions remove this limit. The dilated stencil places the neighbours of the graph network at lesion scale, so that one hop spans 21 columns (§5.4.2). The transport term of the T-FEN moves the state along a learned velocity (§5.4.4). Each is established against its own control, and the two arms lie within about 0.017 of each other as the two highest arms of Table 5.1 (§5.2). That a graph with dilated edges and a transport term on a mesh with the same seven-column spacing reach about the same score is the central architectural result of the survey. The transport comparison does not separate the transport mechanism from the coarser mesh on which it runs. The result is consistent with the reach explanation but does not prove the mechanism.

Spatial context is necessary (§5.4.1), and a local path for the FNO is not established (§5.4.3). Within one architecture class, changing one of these properties moves the score by 0.06 to 0.07; between the three highest classes, the scores differ by at most 0.04. The settings of the framework itself are tested in §5.6.

5.5 Differences Between Folds

The scores of the five folds differ by more than most operators differ from each other (§5.2). The folds differ mainly in the eyes they hold; two properties of an eye, the distance at which new atrophy appears and contact with the window edge, also go with differences between the operators. This section asks why the folds differ, and whether some eyes suit one kind of operator better than another, which links the fold differences to the reach argument of §5.4.5. The analysis uses the five main operators of Table 5.1 at both seeds. Its per-eye values are the growth-region Dice at the one-year anchor.

Figure 5.4 shows the fold values of the five operators. Fold 2 is the highest fold of every operator at both seeds, with one exception: the FNO at seed 42, whose highest fold is fold 4. The folds come from split files supplied with the data, which assign the patients at random, stratified by lesion area (§3.7). With 12 to 20 eyes per fold, a few eyes can still shift a whole fold. Fold 2 holds the largest lesions, with a median baseline area of 7.9 mm^2 against 4.1 to 6.3 mm^2 on the other folds, and slowly growing ones (median growth rate 0.21 mm per year in square-root area, the lowest together with fold 3). Both properties go with a higher score: per eye, the growth-region Dice of the canonical model rises with the baseline lesion area (Spearman $\rho = 0.45$) and falls with the growth rate ($\rho = -0.35$, 75 eyes). Every operator meets the same eyes on a fold, so this common shift cancels when two operators are compared fold by fold.

Beyond this common shift, some folds favour one kind of operator. At both seeds, the dilated-stencil network lies below the mean of the T-FEN, the U-Net and the FNO on folds 1 and 4 and above it on folds 0, 2 and 3; the k-nearest-

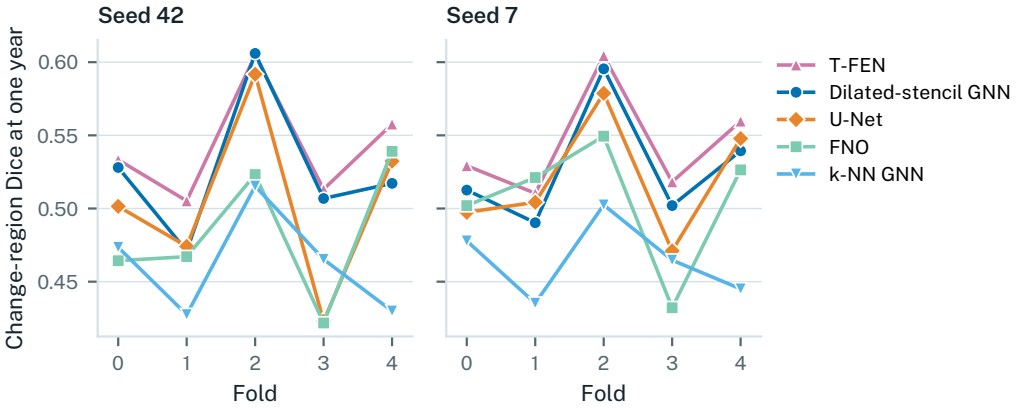


Figure 5.4: Change-region Dice at the one-year anchor on each of the five folds (late-epoch mean) for the five main operators of Table 5.1, at seed 42 (left) and seed 7 (right). Lines connect the values of one operator to make the shift between folds visible; the folds themselves have no order.

neighbour network, below it on most folds, falls furthest behind on fold 4. On fold 3 the U-Net and the FNO fall behind the dilated-stencil network and the T-FEN, at both seeds and on 14 of the fold's 15 eyes. Two properties of the eyes separate the operators.

The first is the distance at which new atrophy appears. For every eye, the share of its new atrophy at the one-year anchor that lies more than 0.24 mm from the baseline lesion was computed; 0.24 mm is about the reach of two message-passing rounds of the dilated stencil (§5.4.2). The larger this share, the lower the dilated-stencil network scores relative to the mean of the T-FEN, the U-Net and the FNO (Figure 5.5; Spearman $\rho = -0.32$ over the 75 eyes). Eyes with little distant atrophy favour the graph network: on the third of the eyes with the least distant atrophy, it scores 0.037 higher than their mean, and its lead shrinks as the share grows, to 0.005 below their mean on the third with the most, within one standard error of zero. Fold 1 holds the most eyes of the second kind, with a median share of 22 % against 5 to 16 % on the other folds, and its three largest deficits of the graph network are eyes of this kind at both seeds. Two of them are the two eyes of one patient. The pattern is consistent with the reach argument of §5.4.5: an operator with limited reach cannot place atrophy that appears beyond it.

The second property is contact with the edge of the window. The dense operators fall behind the local ones most when the lesion touches the edge of the imaged field (§5.1.2), by 0.073 on these 21 eyes against 0.021 on the other 54 ($\rho = -0.37$). A possible reason is how the operators treat the edge: both dense operators see zeros beyond it, the U-Net through the padding of its convolutions and the FNO through the zero padding before its Fourier transform, whereas a graph node at the edge simply has fewer neighbours.

The dips on folds 4 and 3 have partial explanations. On fold 4, both graph networks, the dilated-stencil and the k-NN network, predict too little new atrophy. The fold holds the smallest lesions, which grow most relative to their size: the new atrophy at the one-year anchor is a median 39 % of the baseline area, against 18 % (fold 3) to 35 % (fold 0) on the other folds. Every operator pre-

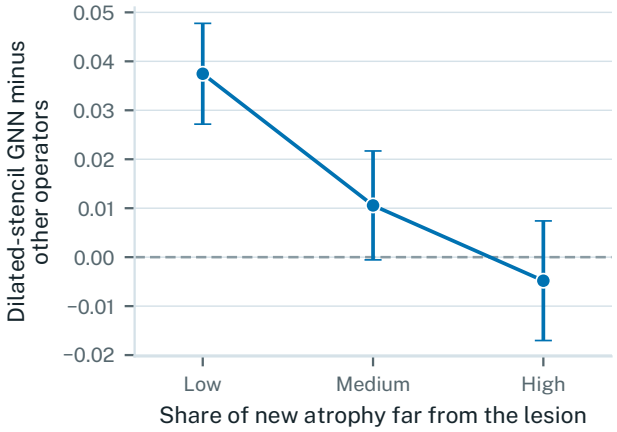


Figure 5.5: Difference in growth-region Dice at the one-year anchor between the dilated-stencil graph network and the mean of the T-FEN, the U-Net and the FNO, per eye and averaged over seeds 42 and 7. The 75 validation eyes are split into thirds by the share of their new atrophy that lies more than 0.24 mm from the baseline lesion (low: 0 to 5 %, medium: 5 to 18 %, high: 19 to 67 %). Points are means over the 25 eyes of a third; bars are standard errors.

dicts too little when growth is large relative to the lesion; on the third of all eyes where it exceeds 40 %, each of them predicts less than half of the true new atrophy. These are mostly small lesions, on which an imprecise outline weighs most, and the shortfall also matches the general tendency of the operators to predict too little change (§6.2). On fold 4, however, the dilated-stencil network falls shorter than the FNO on the same eyes, with a median of 49 % of the true amount against 83 %, and within the fold its deficit follows this extra shortfall (Spearman $\rho = 0.79$, 12 eyes). Relative growth does not explain the extra shortfall: across all eyes, the gap between the two operators' predicted amounts does not depend on it. On fold 3 the new atrophy lies closer to the existing lesion than on any other fold. This is where the dense operators fall behind most, by 0.052 on the half of all eyes with the least distant atrophy against 0.019 on the other half, and within the fold they lose most on its two smallest lesions (0.11 and 0.20 mm^2). Together these properties account for about a fifth of the fold-3 gap.

In short, the folds differ mainly because they contain different eyes. Fold 2 holds large, slowly growing lesions, which every operator forecasts well. Two properties of an eye also go with which operator forecasts it better. Where new atrophy appears far from the existing lesion, the T-FEN, the U-Net and the FNO do better than the dilated-stencil network, whose lead shrinks, which is consistent with its limited reach per step. Where the lesion touches the edge of the window, the graph network does better. The comparisons of the previous sections are taken fold by fold, so the shift that all operators share does not affect them. Whether the remaining dips on folds 3 and 4 come from their validation eyes or from their training sets would need retraining on a different split.

5.6 Negative Results

This section tests the time handling of the update (§5.6.1), the training objective (§5.6.2) and the size of the operators (§5.6.3). Most of these tests are negative results: comparisons whose difference stays within the noise level of §5.1.5. Such a result is a null at the available precision only. It does not show that the tested alternative is equivalent, and it is scoped to the operator, the folds and the seed on which it was measured. Two parts of the objective are the exception: removing them lowers the score clearly (§5.6.2).

5.6.1 Time Handling

The framework handles time in a simple way (§4.2.1). The operator makes one step over the whole interval to the next visit, and Δt enters twice: as an input of the operator and as a multiplier of its output, $\hat{u}_{k+1} = u_k + \Delta t_k \cdot f_\theta(u_k, \Delta t_k, \mathbf{z})$. Because every operator shares this update, two of its details were tested on the canonical dilated-stencil network: the multiplication by Δt, and the single step. The single step was also tested on the FNO, a global operator.

Without the multiplication, the update becomes $u_k + f_\theta(u_k, \Delta t_k, \mathbf{z})$, while Δt remains an input of the operator. The change-region Dice does not change: -0.0006 on average, higher on two of the five folds and on 30 of the 75 eyes. The mask channel explains why. Its two values are a fixed distance apart in normalised units, -0.602 and $+1.660$ (§3.4). At a pixel that changes between two visits, the update must add this distance whatever the interval. With the multiplication, f_θ must therefore output the distance divided by Δt, and the operator learns this division internally. The multiplication changes how the same prediction is parameterised. It is kept because the prediction then falls back to the current state as Δt approaches zero (under the assumption of §4.2.1), and because it is the form MP-PDE uses (Brandstetter et al., 2022). As a consequence, f_θ cannot be read as a growth rate; whether the operator can be read as a continuous-time model is tested in §5.8.1.

Replacing the single step by a classical fourth-order Runge–Kutta step (§4.3.7) evaluates the operator four times within the interval and combines the results. In this setting the operator does not receive Δt as an input (the autonomous field of §4.3.7); time enters only through the integrator. Over the dilated-stencil network, the change-region Dice is 0.012 lower than with the single step, on four of the five folds; over the FNO it is 0.006 higher. Both differences lie within the noise level. For the dilated-stencil network, the Runge–Kutta step costs about 4.8 times the training time of the single step (§5.10). Evaluated with finer steps after training, the Runge–Kutta network scores between 0.016 lower and 0.018 higher, depending on the fold, and 0.004 higher on average (§5.8.1).

Table 5.4 collects the three comparisons. Neither the multiplication by Δt nor the integration scheme changes the result. In every tested variant the length of the interval reaches the operator, either as an input to a single step or through an integrator, and the way it arrives makes no measurable difference. The simple update of the framework is therefore sufficient.

Table 5.4: The time-handling tests. Change-region Dice at the one-year anchor, five folds, seed 42; the mean ± sd is over the five fold values. The difference is against the one-step version of the same operator, as the mean over the five folds of the per-fold difference; the columns Folds and Eyes count the folds and the eyes on which the variant is higher.

Variant	Mean ± sd	Difference	Folds	Eyes
Dilated-stencil network, one Euler step	0.526 ± 0.050	—	—	—
without the multiplication by Δt	0.525 ± 0.043	-0.001	2/5	30/75
RK4, autonomous	0.514 ± 0.050	-0.012	1/5	28/75
FNO, one Euler step	0.483 ± 0.048	—	—	—
RK4, autonomous	0.489 ± 0.040	$+0.006$	4/5	43/75

5.6.2 The Objective

Every operator is trained with the same objective (§4.2.4): a mean squared error in which the mask channel is weighted five times more than each layer channel, plus a soft-Dice term on the mask channel, with no monotonic-growth penalty. Because the objective is shared, three runs on the canonical dilated-stencil network test which of its parts carry the result (Table 5.5). The first uses a plain mean squared error, without mask weighting and without the soft-Dice term. The second removes only the soft-Dice term. The third adds a monotonic-growth penalty.

Table 5.5: The objective tests on the dilated-stencil graph network. Change-region Dice at the one-year anchor, five folds, seed 42; the mean ± sd is over the five fold values. The difference is against the full objective, as the mean over the five folds of the per-fold difference; the columns Folds and Eyes count the folds and the eyes on which the variant is higher. The last column gives, for each fold, the first epoch at which the change-region Dice rises above 0. Standard errors are given in Appendix C.4.

Objective	Mean ± sd	Difference	Folds	Eyes	First epoch with change
Full (mask $\times 5$, soft-Dice)	0.526 ± 0.050	—	—	—	0 on every fold
Without soft-Dice	0.476 ± 0.063	-0.050	0/5	19/75	2, 8, 3, 7, 12
Plain mean squared error	0.000 ± 0.000	-0.526	0/5	0/75	never
With monotonic penalty (2.0)	0.522 ± 0.043	-0.004	2/5	34/75	not recorded

Almost every pixel of the mask stays the same between two visits, so under an unweighted mean squared error a model reaches a low loss by copying its input. Such a model predicts no change and scores 0 on the change-region Dice; the collapse counter (§5.1.2) detects it. Trained with the plain mean squared error, the network stays in this collapse for all 30 epochs. Its change-region Dice is 0 on every fold and at every epoch, and at the last epoch 74 of the 75 validation

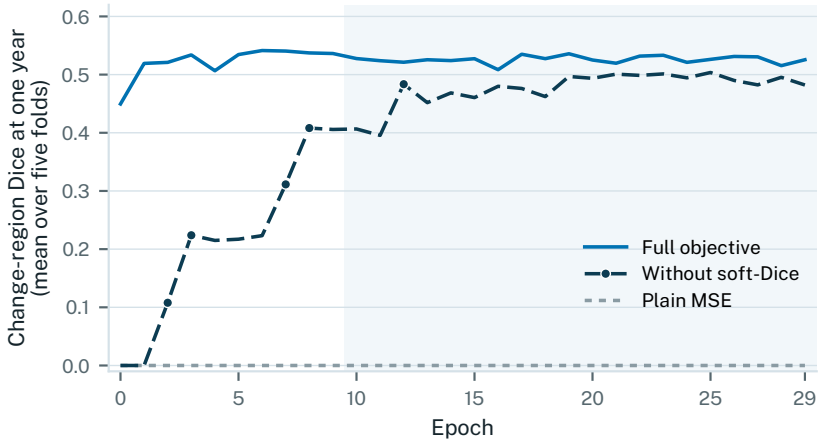


Figure 5.6: Change-region Dice at the one-year anchor over training for the dilated-stencil graph network (seed 42), mean over the five folds, for three objectives: the full objective (late-epoch mean 0.526), the objective without the soft-Dice term (0.476) and a plain unweighted mean squared error, which stays at 0 at every epoch on every fold. Without the soft-Dice term the five folds start to predict change at different epochs (2, 3, 7, 8 and 12). Each dot marks one of these epochs, which is why the dashed line rises in steps. The shaded band marks epochs 10 to 29, over which every reported value is averaged.

eyes have a prediction identical to their baseline mask.

Without the soft-Dice term but with the mask weighting, the network leaves the collapse, though later than with the full objective. On the five folds, its change-region Dice first rises above 0 at epochs 2, 8, 3, 7 and 12, against epoch 0 with the full objective (Figure 5.6). It also ends lower, by 0.050 on average, on all five folds and on 56 of the 75 eyes. Most of the difference comes from fold 4, where the network starts to predict change last: -0.114 there, and -0.012 to -0.044 on the other folds. Because that start falls at epoch 12, inside the late-epoch window, the late-epoch Dice also varies more from epoch to epoch: its standard deviation is 0.043 against 0.015 with the full objective, but on folds 0 to 3 the two are close (0.017 against 0.013). The soft-Dice term has one cost. It stretches the raw mask regression (§4.2.4), so the raw mask error of the free-running rollout is about twice as large with the term as without it (1.10 against 0.51). No thresholded metric is affected.

The third run adds a monotonic-growth penalty. Atrophic tissue does not regenerate, so a penalty on any decrease of the mask where the previous state is atrophic is a natural prior. It is implemented (§4.2.4) but switched off in every other run. At weight 2.0, the change-region Dice changes by -0.004 on average, higher on two of the five folds and on 34 of the 75 eyes. This is within the noise level, and the objective is kept without the penalty.

Two further arguments speak against the penalty. Under the pushforward curriculum (§4.2.5), the previous state on an unrolled step is the model's own earlier prediction. A pixel the model wrongly marked as atrophic can then be corrected only by raising the penalty. A direct computation with the training loss on one such pixel, at weight 2.0, gives a penalty gradient 31.8 times that

of the data terms, so the total gradient points away from the ground truth. The penalty therefore acts differently on steps that start from an observed state and on unrolled steps. In addition, the reference segmentations themselves shrink locally between most pairs of visits (§4.2.1), which a strict growth constraint would contradict.

Taken together, the first two runs show that the network learns change only when the loss emphasises the mask: with the mask weighting it escapes the collapse, without any emphasis it does not. The soft-Dice term makes it predict change from the first epoch and raises the score, and the monotonic penalty adds nothing.

5.6.3 Capacity

The four architecture classes scaled here (the dilated-stencil graph network, the U-Net, the FNO with a local path and the T-FEN) have 67 to 83 thousand parameters at their original size (§4.3.8). The capacity study tests whether the comparisons of §5.2 and §5.4 depend on that size. Each class was scaled in one size setting at a time, with the loss, curriculum, epoch budget and the five folds at seed 42 unchanged. Width is compared across classes, because it adds parameters without changing how far one step reaches. Depth changes the reach of the graph network, so it is read for each class separately; the U-Net and the FNO have no depth setting. The design and a decision rule were fixed before the first run. A width step that raised the score beyond the noise level would have been followed by a further step up, and a step without a gain by a step down, to between about a sixth and a third of the original size. No width step raised the score, so every class was also run at a smaller width; these runs form the remaining rows of Table 5.6.

No width step raises any class beyond the noise level (Table 5.6). The dilated-stencil graph network is flat from a quarter to four times its size (0.520, 0.526, 0.518), and the U-Net does best at or near its original size. The FNO with a local path is lower at both other widths, by 0.019 at width 2 and by 0.018 at width 9, each beyond the variation of retraining (Appendix C.4); its original width is the best tested one. Elsewhere, scaling down costs little: the graph network at 18 219 parameters stays within 0.005 of the canonical configuration, the U-Net at width 2 stays within the noise level, and the T-FEN at width 48 is level with width 96.

Depth does not help where it was tested. Four message-passing rounds instead of two, which also double the reach of the dilated stencil, lower the graph network on every fold, by 0.009 on average, within the noise level. The T-FEN with an eight-layer free-form network trains poorly on every fold: its training error barely falls, and it ends 0.285 lower than at depth four (0.257 against 0.543).

At the shared learning rate and 30-epoch budget (§4.2.6), more parameters improve none of the four classes. The larger T-FEN networks do not train reliably under these settings, so the study does not show whether larger models with retuned optimisation would score higher. The result is scoped to one seed. The failures of the larger T-FEN networks occur in both forms of the transport operator: at width 192 the T-FEN diverges during training on folds 1 and 4 and ties width 96 on the other three, and the eight-layer network trains poorly in

Table 5.6: The capacity study. Change-region Dice at the one-year anchor, five folds, seed 42; the mean is over the five fold values. The ratio is the parameter count divided by that of the arm at its original size. The difference is against that original size, as the mean over the five folds of the per-fold difference; the counts give the folds and the eyes on which the scaled arm is higher. Standard errors are given in Appendix C.4. All T-FEN points use the energy-conserving form of the transport operator (§5.8.2). †Diverged in training on folds 1 and 4; the values cover folds 0, 2 and 3 (51 eyes).

Arm	Setting	Params	Ratio	Mean	Difference	Folds higher	Eyes higher
Dilated-stencil	width 32	18 219	0.27	0.520	−0.005	1/5	30/75
GNN	width 64	67 147	1	0.526	—	—	—
	width 128	257 163	3.83	0.518	−0.008	2/5	22/75
	4 rounds	119 499	1.78	0.517	−0.009	0/5	20/75
U-Net	width 2	13 637	0.16	0.488	−0.016	3/5	40/75
	width 5	83 081	1	0.505	—	—	—
	width 9	267 149	3.22	0.508	+0.004	1/5	38/75
FNO + 3×3	width 2	13 189	0.16	0.479	−0.019	2/5	30/75
	width 5	79 960	1	0.498	—	—	—
	width 9	257 804	3.22	0.481	−0.018	1/5	26/75
T-FEN	width 48	20 455	0.30	0.543	+0.001	3/5	37/75
	width 96	68 503	1	0.543	—	—	—
	width 192†	247 543	3.61	0.552	+0.001	2/3	25/51
	MLP depth 8	142 999	2.09	0.257	−0.285	0/5	2/75

both forms. They therefore come from training the larger network rather than from the form of the operator; their cause was not tested.

Of the settings tested in this section, only the objective changes the result: the loss must emphasise the mask, and the soft-Dice term raises the score by 0.050. The time handling and the size of the operators can stay as they are.

5.7 Additional Experiment: Adaptive Meshes

This section reports one additional experiment that was carried out once and not developed further. It is not an arm of the comparison. It extends one operator, the k-NN graph network, by a second branch on an adaptive mesh, following the moving-mesh solver of Hu et al. (2024). The background of the method and the full design are given in Appendix E. This section explains the idea, what was tested and what it gave.

Classical solvers often use adaptive meshes. A uniform grid spends the same resolution everywhere: smooth regions are resolved more finely than needed, while sharp features such as shock or wave fronts are resolved too coarsely. Moving-mesh methods keep the number of nodes fixed and move the nodes towards the regions where the solution changes sharply (W. Huang and Russell, 2011). The density of nodes is set by a monitor function, which is large where

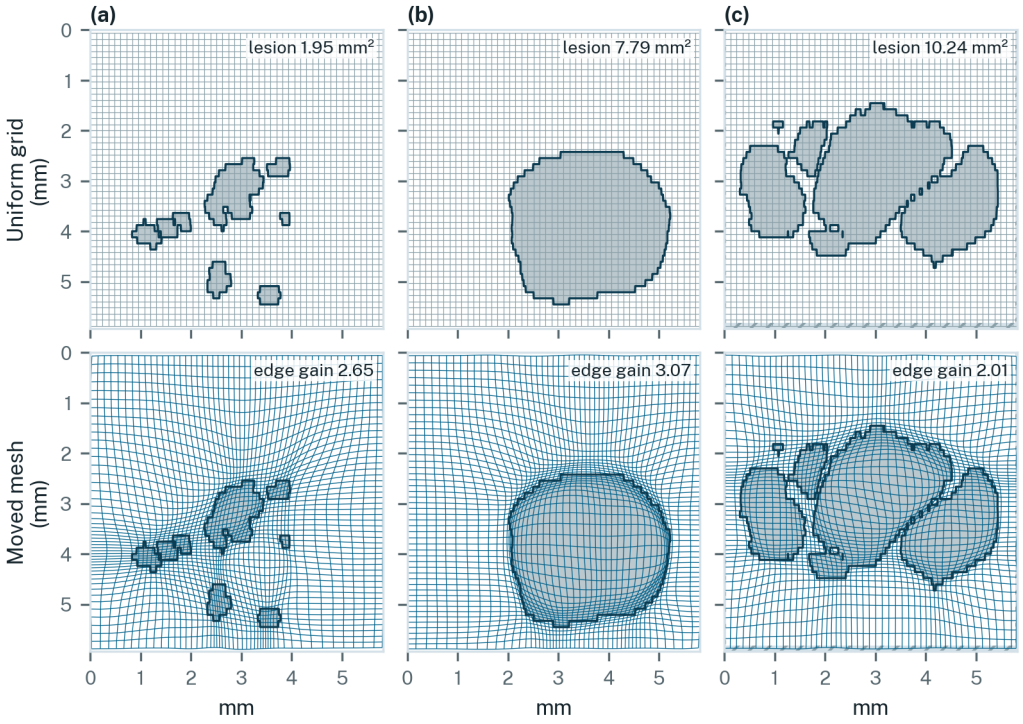


Figure 5.7: Meshes of the selected mesh mover (Appendix E.2.1; seed 42, epoch 500) for the baseline masks of three validation eyes of fold 2, drawn over the GA lesion at the physical aspect ratio of the 5.94×5.82 mm window. Top row: the uniform 49×1024 grid; bottom row: the moved mesh for the same mask. All 49 rows and every 21st column, plus the last, are drawn, so every drawn node is a grid node and a drawn cell measures about 0.12×0.12 mm. The eyes are the lower quartile, the median and the upper quartile of baseline lesion area among the 16 validation eyes of fold 2: (a) 1.95 mm^2, (b) 7.79 mm^2, (c) 10.24 mm^2. Their edge gain (Appendix E.3) is 2.65, 3.07 and 2.01 (1 for the uniform grid), and no cell of the three meshes is folded. The meshes are in-sample, because the mesh mover was trained on 478 masks of the cohort, these three included. The hatched strip at the bottom of (c) is a zero-padded row.

the gradient of the solution is large, and the nodes are placed so that every cell carries the same share of the monitor. For GA, the sharp feature is the lesion border, where the mask jumps from 0 to 1 and where the lesion grows.

Hu et al. (2024) learn this node placement with a network. Their mesh mover is trained without any ground-truth meshes, in a so-called data-free manner, and is therefore called the data-free mesh mover (DMM). Its loss measures only how evenly a mesh distributes the monitor, so no example meshes from a classical solver are needed. For GA, the DMM reads the binary lesion mask and moves the nodes of the 49×1024 grid towards the border of the current lesion. It is trained in advance on the masks of the cohort and kept frozen while the forecasting model is trained. At every step of a rollout it recomputes the mesh from the current mask, which is the model's own predicted mask after the first step.

The moved mesh is used by a dual-branch graph architecture. It extends the

Table 5.7: Change-region Dice at the one-year anchor, mean $\pm$ sd over the five folds, seed 42, for an earlier version of the k-NN graph network than the one in Table 5.1. Parameters are the trainable parameters; the frozen mesh mover is not counted. Cost is the minimum epoch time over the folds and supports order-of-magnitude statements only.

Model	Dice	Parameters	Epoch time
Single branch	0.4515 ± 0.0429	75 339	75 s
Dual branch (moving mesh)	0.4501 ± 0.0326	175 663	759 s
Bypass control (uniform grid)	0.4503 ± 0.0324	175 663	718 s

graph network of §4.3.2 with a second, adaptive branch. A graph network does not need its nodes on a fixed lattice: messages are exchanged along the edges of whatever graph connects the nodes. This is why graph networks are commonly used to learn solutions on adapted meshes (Hu et al., 2024; Pfaff et al., 2021), and the adaptive branch builds its own neighbourhood graph on the moved node positions. The two branches nevertheless work on different points. The state therefore has to be interpolated from the uniform grid onto the moved nodes before the adaptive branch can process it. The result of the adaptive branch must then be interpolated back onto the uniform grid before the two branches can be combined. Both interpolations are learned networks (ItpNet).

The combined update adds the interpolated result of the adaptive branch, scaled by Δt_k, to the update of Equation (4.1):

$$\hat{u}_{k+1} \;=\; u_k + \Delta t_k\, f_\theta(u_k, \Delta t_k, \mathbf{z}) \;+\; \alpha \cdot \tilde{I}\big[\Delta t_k\, g_\theta(u_k)\big], \tag{5.2}$$

where f_θ is the graph network on the uniform grid, g_θ the graph network on the moved mesh and $\tilde{I}$ the interpolation back to the grid. The scalar α is a gate on the adaptive branch. It is initialised to zero, so at the start the adaptive branch is switched off, and the model is identical to the single-branch graph network. The gate is learnable: if the adaptive branch improves the forecast, training should move α away from zero and let its contribution in. How far the gate opens is therefore a direct measure of whether the adaptive branch is used.

One control separates the effect of the mesh from the effect of the extra parameters. The bypass control runs the same second branch and the same interpolations, but on the uniform grid instead of the moved mesh. It has exactly the same parameters as the dual-branch model and differs from it only in the position of the nodes.

Figure 5.7 shows the moved mesh for three eyes. The nodes move from the surrounding retina towards the lesion, most strongly at its border, and no cell folds on any scored mask. The share of nodes in a narrow band around the border is about 2.6 times that of the uniform grid (Appendix E.3). The mesh mover thus places the nodes as intended.

The three models were run with an earlier version of the k-NN graph network than the one in Table 5.1, before the dilated stencil was introduced, and are compared only with each other (Table 5.7). The dual-branch model gives no measurable gain. It is lower than the bypass control by 0.0001 on average and lower than the single branch by 0.0013, both far within the noise level of §5.1.5.

The gate stays shut in both two-branch models: it peaks at 0.06 to 0.09 with the moved mesh and at 0.03 to 0.10 in the bypass control, and ends near zero on every fold. Training the dual-branch model takes about ten times as long per epoch as the single branch.

The adaptive mesh did not improve the forecast, and the adaptive branch was not used by the trained models. The experiment was not developed further; the reason is discussed in §6.1.2. The result concerns this design on this task and is not a verdict on adaptive meshes in general.

5.8 Validity Diagnostics

This section examines two properties of the trained operators that the accuracy numbers do not show: whether an operator can be read as a continuous-time model, and whether the T-FEN stays stable when it predicts far beyond the training horizon. Two tests were designed for this purpose. The step-size test re-evaluates trained models with smaller integration steps (§5.8.1), and a stability criterion checks the free-running rollout (§5.8.2). Their results decide which statements about the operators can be made.

5.8.1 The Step-Size Test

The update of the framework, $u_t + \Delta t \cdot f_\theta$, has the form of one explicit Euler step (§4.2.1). The Runge–Kutta wrapper and the Finite Element Network integrate f_θ over several sub-steps instead (§4.3.7, §4.3.6). As discussed in §4.3.7, both suggest reading f_θ as the right-hand side of an ordinary differential equation $du/dt = g(u)$, in the sense of a Neural ODE (Chen et al., 2018): a vector field that gives the change of the state at every moment. Under that reading, a model would define the state at any time, its predictions for different intervals would be consistent with one another, and f_θ could be interpreted as a rate of growth. A network trained with one fixed discretisation need not have this property. It can fit the observed data while its output depends on the discretisation it was trained with (Krishnapriyan et al., 2023; Ott et al., 2021). The step-size test determines which of the trained models have the property, and therefore for which models continuous-time statements may be made.

The test rests on a basic property of numerical integration. If a model defines an ordinary differential equation $du/dt = g(u)$, every convergent integration scheme applied to g approximates the same solution, with an error that shrinks as the step h shrinks, in proportion to h^p for a scheme of order p (Hairer et al., 1993). The same trained weights, integrated with a smaller step or a scheme of higher order, must therefore approach one prediction. Two criteria make this operational:

- **C1, convergence:** at the two finest steps the prediction has settled; from 22.8 to 11.4 days it changes by less than the run-to-run noise with both schemes, and the two schemes agree to within it at 11.4 days;
- **C2, agreement:** all settings that are at least as fine as the one used in training in both step and order agree to within the run-to-run noise, the stan-

dard deviation of 0.0127 between repeated runs of one configuration (Appendix C.3).

A model is read as a continuous-time model only if it satisfies both.

The test requires no retraining. A trained checkpoint is rebuilt from its own configuration and evaluated with the unmodified evaluation code; only the integration scheme and the step change. The integration wrapper has no parameters (§4.3.7), so every setting uses the same weights. As a control, each model was evaluated with the setting it was trained with, which reproduces its recorded change-region Dice exactly. The step h is fixed in time, at 1, 1/2, 1/4, 1/8, 1/16 and 1/32 year, that is 365 to 11.4 days. Each visit interval is split into the smallest number of equal sub-steps that are no longer than h, so no visit is skipped. Both explicit Euler and classical RK4 are evaluated at every step. Because the median interval between visits is 180 days, steps of one and of half a year leave almost every interval in one piece. Three models were tested, each on all five folds with one seed:

- the canonical dilated-stencil network, trained with one Euler step per interval and with Δt as an input of the operator;
- the same network trained inside the autonomous Runge–Kutta wrapper, with one RK4 step per interval and without a Δt input;
- the T-FEN in its energy-conserving form, trained with RK4 in steps of 45 days and without a Δt input.

Table 5.8 summarises the results per model and fold, Table C.6 lists every setting, and Figure 5.9 shows the change against the step for the two graph networks. The canonical network fails C2 on all five folds. Over the settings at least as fine as its training setting, its Dice spans 0.026 to 0.201, two to sixteen times the run-to-run noise. At the finest step, Euler changes the Dice by between -0.187 and $+0.043$ against the trained setting. On folds 0, 1 and 3 the two schemes approach a common value at the finest steps (C1), but on folds 0 and 1 that value lies far from the prediction of the trained setting; on folds 2 and 4 the prediction still changes by 0.03 to 0.05 per halving of the step (C1 fails). The operator receives Δt as an input, so a smaller step also changes the input on which it is conditioned, and steps of 45 days or less are shorter than any interval between two visits, so the operator never received such a Δt during training. The canonical network is a one-step map conditioned on the length of the step, a jump in the sense of §4.3.7.

The network trained inside the Runge–Kutta wrapper converges on all five folds (C1). From a step of a quarter year on, its RK4 predictions change by less than 0.004, and its Euler predictions approach the same value, to within 0.003 at the finest step. It therefore defines a single vector field. C2 holds on folds 2 and 4. On folds 0, 1 and 3, the converged value differs from the value of the trained setting, one RK4 step per interval, by 0.016 to 0.018, more than the run-to-run noise of 0.0127. On these folds, that step is too coarse to resolve the vector field to within the run-to-run noise. The difference has no consistent sign: finer steps raise the Dice by 0.018 on folds 0 and 1, lower it by 0.016 on fold 3 and change it by at most 0.003 on folds 2 and 4, 0.004 on average. The network fails C2 because one RK4 step per interval integrates its vector field less accurately than finer steps do, not because the finer steps forecast better.

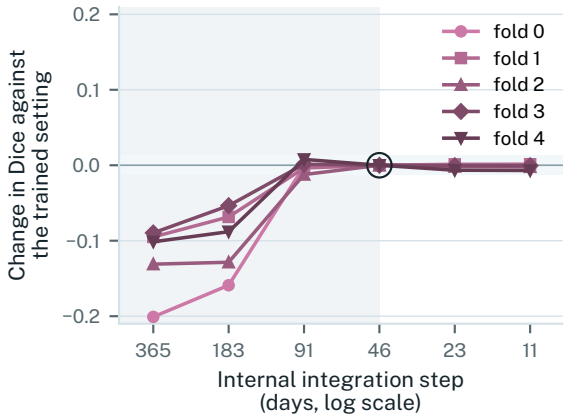


Figure 5.8: The step-size test for the T-FEN in its energy-conserving form. The network was trained with RK4 in steps of 45 days (ring) and is re-evaluated, without retraining, with RK4 at a fixed internal step of 365 to 11.4 days (logarithmic axis, finer steps to the right); every point forecasts the same visits. The vertical axis gives the change of the change-region Dice at the one-year anchor against the trained setting, one line per fold (all five folds, seed 42); the band marks ± 0.0127, the run-to-run standard deviation. Criterion C2 covers the training step and every finer step; there the forecast changes by at most 0.007. Steps coarser than training (shaded) lie outside the test: at 91 days the forecast still agrees with the trained one, at 183 and 365 days it drops by 0.05 to 0.20.

The T-FEN passes both criteria on all five folds. Between its training step of 45 days and the finest step of 11.4 days, its RK4 predictions change by at most 0.007. Steps coarser than its training step are not covered by C2. At a quarter year the predictions still agree with the trained one, to within 0.012; at half a year and one year they drop by 0.05 to 0.20 (Figure 5.8). Explicit Euler amplifies every mode of the transport operator (§4.3.6). At small steps, however, this growth stays small over one year, and the Euler predictions approach the RK4 value with an error that roughly halves with each halving of the step, as expected of a first-order scheme.

The canonical network, and with it every headline result, may therefore not be described as a continuous-time model: no statement about an ordinary differential equation, a learned rate of growth or learned dynamics is made for it. Within the scope of one seed and of the change-region Dice at the one-year anchor, the energy-conserving T-FEN behaves as a continuous-time model; its 45-day training step lies slightly above the RK4 stability limit on the mask channel (§5.8.2), so this reading does not extend to rollouts over several years. The T-FEN is also the only model of the three trained with sub-steps, so the difference between the models may come from the training step rather than from the operator. The network trained inside the Runge–Kutta wrapper defines a vector field as well, but one RK4 step per interval is too coarse to resolve it on three of the five folds, where finer steps change its Dice by 0.016 to 0.018 in either direction. A continuous-time reading does not come with higher accuracy: the Runge–Kutta network scores no higher than the canonical network (§5.6.1), and the advantage of the T-FEN over the free-form FEN comes from

its transport term (§5.4.4).

Table 5.8: The step-size test. Each trained model is re-evaluated, without re-training, with a fixed step of $h = 1$ to $1/32$ year (365 to 11.4 days) and with explicit Euler or classical RK4; all five folds, seed 42. The table gives the change of the change-region Dice at the one-year anchor at the finest step against the setting the model was trained with, the span over all settings at least as fine as the training setting in both step and order, and the two criteria: convergence at the finest steps (C1) and agreement of those settings within the run-to-run noise of 0.0127 (C2). Every setting is listed in Table C.6.

	Change at 11.4 days				
Fold	Euler	RK4	Span	C1	C2
Canonical dilated-stencil network (trained: Euler, one step per interval)					
0	−0.037	−0.028	0.054	passes	fails
1	+0.043	+0.048	0.077	passes	fails
2	−0.187	−0.190	0.201	fails	fails
3	−0.005	−0.008	0.026	passes	fails
4	−0.085	−0.082	0.130	fails	fails
Network trained in the RK4 wrapper (trained: RK4, one step per interval)					
0	+0.018	+0.018	0.020	passes	fails
1	+0.020	+0.018	0.018	passes	fails
2	+0.001	+0.000	0.000	passes	passes
3	−0.016	−0.016	0.017	passes	fails
4	−0.000	+0.003	0.006	passes	passes
T-FEN, energy-conserving form (trained: RK4, 45-day steps)					
0	−0.003	+0.000	0.000	passes	passes
1	−0.004	+0.001	0.001	passes	passes
2	−0.004	−0.002	0.002	passes	passes
3	−0.005	−0.001	0.001	passes	passes
4	−0.010	−0.007	0.007	passes	passes

5.8.2 Numerical Stability of the T-FEN

Stability is judged on the free-running rollout of the evaluation: every model predicts the next visit from its own previous prediction, over each eye's whole follow-up of up to about six years, without any ground truth fed back. A run is called stable if, in this rollout, the root mean squared error of the mask channel and of the layer channels, both in normalised units, stays at or below 5 on at least 19 of its 20 late epochs (§5.1.4). The normalised values of a correct prediction lie within a few units of zero, so an error above 5 means that the rollout has left the range of plausible states; in an unstable run it keeps growing, up to values of several thousand. The criterion was fixed before the energy-conserving form was trained.

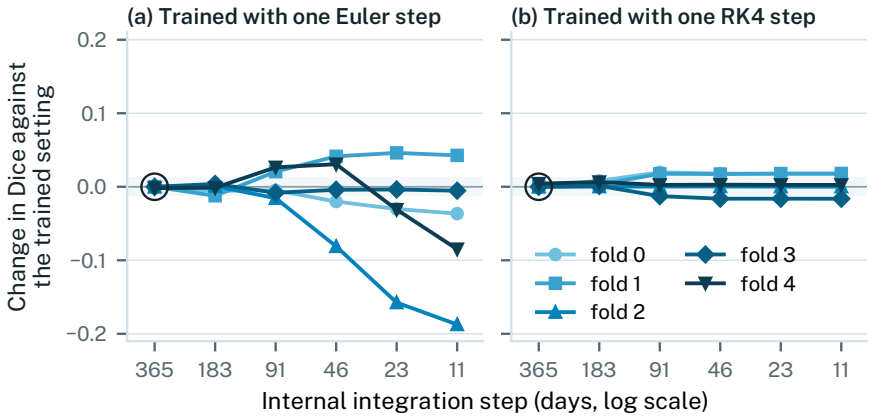


Figure 5.9: The step-size test for the dilated-stencil graph network, trained in two ways (Table 5.8). Each trained network is re-evaluated, without retraining, with the scheme it was trained with at a fixed internal step of 365 to 11.4 days (logarithmic axis, finer steps to the right). Each visit interval is split into equal sub-steps no longer than the step, so every point forecasts the same visits. The vertical axis gives the change of the change-region Dice at the one-year anchor against the trained setting (ring), one line per fold (all five folds, seed 42); the shaded band marks ± 0.0127, the run-to-run standard deviation. *(a)* The canonical network, trained with one Euler step per visit interval and with Δt as an input. *(b)* The same network trained with one RK4 step per interval inside the autonomous wrapper. Its forecast settles 0.016 to 0.018 away from the trained one on folds 0, 1 and 3, just outside the band, and within 0.003 of it on folds 2 and 4.

The T-FEN was first trained with the Galerkin form of its transport operator (§4.3.6). Four of its five runs are unstable in the sense defined above (Table 5.9), and so are most of the runs repeated with seed 7, whereas the free-form FEN, which lacks only the transport term, is stable on every fold. Tracing the rollouts step by step showed where the divergence comes from. It starts at the edge of the crop, in a layer channel, driven by the transport term, after about 1.6 to 2 years. In 14 of the 20 eyes of fold 0, the first layer error above 5 lies within 40 µm of the left or right edge, a band that holds 1.4 % of the grid. A smaller step left the divergence in place, and the velocities at the onset were within the stability limit of RK4. The divergence therefore comes from the defect of the Galerkin operator at the domain edge (§4.3.6, Figure 4.3), not from the step size. Because the divergence typically starts after the one-year anchor and away from the lesion, it changed the change-region Dice of the Galerkin T-FEN at one year little.

To remove the defect at the domain edge, the T-FEN was retrained with the energy-conserving form of the transport operator (Equation 4.7), identical in every other respect, at seed 42 on all five folds. It passes the stability criterion on every fold, with a largest late-epoch error of 2.32 on the mask channel and 1.13 on the layer channels (Table 5.9, Figure 5.10). Repeated with seed 7, it passes on every fold as well. The energy-conserving form is also at least as accurate as the Galerkin form: 0.5428 ± 0.0401 against 0.5337 ± 0.0453, higher on all

five folds, by $+0.009$, which is under the noise level. The difference lies mainly on the lesions that touch the crop border, which is consistent with the location of the defect: their change-region Dice rises from 0.516 to 0.544, while interior lesions are unchanged (0.535 and 0.536). The growth-region Dice beyond three years rises from 0.537 to 0.611, still below the dilated stencil (0.655, §5.3). The energy-conserving T-FEN is therefore the one reported in §5.2 and §5.4.

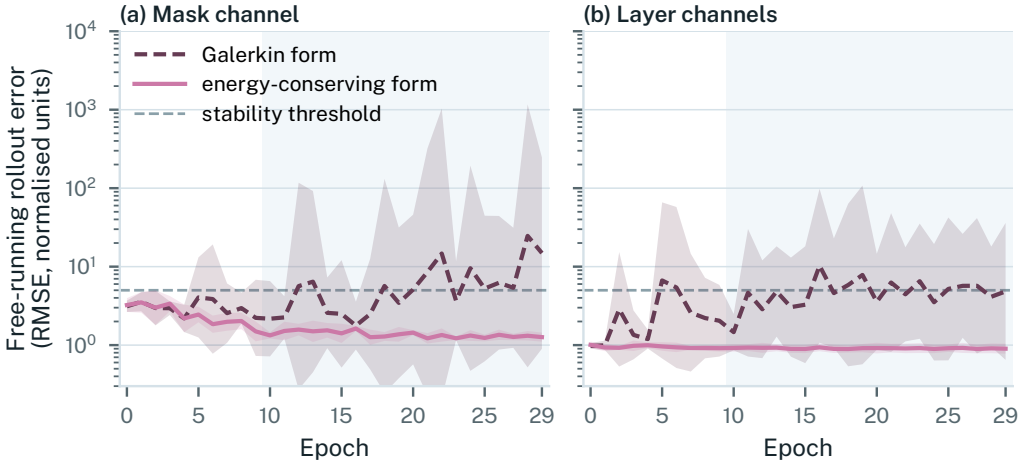


Figure 5.10: Free-running rollout error of the T-FEN per training epoch, for the Galerkin form (dashed) and the energy-conserving form (solid) of its transport operator, five folds, seed 42. The line is the geometric mean over the folds, the shading one standard deviation of the logarithm of the error. *(a)* Mask channel, *(b)* layer channels; root mean squared error in normalised units, on a logarithmic scale. The grey dashed line marks the stability threshold of 5 and the shaded background the late epochs 10 to 29 over which the criterion is evaluated: a run is stable if both errors stay at or below 5 on at least 19 of the 20 late epochs. The criterion is applied to each fold separately (Table 5.9).

The energy-conserving form leaves a small residual with a different cause. Read back over every sub-step of the free-running rollout on fold 0, the Courant number (§4.3.6) of the layer channels stays at or below 1.36, inside the RK4 limit of about 2.4, but that of the mask channel lies slightly above it at the 45-day training step (median 2.64, maximum 2.85). In 9 of the 20 validation eyes a few mask values on the top or bottom row of the grid eventually leave the plausible range: at most 26 mask values per eye, each at most 36 in normalised units. They start around day 1 260 and grow by a factor of about 1.04 per sub-step. This growth matches the step-size limit of RK4. It is a minor effect compared with the defect of the Galerkin form, which produced growth by a factor of about 1.3 per sub-step from around day 585. The run still passes the stability criterion. On fold 0, a re-evaluation with RK4 in 30-day steps, at which the Courant number of the mask channel falls to about 1.8, removes the residual and leaves the change-region Dice unchanged. Only the first, Galerkin form of the T-FEN is therefore unstable; the energy-conserving form reported in this thesis is stable on every fold at both seeds.

Table 5.9: Stability of the T-FEN in the Galerkin and the energy-conserving form of its transport operator, seed 42. For each fold: the number of late epochs (10 to 29) on which the root mean squared errors of the mask and of the layer channels in the free-running rollout stay at or below 5, and the largest late-epoch value of each, in normalised units. A run is stable if both counts are at least 19.

	Galerkin form		Energy-conserving form	
Fold	Epochs $\leqslant 5$ mask / layers	Largest error mask / layers	Epochs $\leqslant 5$ mask / layers	Largest error mask / layers
0	1 / 2	24 418 / 456	20 / 20	2.00 / 1.13
1	12 / 19	196 / 14.9	20 / 20	2.32 / 0.89
2	18 / 19	6.97 / 33.5	20 / 20	1.82 / 1.08
3	20 / 20	1.49 / 3.33	20 / 20	1.73 / 0.82
4	19 / 6	11.9 / 174	20 / 20	2.00 / 0.99

5.9 Comparison with Prior Work

No study reports results for the exact task and data of this thesis, but several report comparable measures. The closest is the study of Mai et al. (2024), who predict GA growth from a single baseline OCT volume, with a Neural ODE integrated by RK4, on the MUW cohort from which the data of this thesis are drawn. Three further studies report values on related tasks (§2.4.2). This section compares the canonical model with them on the two kinds of measure they use: the growth-region Dice by time since baseline, which scores where the lesion grows, and eye-level measures of growth speed, which the Dice does not capture.

The values give an orientation, not a head-to-head result. The study of Mai et al. (2024) differs from this thesis in several respects. It evaluates 184 eyes of 100 patients with a mean follow-up of 32 months, about 2.7 years; this thesis uses 75 eyes of 51 patients with a mean follow-up of 3.3 years. Their model starts from the raw OCT volume, the models here from segmented masks and layer depths. Their paper reports no cropping, whereas the areas here are measured inside the 49×1024 crop, which censors growth at its border (§3.3). The two studies also evaluate different sets of follow-up visits. The measures below follow their definitions as far as these are specified. One difference remains: their paper computes the growth rate from the baseline and the respective follow-up area, whereas here one rate per eye is computed from the baseline and the last visit. The other studies differ further, as stated with each comparison.

Mai et al. (2024) report the growth-region Dice by time since baseline. The same quantity is computed here for the dilated-stencil network, the U-Net and the T-FEN, averaged over the follow-up visits of the rollouts and the late epochs (Table 5.10). All three lie above the values of Mai et al. (2024) in every bin, by 0.18 to 0.29.

Gigon et al. (2021) predicted the progression of RORA from OCT en-face maps of layer thicknesses and reflectivity, an input close to the one used here, and report the growth-region Dice of RORA by year since baseline; their bins match

Table 5.10: Growth-region Dice (newly atrophic region, follow-up minus baseline) by time since baseline, against Table 2 of Mai et al. (2024) and Table 2 of Gigon et al. (2021), whose lesion is RORA and whose last two bins are 3 to 4 and more than 4 years. Values of this thesis: mean over the follow-up visits in each bin and over the late epochs, pooled over the five folds, seed 42. Number of visits per bin in brackets. The values are a reference, not a head-to-head comparison (see text).

Years	Mai et al. (2024)	Gigon et al. (2021)	Dilated-stencil GNN	U-Net	T-FEN
0–1	0.25 (296)	0.46	0.503 (149)	0.458	0.519
1–2	0.38 (260)	0.57	0.600 (149)	0.563	0.602
2–3	0.38 (148)	0.59	0.634 (109)	0.581	0.633
> 3	0.37 (79)	0.65 / 0.72	0.655 (71)	0.555	0.611

those used here up to three years and then split into 3 to 4 and more than 4 years. Their values, 0.46 in the first year rising to 0.72 after four years, lie close to those of the canonical network, 0.50 rising to 0.66.

Salvi et al. (2025) report a Dice of 0.70 on their test set for the one-year region of growth, predicted from fundus autofluorescence images. Their setting differs in two ways that favour a higher value of this metric. Their best model receives two visits, at baseline and after six months, and the lesions of their trial cohort are larger, 2.54 to 17.78 mm^2 at inclusion with a median of 9.03 mm^2, against a median of 6.7 mm^2 here (§3.1), and in this cohort the growth-region Dice rises with the baseline lesion area (§5.5). Their region of growth is also measured from the six-month lesion over the following year rather than from the baseline.

The growth rate of an eye is the change of the square root of its lesion area per year, $(\sqrt{A_{\text{last}}} - \sqrt{A_{\text{base}}})/T$, where T is the time from the baseline visit to the eye's last visit. Taking the square root of the area removes the dependence of the growth rate on the baseline lesion size (Feuer et al., 2013). The rate is computed once from the ground truth and once from the model's rollout, over the eye's whole follow-up, not at the one-year anchor; unlike Mai et al. (2024), one rate per eye is used (see above). Two measures compare the two rates over the cohort. The Pearson correlation r measures how closely the predicted rates follow a linear relation with the true rates across eyes: $r = 1$ is a perfect linear relation and $r = 0$ no linear relation. It does not require the two to agree in scale. The fast-progressor area under the receiver operating characteristic curve (AUC) measures how well the predicted rate separates the fastest-growing eyes, the top 10, 15 or 20 % of true growth rates, from the rest; 0.5 is chance and 1 is perfect separation.

Both measures are computed for the canonical model. Each eye's predicted rate is averaged over the late epochs, the 75 validation eyes of the five folds are pooled, and 95 % intervals come from a bootstrap over eyes. Table 5.11 lists the result.

The correlation of the canonical model is 0.40, and its 95 % bootstrap interval, 0.20 to 0.57, excludes zero (Figure 5.11). The baseline lesion size alone is not correlated with the growth rate in this cohort ($r = -0.10$), as intended by the square-root transform. The model therefore predicts part of the differences

Table 5.11: Eye-level growth-speed measures of the canonical model against the values reported by Mai et al. (2024). Growth rates are computed over each eye's whole follow-up with the square-root transform; the canonical model is pooled over the 75 validation eyes of the five folds, with 95 % bootstrap intervals and, for the AUC, the number of eyes above the cut-off. The values are a reference, not a head-to-head comparison (see text).

Measure	Mai et al. (2024)	This thesis
Pearson r of growth rates	0.61	0.40 [0.20, 0.57]
AUC, top 10 %	0.81	0.74 [0.53, 0.91] (8 eyes)
AUC, top 15 %	0.79	0.70 [0.53, 0.85] (11 eyes)
AUC, top 20 %	0.77	0.70 [0.53, 0.84] (15 eyes)

in growth speed between eyes. Its correlation is lower than that of Mai et al. (2024), whose value of 0.61 lies outside the bootstrap interval. Its AUCs are also lower, but each value of Mai et al. (2024) lies inside the corresponding interval. The AUCs rest on 8 to 15 positive eyes, and their intervals reach down to values close to chance. The fastest eyes are under-predicted: of the 15 eyes with the highest true rates, 14 receive a lower predicted rate, with a median ratio of predicted to true rate of 0.50. The other operators of the survey give similar values (r between 0.30 and 0.47, fast-progressor AUCs between 0.66 and 0.78; Table C.2), so the weaker agreement on growth speed is shared by every operator in the framework.

Two further studies report how well predicted and true growth rates agree. Gigon et al. (2021) report a correlation of 0.52 over the whole follow-up. The four OCT-based models of Yoshida et al. (2025) reach a squared correlation of 0.33 to 0.35 on 442 holdout eyes, an r of about 0.57 to 0.59. Their models were trained on 1219 eyes to predict the growth rate itself, defined as the annual slope of the lesion area in square millimetres, fitted over all visits and measured on fundus autofluorescence images with a larger field of view than the OCT scan. The canonical model, with $r = 0.40$, lies below all three models on growth speed in its point estimate; the values of Gigon et al. (2021) and Yoshida et al. (2025) lie inside or at the upper edge of its interval of 0.20 to 0.57. All three differ from it in the setup as described above.

The two kinds of measure point in different directions. On where the lesion grows, the canonical model lies within the range of prior work: above Mai et al. (2024) on the same cohort, close to Gigon et al. (2021), whose input is similar, and below the FAF model of Salvi et al. (2025), whose setting differs in ways that favour this metric. The higher values against Mai et al. (2024) are not evidence that one approach predicts better. The models of this thesis receive the segmented baseline lesion as input, so its outline is given exactly, whereas the model of Mai et al. (2024) first has to find the lesion in the raw OCT volume. Together with the differences listed above, this means that the comparison mainly shows how much of the spatial forecast the segmented mask already carries. On how fast the lesion grows, the canonical model lies below every comparable model in its point estimate, with $r = 0.40$ against 0.52 to 0.61, and outside its interval only against Mai et al. (2024). Possible reasons are dis-

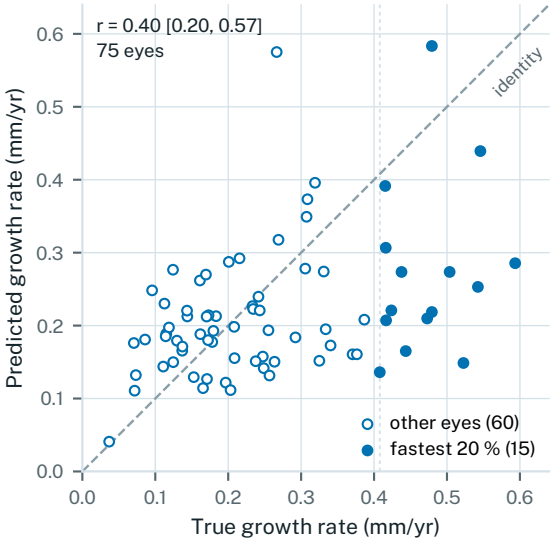


Figure 5.11: Predicted against true growth rate of the canonical dilated-stencil graph network for the 75 validation eyes of the five folds, seed 42. The growth rate is the change of the square root of the lesion area per year over the eye's whole follow-up; the predicted rate is averaged over the late epochs. Filled markers: the 15 eyes with the highest true rates (top 20 %, right of the dotted line), the positives of the fast-progressor AUC. Dashed: identity. The Pearson correlation is given with its 95 % bootstrap interval (2000 resamples over eyes).

cussed in §6.2.

5.10 Computational Cost

This section compares the training cost of the arms. Cost is reported as the minimum epoch time over the five folds, not as the mean. The epoch times of a run carry a nearly constant additive overhead of 17 to 64 s, which would penalise cheap operators far more than expensive ones. The epoch times also differ between folds because the work per epoch differs: an epoch is twenty passes over the fold's training eyes in batches of four, which gives 280, 320, 300, 300 and 320 optimiser steps for folds 0 to 4. Normalised by the number of steps, the minimum epoch times of the five folds agree to about 3 %. All runs used one NVIDIA RTX A6000 GPU and 30 epochs. Inference time per rollout step and memory footprint were not measured. The figures support order-of-magnitude statements only.

Table 5.12 lists the minimum epoch time of every arm. Within this precision, the U-Net is the cheapest spatial operator, at about $0.23\times$ the 91 s of the canonical dilated-stencil graph network. The most expensive arms are those that integrate their operator over several sub-steps: the free-form FEN and the T-FEN, both with RK4 in 45-day steps, and the dilated-stencil network inside the Runge–Kutta wrapper.

How much such an integration adds depends on how much of an epoch the operator itself takes. The Runge–Kutta wrapper evaluates the operator four times

Table 5.12: Training cost of the arms of Table 5.1: the minimum epoch time over the five folds in seconds, on one NVIDIA RTX A6000, and its ratio to the canonical dilated-stencil graph network. The figures support order-of-magnitude statements only.

Arm	s/epoch	Ratio
Per-pixel floor	16	0.18
U-Net, width 5	21	0.23
FNO	23	0.25
FNO with 3×3 local path	29	0.32
One-hop floor	37	0.41
FNO, RK4	45	0.49
Graph network, k-NN, $k = 12$	76	0.84
Graph network, dilated stencil (canonical)	91	1.00
Graph network, k-NN, $k = 20$	108	1.19
FEN, free-form only (RK4, 45-day steps)	180	1.98
T-FEN (RK4, 45-day steps)	367	4.03
Graph network, dilated stencil, RK4	439	4.82

per step. Over the dilated-stencil network, whose forward pass dominates the epoch, the wrapper raises the epoch time from 91 to 439 s, about $4.8\times$. Over the FNO, whose forward pass is cheap, the epoch time rises only from 23 to 45 s, about $2\times$, because the fixed part of the epoch weighs more. The T-FEN costs about $4\times$ the dilated-stencil network.

Set against the results of this chapter, most of the extra cost brings no measurable gain. The Runge–Kutta wrappers are nulls against their single-step counterparts (§5.6.1), wider operators did not improve on their original sizes (§5.6.3). The T-FEN is the only expensive arm with an established effect, its transport term. Its lead over the dilated-stencil network at the one-year anchor is not established (§5.2), however, and that network reaches about the same level at about a quarter of the cost.

5.11 Summary

This chapter answered the second part of the research question. Inside the fixed framework, operators of three different classes reach similar scores at the one-year anchor, and every point of change-region Dice comes from spatial context (§5.2). Placing the neighbours of the graph network at lesion scale and adding an explicit transport term each raise the score by 0.06 to 0.07 against their own controls, and the two arms built in these ways are the two highest (§5.4); this is consistent with an operator needing one update that reaches about as far as the lesion front advances between two visits. Of the framework's own settings, the objective matters: it must emphasise the mask. The time handling and the integration scheme do not change the result, and larger operators do not improve it (§5.6). A single additional experiment with an adaptive mesh, which gathers the nodes of a second graph branch at the lesion border, gave no mea-

surable gain at about ten times the training cost (§5.7). The energy-conserving T-FEN behaves as a continuous-time model up to the one-year anchor, the canonical network does not, and this reading does not come with higher accuracy (§5.8). Against prior work, the canonical model forecasts where the lesion grows within the range of other models, but agrees less well on how fast (§5.9). Chapter 6 interprets these results.

Chapter 6

Discussion

This thesis asked what is needed to forecast the progression of GA from longitudinal OCT. The question had two parts: what kind of framework the forecast needs, and what kind of operator that framework needs (§1.2). The task was to predict, from the lesion mask and ten retinal layer boundaries of one visit, the lesion at the later visits of the same eye. The intervals between visits range from 90 to 720 days and are most often 180 days. The data were 75 eyes of 51 patients from the Medical University of Vienna, evaluated in five-fold cross-validation. Every model was scored by the change-region Dice at one year: the overlap between the pixels the model predicts to change and the pixels that do change. Predicting no change, persistence, scores 0 by construction. Inside the shared framework, operators from unrelated classes reached similar scores. The objective decided whether any change was learned, and among operators with spatial context the reach of one update relative to the advance of the front separated them most. Section 6.1 interprets these results for each part of the question, §6.2 discusses their clinical implications, §6.3 the limitations and §6.4 future work.

6.1 Interpretation of Results

6.1.1 The Framework

The framework is everything that stays fixed while the operator changes (§4.2). It predicts the next visit from the current one by a residual update,

$$\hat{u}_{k+1} = u_k + \Delta t_k \cdot f_\theta(u_k, \Delta t_k, \mathbf{z}),$$

whose operator f_θ starts at zero, so that every model begins at persistence. It trains with a loss that weights the mask channel five times and adds a soft-Dice term, and it unrolls its own predictions over up to 360 days during training. Inside this framework, every operator with spatial context learned to predict change: the eleven such arms reach a change-region Dice of 0.45 to 0.54 at one year (Table 5.1), while the per-pixel floor, which sees no neighbours, stays at 0. One framework thus served operators from unrelated literatures without modification. The answer to the first part of the question is therefore a framework that conditions a residual update on the interval Δt, starts every model at persistence and trains it with a loss that weights the mask channel.

Its parts do not matter equally. The objective matters most. Trained with a plain mean squared error, without the weighting of the mask channel and without

the soft-Dice term, the canonical network learned no change at all: at the last epoch, 74 of the 75 eyes are predicted exactly as at baseline (§5.6.2). With the weighting but without the soft-Dice term, it learned change, but later and to a score lower by 0.050. A penalty against lesion shrinkage added nothing. The time handling, in contrast, can be simplified without loss. Removing the multiplication by Δt left the result unchanged, because the operator learns the division by the interval internally. Replacing the single Euler step by four Runge–Kutta evaluations made no measurable difference for either a graph network or a Fourier Neural Operator (§5.6.1). The update needs the length of the interval, but not in a particular form. These tests were run on the canonical network, or on two operators for the integration scheme, so they describe the framework as it is used here, not in general.

6.1.2 The Operator

Within the framework, the three highest arms come from three unrelated classes: the T-FEN, a finite-element network with a transport term, at 0.543; the dilated-stencil graph network at 0.526; and a U-Net at 0.505 (§5.2). The T-FEN lies above the U-Net at both seeds, and its lead of 0.017 over the graph network is not established. Operators from three unrelated classes thus reach scores within 0.04 of each other. The ablations of §5.4 tested properties one at a time on otherwise unchanged operators:

- Spatial context is required: a graph network that sees only each pixel itself predicts no change.
- The reach of one update along the direction in which the lesion front moves raises the score: replacing the index-space neighbourhood of the graph network by the dilated stencil raises it by 0.064, at the same parameter count.
- An explicit transport term raises the finite-element network by 0.062 against the same network without it; a T-FEN with fewer parameters than that control scores the same.

A local path for the global FNO brought it level with the U-Net at one seed but not at the second, and its own effect is not established.

Reach and transport are the most informative pair. The GA front advances a median of 68 µm (12 columns of the grid) between two visits 180 days apart, and 131 µm (23 columns) at the 90th percentile, while one hop of the index-space graph reaches about 11 µm along a row (§4.3.3). The dilated stencil lets one hop reach about 119 µm, and the transport term moves the state along a learned velocity. These two unrelated constructions each remove the same limit, and the two arms built with them are the two highest of the survey. The limit is the one that the stability condition of Courant et al. (1928) describes for explicit schemes: the stencil of one step must cover the region on which the solution depends (§2.2.2). Section 2.5.1 stated this as an expectation for GA, and the survey is consistent with it. The agreement of the two constructions is suggestive but does not prove a mechanism. It is nevertheless a more specific statement than a ranking of architectures.

The T-FEN is on the verge of being the best operator of the survey. Its lead of 0.017 over the dilated-stencil network has almost the same size at both seeds

and passes the fold-paired test at both, each time just above the bar ($z = 2.1$ against 1.96). Of the conditions of §5.1.5, it misses a single one, and only narrowly: at the first seed its per-eye t is 2.2 against a bar of about 2.5, while at the second seed it clears that bar with 3.1. A third seed or a larger cohort would most likely decide the order. Even so, the lead is small next to the effects of the properties themselves: reach and transport each raise their operator by more than 0.06, almost four times as much. Both arms carry one of these two properties, so for the question of this thesis their order matters little.

The dilated-stencil network nevertheless remains the canonical model, for two reasons. First, it trains about four times faster (§5.10). Second, it is the reference configuration on which the tests of the objective and the time handling were run (§5.6); replacing it would require repeating these tests on the T-FEN. The capacity study and the step-size test already include the T-FEN, and the step-size test gives the two operators different verdicts (§5.8.1).

More reach does not keep helping. Reaches of ± 35 and ± 42 columns score lower than ± 21, the latter beyond the noise level. Widening the stencil across rows lowers the score most. Four message-passing rounds instead of two, which double the reach, lower the canonical network by 0.009 (§5.4.2, §5.6.3). The best tested reach, about 119 µm, lies close to the 90th percentile of the advance between visits. Reach beyond that point may hurt for two reasons. With mean aggregation, far neighbours may dilute the near ones, and the extra reach may bring in tissue unrelated to the next visit's change.

The same property explains why the experiment with adaptive meshes (§5.7) was not developed further. The dilated stencil was introduced so that one update reaches further along the direction in which the front moves. The adaptive mesh does the opposite. The mesh mover gathers nodes at the current lesion border so that the second branch sees the border in finer detail. By construction, one message-passing step on the moved mesh then reaches less far than on the uniform grid. Yet the front advances a median of 12 and, at the 90th percentile, 23 columns between visits (§4.3.3), and new atrophy appears outside the current border. A second branch adds something only if it sees what the first one does not. On the moved mesh it sees more detail over a shorter distance; in the bypass control it sees the same as the uniform branch. In neither case did its gate open. Concentrating nodes and reaching far work against each other, and with the dilated stencil the uniform branch already reaches the front on its own. The extension also more than doubles the trainable parameters, adds a frozen mesh mover and takes about ten times as long to train, for no measurable gain. Finally, a moving mesh is not available to the regular-grid operators, the U-Net and the FNO.

Global context behaves differently. The FNO sees the whole grid in every layer but lies below the U-Net at one year, beyond the noise level at one of the two seeds. Beyond three years it lies above the U-Net (§5.3), but the FNO with a local path, which has the same global context, does not, so the horizon values do not show that global context matters more for long forecasts.

The answer to the second part of the question is therefore an operator with spatial context whose single update reaches about as far as the lesion front moves between two visits, through the placement of its neighbours or through an explicit transport term. Operators of different classes that meet this condition reach similar scores.

6.1.3 Capacity

Capacity was not a lever in any of the four classes tested (§5.6.3): no width step raised a class above the noise level, and the smaller versions lost little. With 75 eyes, this is what the low-data argument of §2.5.1 leads one to expect: the assumptions built into an operator weigh more than its size. The capacity study shows that size did not help; that the built-in assumptions are the reason is an interpretation, supported by the ablation studies.

6.1.4 A Missing Counterpart to the Equation Features

A solver trained on a family of equations predicts best when it is told which member of the family it faces. In MP-PDE, a vector of equation features θ_{PDE}, the coefficients and boundary conditions of the equation being solved, is passed to every layer, and removing it worsens the predictions on unseen equations, most when all parameters vary (Brandstetter et al., 2022). GA has no governing equation, and so no such vector exists. In this thesis the slot is filled by Δt and the two patient covariates (§4.2.3). The covariates are two numbers per visit, shared by every grid position and nearly constant along a trajectory, so they can shift a prediction as a whole but cannot say anything about a particular eye's lesion. The learned stand-in that the code provides, a layer encoder that summarises the retinal structure into one vector, was tested in an earlier configuration and in one design only, without a detectable benefit.

What is lost is an eye-specific description of how this eye progresses. Two eyes with the same lesion shape but different speeds receive the same conditioning, and the operator can only infer the speed from the current state. This is a plausible contributor to the weaker prediction of growth speed discussed in §6.2. Candidates for the slot are descriptors that do vary between eyes: lesion-level quantities such as size, shape or focality, a learned summary of the observed layers, or — with trajectory memory — the eye's own past growth.

6.1.5 Predictions Between Visits

Every operator of the survey is trained to predict the state at the next visit, so its predictions between visits are not constrained by the data. Two models nevertheless define a prediction at any time. The T-FEN in its energy-conserving form passes the step-size test on all five folds and can therefore be read as a continuous-time model (§5.8.1). The network trained inside the Runge–Kutta wrapper also defines a vector field, but its training step, one RK4 step per interval, resolves that field on only two of the five folds. Neither is more accurate than the canonical network at one year, and the wrapper costs about $4.8\times$ the training time. If predictions between visits are needed clinically, the T-FEN provides them consistently with its predictions at the visits. The wrapper network does so only when it is integrated with a finer step than in training, which changes its change-region Dice at one year by up to 0.018 on single folds and by 0.004 on average. Whether these predictions are correct cannot be checked, because no ground truth exists between two observations. An earlier attempt to constrain them, a regulariser that penalised predictions between visits against the linear interpolation of the two visits, brought no benefit and was removed

before the experiments of Chapter 5. Predictions between visits are therefore available from the T-FEN, but their correctness cannot be assessed with these data.

6.1.6 Implications for Modelling

Among operators with spatial context, the largest effects came from two changes to the operator: extending the reach of one update along the direction in which the front moves, and adding a mechanism that transports the state. Larger models, a higher integration order, a penalty against lesion shrinkage and the single experiment with adaptive meshes (§5.7), whose adaptive branch never engaged, did not improve accuracy at the available precision. In the framework, the objective decided whether the model learned any change at all. For a new disease-forecasting task of this kind, two questions therefore come first: whether the loss puts enough weight on the region that changes, and how far one update of the operator reaches compared with how far the disease moves between visits.

6.2 Clinical Implications

The survey was designed around the change-region Dice at one year, which measures whether a model predicts change where change actually happens. On this measure the operators predict change where it happens, which persistence cannot. The speed of growth of an individual eye over its whole follow-up is captured less well. The correlation between predicted and true growth rates is 0.40 for the canonical model, against 0.52 to 0.61 for the comparable models of §5.9, and its fast-progressor AUCs are 0.70 to 0.74 against 0.77 to 0.81 reported by Mai et al. (2024) on the full cohort from which the data here are drawn. The other operators of the survey give similar values, so the weaker agreement on growth speed is shared by every operator in the framework.

The comparison is not like-for-like (§5.9). It is still notable that a model given the segmented lesion as input is weaker on growth speed than one that must find the lesion itself.

Several explanations are possible:

- A local model applied autoregressively over several years can drift, and the growth rate is taken at the last visit, often three or more years after baseline.
- Large, fast-growing lesions are the ones most often cut by the crop, which lowers their measured growth.
- The operators see one visit and have no memory of how fast the eye has grown so far.
- The conditioning carries no information on how fast this particular eye progresses (§6.1.4).
- The operators are not trained on the growth rate. It is computed afterwards from their predicted maps, whereas some models of §5.9 regress the rate directly.

- The canonical model predicts less change than occurs in most eyes, 35 of 48 on folds 0 to 2 at a median of 0.73 of the true change (§5.2), and about half the true rate for the fastest eyes (§5.9). A deterministic model trained with a pixel-wise loss may favour such cautious predictions.
- The inputs carry no image intensity. The models of §5.9 with stronger growth-speed values use reflectivity maps or the raw OCT volume, and for fundus autofluorescence images the tissue around the lesion carries the most information on growth speed (§2.4). Masks and layer depths may hold less of that information.

The distinction is clinically relevant. Complement inhibitors slow GA growth but do not stop it (§1.1), so the benefit of treatment relative to its burden is largest in the eyes that grow fastest. The decision to treat therefore depends on growth speed, the weaker of the two capabilities. A forecast of where the lesion will expand is useful in a different way, for example to judge whether growth is heading towards the fovea, but that use was not evaluated here.

A second clinical question is how far ahead the forecast remains useful. All tested comparisons between operators (§5.1.5) are made at the one-year anchor, the first visit about one year after baseline; beyond it, only descriptive values exist (§5.3). Beyond one year, the growth-region Dice of the canonical model, which scores only new atrophy, does not fall; it lies at about 0.60 to 0.66 in the second, third and later years (Table 5.2). These values are descriptive and pooled over rollout steps. Whether any of these levels is clinically sufficient cannot be judged from the data: no clinically established Dice threshold for GA progression forecasts is known to the author.

A scale for these values comes from the reference outlines themselves. When two trained graders outline the same fundus autofluorescence images, their outlines of the region that became atrophic within one year agree with a Dice of only about 0.71 to 0.73 (Salvi et al., 2025). The outlines that serve as ground truth therefore carry considerable uncertainty of their own, and a Dice of 1 is not reachable in practice. On this scale, the first-year growth-region Dice of about 0.5 obtained here (Table 5.10) lies closer to the attainable level than its distance from 1 suggests. The grader value is a reference point, neither a target nor an upper limit for the models of this thesis. It was measured on a different imaging modality and a different cohort, and the graders outlined existing images rather than predicting a future one.

Finally, because every model starts from segmented masks and layer depths, a clinical use would place it behind a segmentation model. Errors of that segmentation would pass into the forecast, and this thesis does not model them. Forecast failures were observed where new atrophy appears far from the existing lesion or as new separate lesions (§5.5, Appendix B), and for fast progressors, whose growth is under-predicted (§5.9). Lesions that touch the edge of the imaged field are censored in the reference itself. Whether long intervals between visits degrade the forecast was not analysed.

In summary, the forecast locates growth along the lesion border at a level within the range of prior work, but predicts the growth speed of an individual eye less well than comparable models and was evaluated on one cohort only. It would need external validation and a measure of its own uncertainty (§6.4) before it could support treatment decisions.

6.3 Limitations

The conclusions of this thesis are limited by the data, the evaluation and the scope of the models. The cohort is small and comes from a single site: 75 eyes of 51 patients, with 12 to 20 validation eyes per fold. Every result is a validation result of a five-fold cross-validation; there is no held-out test split (§3.7), and no external cohort was available for testing. Twenty-four patients contribute both eyes, and fellow eyes are not independent, so the per-eye statistics are somewhat optimistic.

The fixed 49×1024 window censors growth at its border, in the training targets and in the metrics, without any flag (§3.3). On the 54 eyes whose lesion does not touch the crop border, the arms keep the order of Table 5.1 (§5.2).

Every area in mm^2 rests on a pixel spacing that is treated as constant across the cohort. It was checked only against the cohort's own transform files, to about 2 %, and not against the original DICOM spacing fields. The GA reference was annotated on fundus autofluorescence and registered onto the OCT grid (§3.1).

The reference masks also shrink between visits. The total lesion area decreases in 5 % of the 478 pairs of consecutive visits, by a median of 2.6 %, but almost every pair (96 %) loses some atrophic pixels along the border, a median of 2.2 % of the lesion. This pattern is more consistent with annotation and registration noise at the boundary than with regression of the atrophy, although the masks alone cannot separate the two. Boundary noise of this size is plausible: on the region that becomes atrophic, two graders reach a growth-region Dice of only 0.71 to 0.73 (§6.2). The observed shrinkage is one reason why no monotonic-growth constraint was imposed.

The evaluation has limits of its own. The one-year anchor is the first visit at or after about one year, and its time has no upper bound, so a few eyes are scored later, where more change has accumulated. Several findings rest on one seed. The training curriculum and the zero-initialised output are used in every run, so their effect on the forecast is not measured. No non-learned growth model, such as a uniform dilation of the baseline lesion, was evaluated, so the scores do not show how much the operators add beyond the average growth of the cohort.

The scope of the models is limited as well. Every operator sees one visit and has no memory of the trajectory so far, although each eye has 5 to 13 visits. An eye's own past growth, the most direct evidence of its speed, is therefore never used. The patient covariates are enabled in every operator, but whether they carry useful information is not settled; two numbers shared by every grid position cannot localise anything. The pipeline starts from segmented masks, so it neither benefits from nor accounts for the information in the raw OCT volume.

Two directions lie outside the survey. Physics-informed neural networks were deliberately not run, because the governing equation is unknown and the equation residual would have to be taken on a binary mask. The second is mesh adaptation: all operators of the comparison work on the uniform 49×1024 grid. Mesh adaptation was tried only in the single experiment of §5.7, so no general conclusion about it is drawn.

6.4 Future Work

The following directions follow from the results of §6.1 and the limitations of §6.3.

Spatial pre-processing. The fixed centre crop censors growth at the window border (§3.3). A larger or adaptive window with a mask for padded positions, or a crop that follows the registration to the fundus image, would reduce this censoring and make areas comparable with whole-field studies.

A continuous representation of the lesion. The lesion enters the state as a binary mask, so the normalised state has a hard step at the lesion edge (§3.4), and a pixel that becomes atrophic has to jump the full distance between the two mask values in a single update (§5.6.1). A signed distance to the lesion boundary would replace the step by a smooth field whose zero level marks the boundary. Growth would then appear as a gradual change of this field rather than as jumps at single pixels, and the binary forecast would be recovered by thresholding at zero. Whether this makes the forecast easier to learn, including the growth speed that the binary state predicts less well (§6.2), remains open.

Several scales of reach. The graph network uses one neighbourhood with a fixed reach. A graph that combines edges at several scales, or a pyramid of coarsened graphs, would let one update see both the immediate surroundings of a node and the distance the front moves between visits, as the U-Net does on the grid.

Trajectory memory. Giving the operator several past visits, or a recurrent state carried along the rollout, would let it use the eye's own growth history. This would address the growth-speed gap of §6.2 directly. It is also the setting in which sequence models such as recurrent networks and transformers become applicable.

An eye-level descriptor. A learned descriptor could take the place of the missing equation features (§6.1). The one design tried here saw predicted rather than observed layers during the rollout, corrected only about $2\times$ of the grid's 21:1 anisotropy, and conditioned the operator globally. A next design should see the observed layers of the input visit, respect the physical spacing, and be able to condition locally.

A bounded transport model. The transport term is established, but the velocity field of the T-FEN is not interpreted in this thesis. A next step is a velocity parameterisation that is bounded by construction and builds on the energy-conserving form. Such a velocity would also keep the Courant number below the stability limit of the integrator at any step (§5.8.2), and the velocity field could then be read as a direction and speed of growth.

From raw OCT, in 3-D. One way to address the gap to Mai et al. (2024) is to learn from the raw OCT volume, either end to end or jointly with the segmentation. A further step could be to work on the volumetric OCT rather than on en-face maps.

Clinical use. A forecast used for treatment decisions needs a measure of its own uncertainty, which none of the operators provides. Joint training across several retinal diseases could also ease the data limitation.

Chapter 7

Conclusion

This thesis set out to answer one question: *What is needed to forecast the progression of Geographic Atrophy from longitudinal OCT?* It was split into two parts: (1) What kind of framework does the forecast need? (2) What kind of operator does that framework need?

For the first part, one framework was sufficient for every operator with spatial context: the framework of Chapter 4, with a residual update conditioned on the elapsed time Δt, a zero-initialised output so that every model starts from persistence, a training curriculum measured in elapsed days, and one loss and one evaluation for all models. It served operators from unrelated literatures without modification, and every operator with spatial context learned to predict change. Of its parts, the emphasis of the loss on the mask was shown to be necessary; the curriculum and the zero initialisation were not ablated. Trained with a plain mean squared error, the canonical operator learned no change. A soft-Dice term on the mask added about 0.05 to the change-region Dice. The interval length has to reach the operator, but the form in which it does so mattered little: a fourth-order Runge–Kutta step gave no measurable gain over a single Euler step.

For the second part, the score of an operator did not follow from its architecture class. The three strongest operators (a finite-element network, a graph network and a U-Net) come from unrelated classes. At one year they reach a change-region Dice of about 0.50 to 0.54, against zero for persistence, and their scores differ by at most about 0.04. The finite-element network scores highest: above the U-Net at both seeds, and slightly above the graph network, a lead that is not established. The finite-element network carries the transport term and the graph network the extended reach described below. A few properties, each measured one at a time, separate the operators:

- **Spatial context.** A model without spatial context predicts no change; one ring of neighbours raises the change-region Dice from 0 to about 0.45.
- **Physical reach at the scale of the lesion front.** Placing the neighbours of a graph network so that one step reaches as far as the front moves between visits raised the change-region Dice by 0.064, at an unchanged parameter count.
- **An explicit transport term.** Adding a learned transport term to a Finite Element Network raised the change-region Dice by 0.062 over the same network without the term, and a smaller T-FEN with fewer parameters than that network scored the same.

Reach and transport are two different ways to correct the same deficit: a single update that cannot see as far as the disease moves between two visits. Both

routes lead to about the same score, and this agreement is the central architectural result of this thesis. A third candidate, a local path for the global FNO, is not established. The FNO scores below the U-Net at one year at both seeds, and a small local path brought it level at one seed but not at the second.

Several other changes did not improve the forecast. Larger models did not help in any of the four classes tested, and neither did a penalty against lesion shrinkage. A single experiment with an adaptive moving mesh (§5.7) gave no measurable gain: its adaptive branch never engaged, and the mesh, by gathering nodes at the border, reduces the reach the operators need. The validity diagnostics also restrict how the canonical model may be described: it is a one-step map conditioned on the step length and does not behave as a continuous-time model. A model that does behave as one, the energy-conserving T-FEN, can be trained, but this property does not make it more accurate.

Clinically, the operators predict where the lesion grows better than how fast an individual eye grows. On where it grows, the growth-region Dice of the canonical model lies within the range of prior work (§5.9). On how fast, the model reaches a correlation of 0.40 with the true growth rates, against 0.61 for the model of Mai et al. (2024), which starts from the raw OCT volume instead of segmented masks, 0.52 for Gigon et al. (2021) and about 0.57 to 0.59 for Yoshida et al. (2025).

Every conclusion drawn here is limited by the small cohort, the censoring by the fixed crop and the lack of memory of past visits. Within these limits, operators of different families reached similar scores once they shared one framework. Among the operators with spatial context, the property with the largest effect was whether one update can see as far as the disease moves between two visits.

Bibliography

Abràmoff, Michael D., Mona K. Garvin, and Milan Sonka (2010). "Retinal Imaging and Image Analysis". In: *IEEE Reviews in Biomedical Engineering* 3, pp. 169–208. DOI: 10.1109/RBME.2010.2084567.

Ba, Jimmy Lei, Jamie Ryan Kiros, and Geoffrey E. Hinton (2016). *Layer Normalization*. Preprint. arXiv: 1607.06450.

Banerjee, Chayan et al. (2025). "Physics-informed Machine Learning for Medical Image Analysis". In: *ACM Computing Surveys* 58.4. arXiv:2408.01026, preprint title "PINNs for Medical Image Analysis: A Survey", pp. 1–35. DOI: 10.1145/3768158.

Bar-Sinai, Yohai et al. (2019). "Learning Data-Driven Discretizations for Partial Differential Equations". In: *Proceedings of the National Academy of Sciences* 116.31, pp. 15344–15349. DOI: 10.1073/pnas.1814058116.

Bartels, Sören (2016). *Numerical Approximation of Partial Differential Equations*. Vol. 64. Texts in Applied Mathematics. Cham: Springer. DOI: 10.1007/978-3-319-32354-1.

Battaglia, Peter W. et al. (2018). *Relational Inductive Biases, Deep Learning, and Graph Networks*. Preprint. arXiv: 1806.01261.

Bogunović, Hrvoje et al. (2017). "Machine Learning of the Progression of Intermediate Age-Related Macular Degeneration Based on OCT Imaging". In: *Investigative Ophthalmology & Visual Science* 58.6, BIO141–BIO150. DOI: 10.1167/iovs.17-21789.

Boopathiraj, Nithya et al. (2024). "Recent Updates on the Diagnosis and Management of Age-Related Macular Degeneration". In: *Mayo Clinic Proceedings: Innovations, Quality & Outcomes* 8.4, pp. 364–374. DOI: 10.1016/j.mayocpiqo.2024.05.003.

Boyer, David S. et al. (2017). "The Pathophysiology of Geographic Atrophy Secondary to Age-Related Macular Degeneration and the Complement Pathway as a Therapeutic Target". In: *Retina* 37.5, pp. 819–835. DOI: 10.1097/IAE.0000000000001392.

Brandstetter, Johannes, Daniel E. Worrall, and Max Welling (2022). "Message Passing Neural PDE Solvers". In: *International Conference on Learning Representations (ICLR)*. arXiv: 2202.03376.

Butcher, J. C. (1987). *The Numerical Analysis of Ordinary Differential Equations: Runge-Kutta and General Linear Methods*. Chichester: Wiley. ISBN: 0-471-91046-5.

Chen, Ricky T. Q. et al. (2018). "Neural Ordinary Differential Equations". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 31. arXiv: 1806.07366.

Chu, Zhongdi et al. (2022). "Optical Coherence Tomography Measurements of the Retinal Pigment Epithelium to Bruch Membrane Thickness Around Geographic Atrophy Correlate With Growth". In: *American Journal of Ophthalmology* 236, pp. 249–260. DOI: 10.1016/j.ajo.2021.10.032.

Cluceru, Julia et al. (2024). "Topographic Clinical Insights From Deep Learning–Based Geographic Atrophy Progression Prediction". In: *Translational Vision Science & Technology* 13.8, p. 6. DOI: 10.1167/tvst.13.8.6.

Courant, Richard (1943). "Variational Methods for the Solution of Problems of Equilibrium and Vibrations". In: *Bulletin of the American Mathematical Society* 49.1, pp. 1–23. DOI: 10.1090/S0002-9904-1943-07818-4.

Courant, Richard, Kurt Friedrichs, and Hans Lewy (1928). "Über die partiellen Differenzengleichungen der mathematischen Physik". In: *Mathematische Annalen* 100.1, pp. 32–74. DOI: 10.1007/BF01448839.

Ebneter, Andreas et al. (2016). "Relationship Between Presumptive Inner Nuclear Layer Thickness and Geographic Atrophy Progression in Age-Related Macular Degeneration". In: *Investigative Ophthalmology & Visual Science* 57.9, OCT299–OCT306. DOI: 10.1167/iovs.15-18865.

European Medicines Agency (2024a). *Izelvay (avacincaptad pegol): Withdrawal of the marketing authorisation application*. Application withdrawn on 24 October 2024. URL: https://www.ema.europa.eu/en/medicines/human/EPAR/izelvay (visited on 10/14/2026).

– (2024b). *Syfovre (pegcetacoplan): Refusal of the marketing authorisation*. Final opinion after re-examination, 19 September 2024. URL: https://www.ema.europa.eu/en/medicines/human/EPAR/syfovre (visited on 10/14/2026).

Ezhov, Ivan, Tudor Mot, et al. (2022). "Geometry-Aware Neural Solver for Fast Bayesian Calibration of Brain Tumor Models". In: *IEEE Transactions on Medical Imaging* 41.5. arXiv:2009.04240, pp. 1269–1278. DOI: 10.1109/TMI.2021.3136582.

Ezhov, Ivan, Kevin Scibilia, et al. (2023). "Learn-Morph-Infer: A New Way of Solving the Inverse Problem for Brain Tumor Modeling". In: *Medical Image Analysis* 83. arXiv:2111.04090, p. 102672. DOI: 10.1016/j.media.2022.102672.

Feuer, William J. et al. (2013). "Square Root Transformation of Geographic Atrophy Area Measurements to Eliminate Dependence of Growth Rates on Baseline Lesion Measurements: A Reanalysis of Age-Related Eye Disease Study Report No. 26". In: *JAMA Ophthalmology* 131.1, pp. 110–111. DOI: 10.1001/jamaophthalmol.2013.572.

Fey, Matthias and Jan Eric Lenssen (2019). "Fast Graph Representation Learning with PyTorch Geometric". In: *ICLR Workshop on Representation Learning on Graphs and Manifolds*. arXiv: 1903.02428.

Flaxman, Seth R. et al. (2017). "Global Causes of Blindness and Distance Vision Impairment 1990–2020: A Systematic Review and Meta-analysis". In: *The Lancet Global Health* 5.12, e1221–e1234. DOI: 10.1016/S2214-109X(17)30393-5.

Garvin, Mona K. et al. (2009). "Automated 3-D Intraretinal Layer Segmentation of Macular Spectral-Domain Optical Coherence Tomography Images". In: *IEEE Transactions on Medical Imaging* 28.9, pp. 1436–1447. DOI: 10.1109/TMI.2009.2016958.

Gigon, Anthony et al. (2021). "Personalized Atrophy Risk Mapping in Age-Related Macular Degeneration". In: *Translational Vision Science & Technology* 10.13, p. 18. DOI: 10.1167/tvst.10.13.18.

Gilmer, Justin et al. (2017). "Neural Message Passing for Quantum Chemistry". In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. Vol. 70. Proceedings of Machine Learning Research, pp. 1263–1272. arXiv: 1704.01212.

Gottlieb, David and Steven A. Orszag (1977). *Numerical Analysis of Spectral Methods: Theory and Applications*. Vol. 26. CBMS-NSF Regional Conference Series in Applied Mathematics. Philadelphia: SIAM. DOI: 10.1137/1.9781611970425.

Gupta, Jayesh K. and Johannes Brandstetter (2023). "Towards Multi-spatiotemporal-scale Generalized PDE Modeling". In: *Transactions on Machine Learning Research*. arXiv: 2209.15616.

Hairer, Ernst, Syvert P. Nørsett, and Gerhard Wanner (1993). *Solving Ordinary Differential Equations I: Nonstiff Problems*. 2nd ed. Vol. 8. Springer Series in Computational Mathematics. Berlin: Springer. DOI: 10.1007/978-3-540-78862-1.

Hu, Peiyan, Yue Wang, and Zhi-Ming Ma (2024). "Better Neural PDE Solvers Through Data-Free Mesh Movers". In: *International Conference on Learning Representations (ICLR)*. arXiv: 2312.05583.

Huang, David et al. (1991). "Optical Coherence Tomography". In: *Science* 254.5035, pp. 1178–1181. DOI: 10.1126/science.1957169.

Huang, Weizhang and Robert D. Russell (2011). *Adaptive Moving Mesh Methods*. Vol. 174. Applied Mathematical Sciences. New York: Springer. DOI: 10.1007/978-1-4419-7916-2.

Ioffe, Sergey and Christian Szegedy (2015). "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift". In: *Proceedings of the 32nd International Conference on Machine Learning (ICML)*. Vol. 37. Proceedings of Machine Learning Research, pp. 448–456.

Jiang, Guang-Shan and Chi-Wang Shu (1996). "Efficient Implementation of Weighted ENO Schemes". In: *Journal of Computational Physics* 126.1, pp. 202–228. DOI: 10.1006/jcph.1996.0130.

Kochkov, Dmitrii et al. (2021). "Machine Learning–Accelerated Computational Fluid Dynamics". In: *Proceedings of the National Academy of Sciences* 118.21, e2101784118. DOI: 10.1073/pnas.2101784118.

Koksaldi, Seher et al. (2025). "Clinical Anatomy of the Macula". In: *Medical Hypothesis, Discovery & Innovation Ophthalmology Journal* 14.2, pp. 17–27. DOI: 10.51329/mehdiophthal1520.

Krishnapriyan, Aditi S. et al. (2023). "Learning Continuous Models for Continuous Physics". In: *Communications Physics* 6, p. 319. DOI: 10.1038/s42005-023-01433-4.

Kutta, Wilhelm (1901). "Beitrag zur näherungsweisen Integration totaler Differentialgleichungen". In: *Zeitschrift für Mathematik und Physik* 46, pp. 435–453.

Lachinov, Dmitrii et al. (2024). "Learning Spatio-Temporal Model of Disease Progression With NeuralODEs From Longitudinal Volumetric Data". In: *IEEE Transactions on Medical Imaging* 43.3. arXiv:2211.04234, pp. 1165–1179. DOI: 10.1109/TMI.2023.3330576.

Lad, Eleonora M. and Usha Chakravarthy (2023). "The Issue of End Point Discordance in Dry Age-Related Macular Degeneration: How Might Clinical Trials Demonstrate a Functional Benefit?" In: *Ophthalmology* 130.9, pp. 890–892. DOI: 10.1016/j.ophtha.2023.05.020.

Lam, Remi et al. (2023). "Learning Skillful Medium-Range Global Weather Forecasting". In: *Science* 382.6677, pp. 1416–1421. DOI: 10.1126/science.adi2336.

Langtangen, Hans Petter and Kent-Andre Mardal (2019). *Introduction to Numerical Methods for Variational Problems*. Vol. 21. Texts in Computational Science and Engineering. Cham: Springer. DOI: 10.1007/978-3-030-23788-2.

LeVeque, Randall J. (2002). *Finite Volume Methods for Hyperbolic Problems*. Cambridge Texts in Applied Mathematics. Cambridge: Cambridge University Press. DOI: 10.1017/CBO9780511791253.

– (2007). *Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems*. Philadelphia: SIAM. DOI: 10.1137/1.9780898717839.

Li, Guohao et al. (2019). "DeepGCNs: Can GCNs Go as Deep as CNNs?" In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9266–9275. DOI: 10.1109/ICCV.2019.00936. arXiv: 1904.03751.

Li, Kang et al. (2006). "Optimal Surface Segmentation in Volumetric Images – A Graph-Theoretic Approach". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 28.1, pp. 119–134. DOI: 10.1109/TPAMI.2006.19.

Li, Zongyi et al. (2021). "Fourier Neural Operator for Parametric Partial Differential Equations". In: *International Conference on Learning Representations (ICLR)*. arXiv: 2010.08895.

Lienen, Marten and Stephan Günnemann (2022). "Learning the Dynamics of Physical Systems from Sparse Observations with Finite Element Networks". In: *International Conference on Learning Representations (ICLR)*. arXiv: 2203.08852.

Lindner, Moritz et al. (2018). "Local Progression Kinetics of Geographic Atrophy in Age-Related Macular Degeneration Are Associated With Atrophy Border Morphology". In: *Investigative Ophthalmology & Visual Science* 59.4, AMD12–AMD18. DOI: 10.1167/iovs.17-23203.

Lipková, Jana et al. (2019). "Personalized Radiotherapy Design for Glioblastoma: Integrating Mathematical Tumor Models, Multimodal Scans, and Bayesian Inference". In: *IEEE Transactions on Medical Imaging* 38.8. arXiv:1807.00499, pp. 1875–1884. DOI: 10.1109/TMI.2019.2902044.

Liu, Chen et al. (2025). "ImageFlowNet: Forecasting Multiscale Image-Level Trajectories of Disease Progression with Irregularly-Sampled Longitudinal Medical Images". In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. arXiv:2406.14794.

Liu, Xu-Dong, Stanley Osher, and Tony Chan (1994). "Weighted Essentially Non-oscillatory Schemes". In: *Journal of Computational Physics* 115.1, pp. 200–212. DOI: 10.1006/jcph.1994.1187.

Liu, Rosanne et al. (2018). "An Intriguing Failing of Convolutional Neural Networks and the CoordConv Solution". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 31. arXiv: 1807.03247.

Liu-Schiaffini, Miguel et al. (2024). "Neural Operators with Localized Integral and Differential Kernels". In: *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. Proceedings of Machine Learning Research, pp. 32576–32594. arXiv: 2402.16845.

Loshchilov, Ilya and Frank Hutter (2019). "Decoupled Weight Decay Regularization". In: *International Conference on Learning Representations (ICLR)*. arXiv: 1711.05101.

Lu, Lu et al. (2021). "Learning Nonlinear Operators via DeepONet Based on the Universal Approximation Theorem of Operators". In: *Nature Machine Intelligence* 3.3, pp. 218–229. DOI: 10.1038/s42256-021-00302-5.

Mai, Julia et al. (2024). "Deep Learning-Based Prediction of Individual Geographic Atrophy Progression from a Single Baseline OCT". In: *Ophthalmology Science* 4.4, p. 100466. DOI: 10.1016/j.xops.2024.100466.

Milletari, Fausto, Nassir Navab, and Seyed-Ahmad Ahmadi (2016). "V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation". In: *Proceedings of the Fourth International Conference on 3D Vision (3DV)*, pp. 565–571. DOI: 10.1109/3DV.2016.79.

Nair, Archana A. et al. (2021). "Determining the Location of the Fovea Centralis Via En-Face SLO and Cross-Sectional OCT Imaging in Patients Without Retinal Pathology". In: *Translational Vision Science & Technology* 10.2, p. 25. DOI: 10.1167/tvst.10.2.25.

Ott, Katharina et al. (2021). "ResNet After All: Neural ODEs and Their Numerical Solution". In: *International Conference on Learning Representations (ICLR)*. arXiv: 2007.15386.

Paszke, Adam et al. (2019). "PyTorch: An Imperative Style, High-Performance Deep Learning Library". In: *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pp. 8024–8035. arXiv: 1912.01703.

Pfaff, Tobias et al. (2021). "Learning Mesh-Based Simulation with Graph Networks". In: *International Conference on Learning Representations (ICLR)*. arXiv: 2010.03409.

Pilotto, Elisabetta et al. (2015). "En Face Optical Coherence Tomography to Detect and Measure Geographic Atrophy". In: *Investigative Ophthalmology & Visual Science* 56.13, pp. 8120–8124. DOI: 10.1167/iovs.15-17366.

Puglisi, Lemuel, Daniel C. Alexander, and Daniele Ravì (2024). "Enhancing Spatiotemporal Disease Progression Models via Latent Diffusion and Prior Knowledge". In: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2024*. Lecture Notes in Computer Science. arXiv:2405.03328. Springer, pp. 173–183. DOI: 10.1007/978-3-031-72069-7_17.

Ravì, Daniele et al. (2022). "Degenerative Adversarial Neuroimage Nets for Brain Scan Simulations: Application in Ageing and Dementia". In: *Medical Image Analysis* 75. arXiv:1912.01526, p. 102257. DOI: 10.1016/j.media.2021.102257.

Ronneberger, Olaf, Philipp Fischer, and Thomas Brox (2015). "U-Net: Convolutional Networks for Biomedical Image Segmentation". In: *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. Vol. 9351. Lecture Notes in Computer Science. Springer, pp. 234–241. DOI: 10.1007/978-3-319-24574-4_28.

Rubanova, Yulia, Ricky T. Q. Chen, and David Duvenaud (2019). "Latent Ordinary Differential Equations for Irregularly-Sampled Time Series". In: *Advances in Neural Information Processing Systems*. Vol. 32. arXiv:1907.03907.

Runge, Carl (1895). "Ueber die numerische Auflösung von Differentialgleichungen". In: *Mathematische Annalen* 46.2, pp. 167–178. DOI: 10.1007/BF01446807.

Sadda, Srinivas R. et al. (2018). "Consensus Definition for Atrophy Associated with Age-Related Macular Degeneration on OCT: Classification of Atrophy Report 3". In: *Ophthalmology* 125.4, pp. 537–548. DOI: 10.1016/j.ophtha.2017.09.028.

Salvi, A. et al. (2025). "Deep Learning to Predict the Future Growth of Geographic Atrophy from Fundus Autofluorescence". In: *Ophthalmology Science* 5.2, p. 100635. DOI: 10.1016/j.xops.2024.100635.

Sanchez-Gonzalez, Alvaro et al. (2020). "Learning to Simulate Complex Physics with Graph Networks". In: *Proceedings of the 37th International Conference on Machine Learning (ICML)*. Vol. 119. Proceedings of Machine Learning Research, pp. 8459–8468.

Scarselli, Franco et al. (2009). "The Graph Neural Network Model". In: *IEEE Transactions on Neural Networks* 20.1, pp. 61–80. DOI: 10.1109/TNN.2008.2005605.

Schmidt-Erfurth, Ursula et al. (2018). "Prediction of Individual Disease Conversion in Early AMD Using Artificial Intelligence". In: *Investigative Ophthalmology & Visual Science* 59.8, pp. 3199–3208. DOI: 10.1167/iovs.18-24106.

Singh, Rishi P. et al. (2025). "Consideration of Patient Phenotypes in Geographic Atrophy Due to Age-Related Macular Degeneration". In: *BMJ Open Ophthalmology* 10, e002105. DOI: 10.1136/bmjophth-2024-002105.

Song, Mi Yeon et al. (2025). "Prevalence and Risk Factors of Age-Related Macular Degeneration in South Korea: Korea National Health and Nutrition Examination Survey". In: *Ophthalmic Epidemiology* 32.1, pp. 34–43. DOI: 10.1080/09286586.2024.2321892.

Trefethen, Lloyd N. (2000). *Spectral Methods in MATLAB*. Philadelphia: SIAM. DOI: 10.1137/1.9780898719598.

Trincão-Marques, José et al. (2024). "Gene and Cell Therapy for Age-Related Macular Degeneration: A Review". In: *Survey of Ophthalmology* 69.5, pp. 665–676. DOI: 10.1016/j.survophthal.2024.05.002.

Vallino, Veronica et al. (2024). "Structural OCT and OCT Angiography Biomarkers Associated With the Development and Progression of Geographic Atrophy in AMD". In: *Graefe's Archive for Clinical and Experimental Ophthalmology* 262, pp. 3421–3436. DOI: 10.1007/s00417-024-06497-8.

Vogl, Wolf-Dieter et al. (2021). "Spatio-temporal Alterations in Retinal and Choroidal Layers in the Progression of Age-Related Macular Degeneration (AMD) in Optical Coherence Tomography". In: *Scientific Reports* 11, p. 5743. DOI: 10.1038/s41598-021-85110-y.

Wong, Wan Ling et al. (2014). "Global Prevalence of Age-Related Macular Degeneration and Disease Burden Projection for 2020 and 2040: A Systematic Review and Meta-Analysis". In: *The Lancet Global Health* 2.2, e106–e116. DOI: 10.1016/S2214-109X(13)70145-1.

Wu, Yuxin and Kaiming He (2018). "Group Normalization". In: *Computer Vision – ECCV 2018*. Lecture Notes in Computer Science. Springer, pp. 3–19. DOI: 10.1007/978-3-030-01261-8_1.

Yehoshua, Zohar et al. (2011). "Progression of Geographic Atrophy in Age-Related Macular Degeneration Imaged with Spectral Domain Optical Coherence Tomography". In: *Ophthalmology* 118.4, pp. 679–686. DOI: 10.1016/j.ophtha.2010.08.018.

Yoshida, Kenta et al. (2025). "Deep Learning Approaches to Predict Geographic Atrophy Progression Using Three-Dimensional OCT Imaging". In: *Translational Vision Science & Technology* 14.2, p. 11. DOI: 10.1167/tvst.14.2.11.

Yu, Fisher and Vladlen Koltun (2016). "Multi-Scale Context Aggregation by Dilated Convolutions". In: *International Conference on Learning Representations (ICLR)*. arXiv: 1511.07122.

Appendix A

Dataset Details

This appendix summarises the visit schedule, the lesion scale and the spatial extent of the cohort described in Chapter 3.

A.1 Visit Schedule and Lesion Scale

The visit schedule and the lesion scale of the cohort are summarised in Figure A.1. Most eyes were seen six or seven times, three in four intervals between visits are 180 days long, and the baseline lesion area ranges from 0.11 to 20.8 mm^2, with a median of 6.7 mm^2.

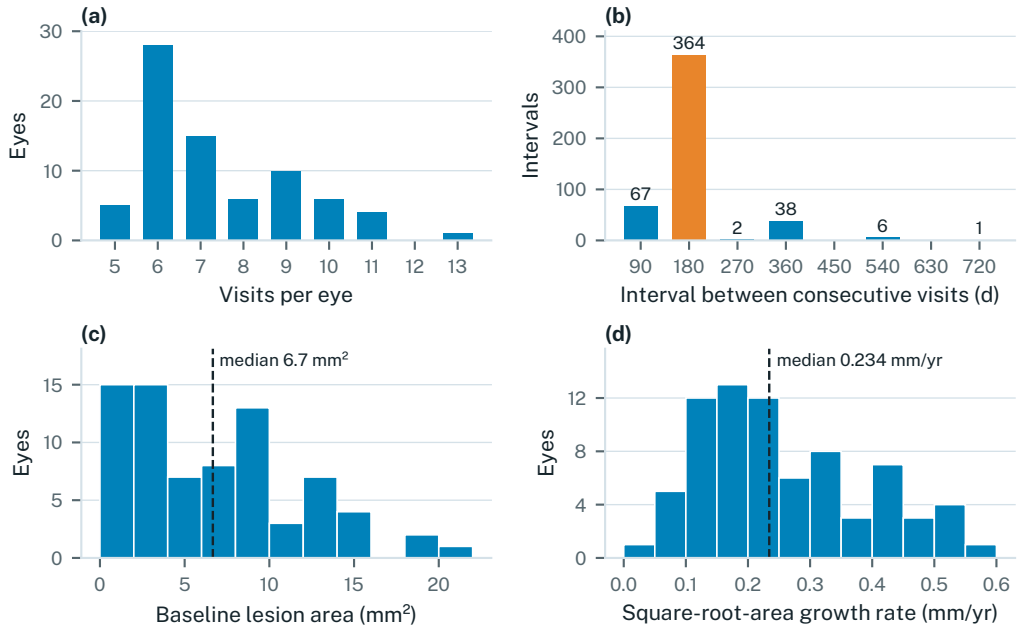


Figure A.1: Visit schedule and lesion scale of the 75 eyes of the cohort. *(a)* Number of visits per eye. *(b)* Intervals between consecutive visits over the 478 windows; every interval is a multiple of 90 days, and the most common interval, 180 days, is highlighted. *(c)* Lesion area at the baseline visit, on the 49×1024 modelling grid. *(d)* Growth rate of the square root of the lesion area from the baseline to the last visit, on the same grid. Dashed lines mark the medians (6.7 mm^2 and 0.234 mm per year). Areas assume the constant pixel spacing stated in §3.1.

A.2 Native Grids and the Crop

Figure A.2 relates the native grids of the eyes to the modelling window of §3.3; how much lesion area the window removes is given there.

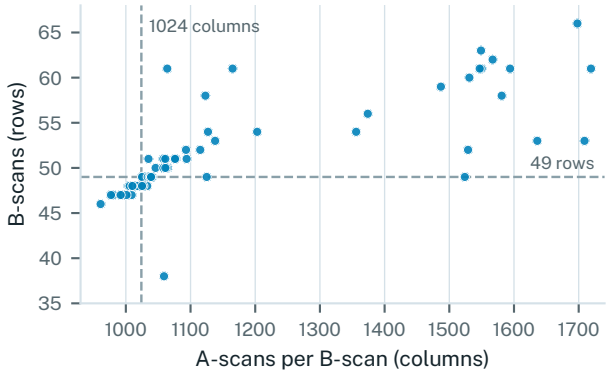


Figure A.2: Native en-face grid of each of the 75 eyes, in B-scans (rows) and A-scans per B-scan (columns), against the 49×1024 modelling window. The grid is the same at every visit of an eye, and eyes with identical grids overlap (67 distinct grids). Dashed lines mark the window size: an eye left of or below a line is zero-padded along that axis, and an eye right of or above it is centre-cropped.

Appendix B

Additional Rollout Figures

Figure B.1 shows rollouts of the canonical dilated-stencil graph network for four further eyes. They were chosen by fixed rules to show different kinds of lesion on which the model works: a small and a large lesion, a lesion made of several parts, and a fast-growing one, each among the eyes whose change-region Dice at the one-year anchor lies at or above the median. In all four, the predicted change lies along the lesion border where the true change occurs. Three weaknesses are also visible: in (a) the prediction runs ahead of the true growth after about four years, new separate lesions away from the existing ones are not predicted (c), and in the fastest-growing eye the predicted growth is smaller than the true growth (d). Where the model fails more clearly is discussed in §5.5 and §5.9.

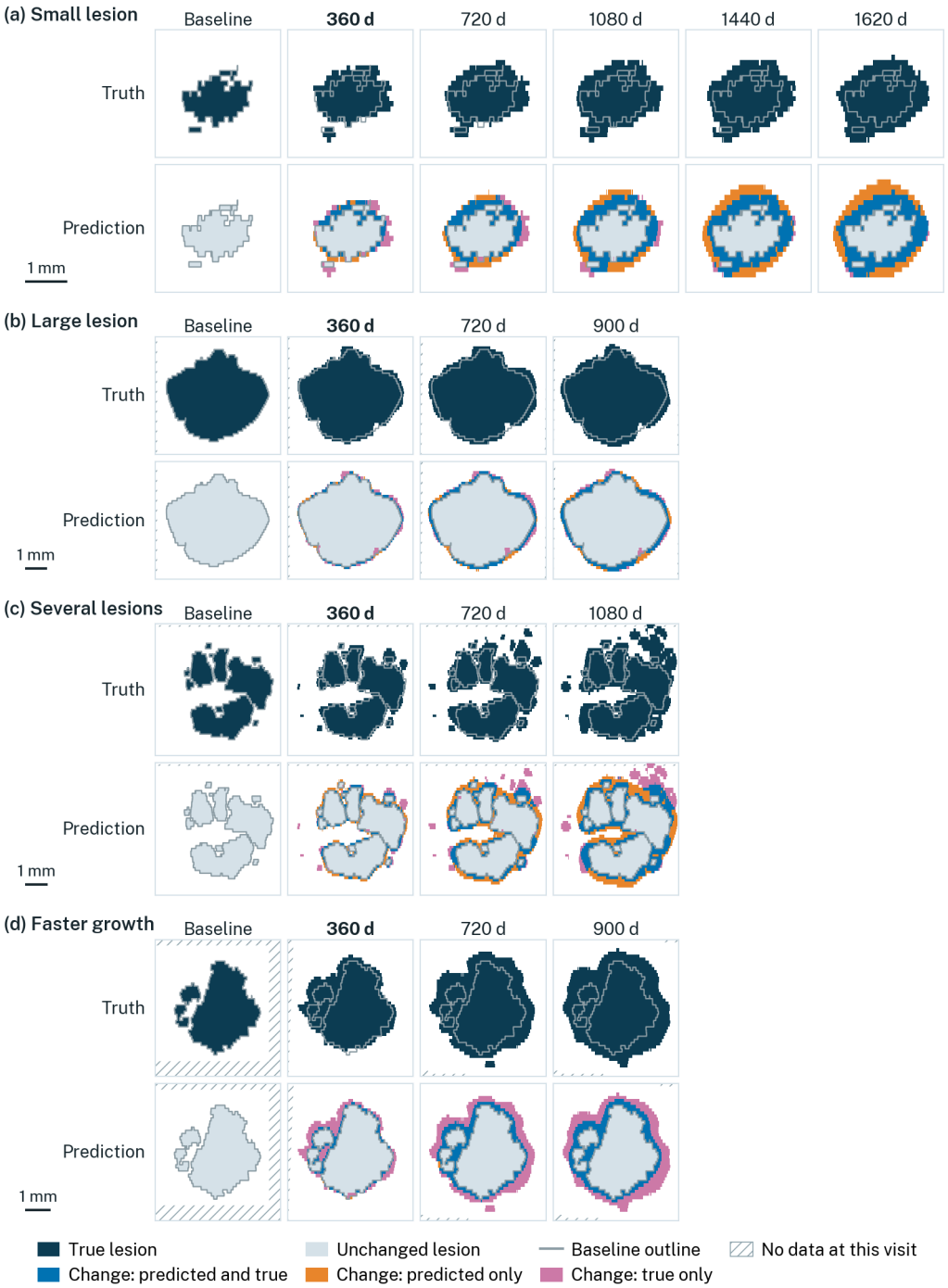


Figure B.1: Additional rollouts of the dilated-stencil graph network (weights after the last training epoch) for four validation eyes, each shown as ground truth (upper row) and prediction (lower row) at the baseline and at the follow-up visits closest to each full year. The eyes show different kinds of lesion on which the model works: *(a)* a small lesion (1.3 mm^2 at baseline), *(b)* a large lesion (13.2 mm^2), *(c)* seven separate lesions, *(d)* a fast-growing lesion (0.42 mm per year on the square-root area scale). Each eye is cropped to its lesion and has its own 1 mm scale bar. Column headers give the time since baseline in days; the one-year anchor is set in bold.

Appendix C

Ablation Details

This appendix collects the material behind Chapter 5: the scores on border-touching and interior eyes (§C.1), the growth-speed measures of every arm (§C.2), the criteria and statistics of all comparisons (§C.3, §C.4) and every setting of the step-size test (§C.5).

C.1 Border-Touching and Interior Eyes

Table C.1 shows whether the order of the arms changes when the eyes whose lesion touches the crop border are separated from the others.

Table C.1: Change-region Dice at the one-year anchor, seed 42, split by whether the eye's lesion touches the border of the fixed crop (§3.3). Of the 75 validation eyes, 21 touch the border and 54 do not; per fold the border group holds 1 to 6 eyes. Each value is the mean over the five folds of the late-epoch mean within that group. Because the folds are weighted equally rather than by their number of eyes, the column for all eyes need not lie between the two group columns. The values are descriptive; the groups were not tested against each other.

Arm	All eyes	Border (21)	Interior (54)
T-FEN	0.543	0.544	0.536
Dilated-stencil GNN	0.526	0.516	0.522
U-Net (width 5)	0.505	0.483	0.511
FNO + 3×3 local path	0.498	0.479	0.506
FNO	0.483	0.452	0.495
FEN, free-form only	0.481	0.466	0.480
k-NN GNN	0.462	0.464	0.455
One-hop floor	0.452	0.464	0.440

The global operators, the U-Net and both FNO arms, score lower on the border-touching eyes than on the interior ones, whereas the local operators score about the same on both groups (§5.5).

C.2 Growth-Speed Measures for All Arms

Table C.2 shows whether the arms differ in how well they predict the growth speed of an eye, which §5.9 reports for the canonical model only.

Table C.2: Eye-level growth-speed measures for every arm of Table 5.1, computed as for the canonical model in §5.9: growth rate on the square-root area scale over each eye's whole follow-up, predicted rate averaged over the late epochs, 75 validation eyes of the five folds pooled, seed 42. The AUC separates the top 10, 15 and 20 % of true growth rates (8, 11 and 15 eyes) from the rest. Only point estimates are given; the 95 % bootstrap intervals are about ± 0.2 wide for every arm (see Table 5.11 for the canonical model), so the arms cannot be told apart on these measures.

Arm	Pearson r	AUC 10 %	AUC 15 %	AUC 20 %
T-FEN	0.40	0.72	0.70	0.74
Dilated-stencil GNN	0.40	0.74	0.70	0.70
Dilated-stencil GNN + RK4	0.30	0.74	0.68	0.69
U-Net (width 5)	0.43	0.68	0.69	0.76
FNO + 3×3 local path	0.33	0.75	0.77	0.78
FNO + RK4	0.31	0.78	0.74	0.72
FNO	0.33	0.72	0.67	0.70
FEN, free-form only	0.39	0.66	0.70	0.72
k-NN GNN	0.41	0.74	0.74	0.73
k-NN GNN, $k = 20$	0.41	0.74	0.74	0.73
One-hop floor	0.47	0.76	0.71	0.71

C.3 Statistical Criteria

Retraining the same model on the same fold changes its late-epoch change-region Dice by a standard deviation of $\sigma \approx 0.0127$ for a graph network, estimated from seven pairs of repeated runs, and of $\sigma \approx 0.019$ for the parameter-matched U-Net, estimated from its two seeds on the five folds.

Two arms are compared in two ways. The fold-paired test averages the differences of the two arms over the five folds. If the two arms performed identically on every fold, this mean $\bar{\Delta}$ would vary only through retraining, with a standard deviation $\sigma_\Delta = \sqrt{\sigma_A^2 + \sigma_B^2}/\sqrt{n}$ over n folds: about 0.008 for two graph networks and about 0.010 when one arm is the U-Net. The tables give $z = \bar{\Delta}/\sigma_\Delta$. The z statistic therefore tests the difference on these five folds against retraining noise. It does not include the variation of the difference between folds, which the SE column of the tables shows to be larger for most comparisons; that variation is covered by the fold count, the per-eye test and the leave-one-fold-out condition. A value of $|z| > 1.96$ ($p < 0.05$) corresponds to a mean difference of about 0.016, or 0.020 with a U-Net, and to about 0.036 for a difference on a single fold. The per-eye test compares the two arms on each of the 75 validation

eyes, using the growth-region Dice at the eye's first rollout step at or beyond 0.95 years (about 347 days), averaged over the late epochs. The tables give its paired t statistic.

A difference is counted as established when $|z| > 1.96$, the per-eye $|t|$ is about 2.5 or larger, neither changes sign when any one fold is left out, and, for the comparisons run at both seeds, it holds at both. At the 5 % level, the bar on t would be about 2.0. It is set higher because the 75 eyes come from 51 patients: the two eyes of one patient are not independent, which makes t too large. The fold-paired test is not affected, because the folds are split by patient. The higher bar also allows for the about fifty comparisons tested, of which one or two may pass the 5 % level by chance.

C.4 Statistics of the Comparisons

Tables C.3 and C.4 give the comparisons of Chapter 5 that were run at both seeds, Table C.5 those run at seed 42 only. The criteria are those of §C.3.

Table C.3: Comparisons of §5.2 and §5.4.1–§5.4.2 at seed 42 and at the second seed 7. Fold-paired: mean ± SE of the differences of the late-epoch change-region Dice over the five folds, with the number of folds on which the first arm is higher and the z statistic of §C.3 ($|z| > 1.96$ corresponds to $p < 0.05$). Per eye: mean ± SE of the differences of the growth-region Dice at the one-year anchor, with the number of eyes on which the first arm is higher and the paired t statistic.

Comparison	Seed	Fold-paired	Per eye
T-FEN − dilated stencil			
	42	$+0.0170 \pm 0.0085$ (4/5, $z = 2.1$)	$+0.0136 \pm 0.0062$ (46/75, 2.2)
	7	$+0.0165 \pm 0.0021$ (5/5, $z = 2.1$)	$+0.0152 \pm 0.0048$ (49/75, 3.1)
	Reading: small lead, just beyond retraining noise at both seeds; not established		
Dilated stencil − U-Net			
	42	$+0.0212 \pm 0.0172$ (3/5, $z = 2.1$)	$+0.0248 \pm 0.0081$ (50/75, 3.1)
	7	$+0.0081 \pm 0.0084$ (3/5, $z = 0.8$)	$+0.0103 \pm 0.0074$ (42/75, 1.4)
	Reading: tie (beyond retraining noise at seed 42, not at seed 7)		
T-FEN − U-Net			
	42	$+0.0383 \pm 0.0134$ (5/5, $z = 3.7$)	$+0.0383 \pm 0.0075$ (51/75, 5.1)
	7	$+0.0246 \pm 0.0073$ (5/5, $z = 2.4$)	$+0.0255 \pm 0.0074$ (52/75, 3.4)
	Reading: T-FEN higher		
One round − two rounds (k-NN)			
	42	-0.0108 ± 0.0061 (1/5, $z = -1.3$)	-0.0122 ± 0.0051 (33/75, -2.4)
	7	$+0.0036 \pm 0.0049$ (2/5, $z = 0.4$)	$+0.0029 \pm 0.0033$ (40/75, 0.9)
	Reading: null		
Reach: dilated stencil − k-NN			
	42	$+0.0635 \pm 0.0106$ (5/5, $z = 7.9$)	$+0.0647 \pm 0.0062$ (70/75, 10.4)
	7	$+0.0629 \pm 0.0131$ (5/5, $z = 7.8$)	$+0.0620 \pm 0.0071$ (66/75, 8.7)
	both	—	$+0.0633 \pm 0.0047$ (136/150 eye-seed pairs)
	Reading: established		
U-Net − k-NN			
	42	$+0.0423 \pm 0.0246$ (4/5, $z = 4.1$)	$+0.0399 \pm 0.0102$ (53/75, 3.9)
	7	$+0.0548 \pm 0.0181$ (5/5, $z = 5.3$)	$+0.0517 \pm 0.0109$ (59/75, 4.7)
	Reading: established		

Table C.4: Comparisons of §5.4.3–§5.4.4 at seed 42 and at the second seed 7, as in Table C.3.

Comparison	Seed	Fold-paired	Per eye
FNO − k-**NN**			
	42	$+0.0208 \pm 0.0258$ (3/5, $z = 2.6$)	$+0.0120 \pm 0.0135$ (41/75, 0.9)
	7	$+0.0412 \pm 0.0216$ (4/5, $z = 5.1$)	$+0.0345 \pm 0.0149$ (40/75, 2.3)
	Reading: not established (at seed 42 the sign changes without fold 4)		
FNO − **U-Net**			
	42	-0.0215 ± 0.0139 (1/5, $z = -2.1$)	-0.0279 ± 0.0111 (30/75, -2.5)
	7	-0.0136 ± 0.0105 (2/5, $z = -1.3$)	-0.0172 ± 0.0104 (27/75, -1.7)
	Reading: FNO lower at both seeds; beyond the threshold at seed 42 only		
FNO $+3\times3$ − k-**NN**			
	42	$+0.0361 \pm 0.0166$ (4/5, $z = 4.5$)	$+0.0310 \pm 0.0111$ (49/75, 2.8)
	7	$+0.0355 \pm 0.0206$ (4/5, $z = 4.4$)	$+0.0291 \pm 0.0117$ (47/75, 2.5)
	Reading: established		
Local path: FNO $+3\times3$ − **FNO**			
	42	$+0.0153 \pm 0.0110$ (4/5, $z = 1.9$)	$+0.0190 \pm 0.0070$ (48/75, 2.7)
	7	-0.0057 ± 0.0105 (3/5, $z = -0.7$)	-0.0054 ± 0.0095 (35/75, -0.6)
	Reading: not established		
FNO $+3\times3$ − **U-Net**			
	42	-0.0061 ± 0.0108 (2/5, $z = -0.6$)	-0.0089 ± 0.0093 (33/75, -1.0)
	7	-0.0193 ± 0.0051 (0/5, $z = -1.9$)	-0.0227 ± 0.0091 (26/75, -2.5)
	Reading: tie at seed 42; lower at seed 7, within retraining noise		
Transport: T-FEN − **free-form, same width**			
	42	$+0.0618 \pm 0.0040$ (5/5, $z = 7.7$)	$+0.0640 \pm 0.0058$ (68/75, 11.0)
	7	$+0.0897 \pm 0.0142$ (5/5, $z = 11.2$)	$+0.0878 \pm 0.0071$ (71/75, 12.4)
	Reading: established		
Transport, Galerkin form: T-FEN − **free-form, same width**			
	42	$+0.0527 \pm 0.0065$ (5/5, $z = 6.6$)	$+0.0555 \pm 0.0057$ (65/75, 9.7)
	7	$+0.0758 \pm 0.0092$ (5/5, $z = 9.4$)	$+0.0761 \pm 0.0068$ (68/75, 11.3)
	Reading: established		

Table C.5: Comparisons of Chapter 5 that were run at seed 42 only, as in Table C.3. Reach-ladder rows are against the default dilated stencil, capacity rows against the arm's own original size (Table 5.6).

Comparison	Fold-paired	Per eye
Reach ladder (§5.4.2)		
Columns up to ± 12	-0.0193 ± 0.0083 (1/5, $z = -2.4$)	-0.0220 ± 0.0042 (20/75, -5.2)
Columns up to ± 35	-0.0096 ± 0.0031 (1/5, $z = -1.2$)	-0.0103 ± 0.0037 (19/75, -2.8)
Columns up to ± 42	-0.0256 ± 0.0040 (0/5, $z = -3.2$)	-0.0279 ± 0.0048 (16/75, -5.8)
Rows ± 2	-0.0390 ± 0.0047 (0/5, $z = -4.9$)	-0.0411 ± 0.0048 (9/75, -8.5)
k-NN, $k = 20 - k = 12$	-0.0104 ± 0.0051 (1/5, $z = -1.3$)	-0.0114 ± 0.0028 (20/75, -4.1)
Time handling (§5.6.1)		
No multiplication by Δt	-0.0006 ± 0.0061 (2/5, $z = -0.1$)	-0.0028 ± 0.0037 (30/75, -0.8)
RK4 − Euler, dilated stencil	-0.0119 ± 0.0037 (1/5, $z = -1.5$)	-0.0132 ± 0.0055 (28/75, -2.4)
RK4 − Euler, FNO	$+0.0062 \pm 0.0056$ (4/5, $z = 0.8$)	$+0.0064 \pm 0.0069$ (43/75, 0.9)
Objective (§5.6.2)		
Plain mean squared error	-0.5258 ± 0.0222 (0/5, $z = -65.5$)	-0.5524 ± 0.0138 (0/75, -40.0)
Without soft-Dice term	-0.0496 ± 0.0172 (0/5, $z = -6.2$)	-0.0460 ± 0.0091 (19/75, -5.1)
Monotonic penalty, weight 2.0	-0.0036 ± 0.0037 (2/5, $z = -0.4$)	-0.0042 ± 0.0034 (34/75, -1.3)
Capacity (§5.6.3)		
Dilated stencil, width 32	-0.0053 ± 0.0063 (1/5, $z = -0.7$)	-0.0050 ± 0.0034 (30/75, -1.5)
Dilated stencil, width 128	-0.0081 ± 0.0056 (2/5, $z = -1.0$)	-0.0091 ± 0.0036 (22/75, -2.5)
Dilated stencil, 4 rounds	-0.0090 ± 0.0021 (0/5, $z = -1.1$)	-0.0097 ± 0.0037 (20/75, -2.6)
U-Net, width 2	-0.0163 ± 0.0290 (3/5, $z = -1.3$)	-0.0186 ± 0.0155 (40/75, -1.2)
U-Net, width 9	$+0.0036 \pm 0.0126$ (1/5, $z = 0.3$)	$+0.0041 \pm 0.0082$ (38/75, 0.5)
FNO $+3\times3$, width 2	-0.0192 ± 0.0175 (2/5, $z = -2.4$)	-0.0205 ± 0.0101 (30/75, -2.0)
FNO $+3\times3$, width 9	-0.0177 ± 0.0075 (1/5, $z = -2.2$)	-0.0207 ± 0.0070 (26/75, -2.9)
T-FEN, width 48	$+0.0006 \pm 0.0026$ (3/5, $z = 0.1$)	-0.0003 ± 0.0020 (37/75, -0.2)
T-FEN, width 192*	$+0.0014 \pm 0.0040$ (2/3, $z = 0.1$)	$+0.0002 \pm 0.0035$ (25/51, 0.1)
T-FEN, MLP depth 8	-0.2854 ± 0.0656 (0/5, $z = -35.5$)	-0.2886 ± 0.0192 (2/75, -15.1)

T-FEN in the energy-conserving form. *Folds 0, 2 and 3; diverged on 1 and 4.

C.5 The Step-Size Test

Table C.6 lists every setting of the step-size test of §5.8.1. Each trained model is re-evaluated, without retraining, with a fixed step h and with explicit Euler or classical RK4. Each visit interval is split into the smallest number of equal sub-steps no longer than h. Because the median interval between visits is 180 days, the steps of one and of half a year leave almost every interval in one piece: on folds 0 to 4, only 2, 0, 3, 0 and 2 of 135, 65, 100, 92 and 86 intervals are split at one year, and 23, 4, 9, 7 and 4 at half a year.

Table C.6: Every setting of the step-size test, all five folds, seed 42. Each cell is the change of the change-region Dice at the one-year anchor against the setting the model was trained with, when the same weights are evaluated with a fixed step h (in years, with days below) and with explicit Euler or classical RK4. The trained setting itself reproduces the change-region Dice recorded at the end of training on every fold. Rows that change only the step of the trained scheme start at or within 0.004 of 0 for the graph networks, whose trained setting is one step per interval and coincides with $h = 1$ year for almost every interval.

h (year)	1	1/2	1/4	1/8	1/16	1/32
h (days)	365	182.5	91.3	45.6	22.8	11.4
Canonical dilated-stencil network (trained: Euler, one step per interval)						
Fold 0, Euler	+0.000	−0.006	−0.005	−0.020	−0.030	−0.037
Fold 0, RK4	+0.017	+0.013	+0.012	+0.001	−0.017	−0.028
Fold 1, Euler	+0.000	−0.012	+0.021	+0.041	+0.046	+0.043
Fold 1, RK4	+0.027	+0.017	+0.049	+0.065	+0.057	+0.048
Fold 2, Euler	+0.000	+0.001	−0.016	−0.081	−0.158	−0.187
Fold 2, RK4	+0.010	+0.011	−0.017	−0.090	−0.162	−0.190
Fold 3, Euler	+0.000	+0.004	−0.008	−0.004	−0.004	−0.005
Fold 3, RK4	−0.004	−0.003	−0.022	−0.015	−0.010	−0.008
Fold 4, Euler	−0.002	−0.001	+0.026	+0.031	−0.031	−0.085
Fold 4, RK4	+0.032	+0.035	+0.045	+0.040	−0.028	−0.082
Network trained in the RK4 wrapper (trained: RK4, one step per interval)						
Fold 0, Euler	−0.113	−0.063	+0.003	+0.016	+0.018	+0.018
Fold 0, RK4	+0.000	+0.008	+0.020	+0.018	+0.018	+0.018
Fold 1, Euler	−0.043	−0.046	+0.015	+0.028	+0.023	+0.020
Fold 1, RK4	+0.000	+0.003	+0.018	+0.017	+0.018	+0.018
Fold 2, Euler	−0.020	−0.016	+0.004	+0.004	+0.003	+0.001
Fold 2, RK4	+0.000	+0.000	+0.000	+0.000	+0.000	+0.000
Fold 3, Euler	−0.015	−0.013	−0.004	−0.010	−0.014	−0.016
Fold 3, RK4	+0.000	+0.001	−0.013	−0.016	−0.016	−0.016
Fold 4, Euler	−0.075	−0.072	−0.012	−0.006	−0.002	−0.000
Fold 4, RK4	+0.004	+0.006	+0.003	+0.003	+0.003	+0.003
T-FEN, energy-conserving form (trained: RK4, 45-day steps)						
Fold 0, Euler	−0.258	−0.248	−0.088	−0.014	−0.006	−0.003
Fold 0, RK4	−0.201	−0.159	−0.001	+0.000	+0.000	+0.000
Fold 1, Euler	−0.175	−0.168	−0.067	−0.024	−0.009	−0.004
Fold 1, RK4	−0.095	−0.068	−0.004	+0.000	+0.001	+0.001
Fold 2, Euler	−0.165	−0.158	−0.081	−0.022	−0.007	−0.004
Fold 2, RK4	−0.131	−0.129	−0.012	+0.000	−0.002	−0.002
Fold 3, Euler	−0.127	−0.119	−0.030	−0.006	−0.006	−0.005
Fold 3, RK4	−0.090	−0.054	+0.001	+0.000	−0.001	−0.001
Fold 4, Euler	−0.181	−0.183	−0.074	−0.018	−0.013	−0.010
Fold 4, RK4	−0.101	−0.088	+0.008	+0.000	−0.007	−0.007

Appendix D

Code Structure

The code is kept in a private repository and is available from the author on request. Its main parts are listed in Table D.1.

Table D.1: Main parts of the code repository.

Path	Content
`tools/precompute_graphs_ga.py`	Preprocessing: crop or pad to 49×1024, normalisation, covariates, training windows, one data variant per fold (Chapter 3)
`GraphPDE/train.py`, `GraphPDE/train_helper.py`	The shared framework: loss, curriculum, optimisation, evaluation and logging (§4.2, §5.1)
`GraphPDE/gnn.py`	The message-passing graph network (§4.3.2)
`GraphPDE/baselines/`	The other operators: dilated stencil, U-Net, FNO, FEN and the Runge–Kutta wrapper (§4.3)
`GraphPDE/mesh/`, `interpolate.py`	The moving-mesh extension (Appendix E)
`utils/utils.py`	Dataset and window sampling
`jobs/`	Slurm scripts; one five-fold experiment is one job array
`Dockerfile`	Local environment; the cluster runs use a fixed container with PyTorch 2.4.1 (§4.4)

Run identity. Every run writes its complete configuration into its run directory at the start. The name of the directory encodes the key settings, the fold and the data variant. Because every fold is mounted at the same path on the cluster, each run also records which fold and data variant it loaded. Resuming an interrupted run is disabled, so every reported run was trained in one piece. The per-epoch metrics of all runs are collected into one version-controlled results table, and every training result in this thesis is read from it. Table D.2 gives the experiment name behind each arm; the five folds are the runs `<name>_f0` to `<name>_f4`, and the second seed adds the suffix `s7`.

Table D.2: Experiment names of the arms of Table 5.1, seed 42.

Arm	Experiment name
T-FEN	`FEN_final_w96ftp7rk4d45skew`
Graph network, dilated stencil	`ANISOGNN_final_dilmean`
Graph network, dilated stencil, RK4	`ANISOGNN_final_oderk4n1`
U-Net, width 5	`UNET_final_w5`
FNO with 3×3 local path	`FNO_final_w5k3`
FNO, RK4	`FNO_final_w5rk4`
FNO	`FNO_final_w5`
FEN, free-form only	`FEN_final_w96Frk4d45`
Graph network, k-NN, $k = 12$	`MPPDE_final_aggrmean`
Graph network, k-NN, $k = 20$	`MPPDE_final_k20`
One-hop floor	`MPPDE_final_l1mean`
Per-pixel floor	`MPPDE_final_pxmean`

Appendix E

The Moving-Mesh Extension (MM-PDE)

This appendix gives the background and the full design of the experiment with adaptive meshes of §5.7, which extends the graph network by the moving-mesh solver of Hu et al. (2024). The main text states the idea and the result; the details are collected here. Section E.1 summarises the original method, §E.2 its adaptation to GA, §E.3 the quality of the meshes, §E.4 further details of the result and §E.5 an outlook.

E.1 Background: MM-PDE

Hu et al. (2024) extend message-passing neural solvers with an adaptive moving mesh. A uniform mesh spends the same resolution everywhere: it resolves smooth regions more finely than needed and sharp gradients, such as wave fronts, too coarsely.

Mesh adaptation methods fall into two families. h-adaptation adds and removes vertices, which changes the topology and the number of degrees of freedom from step to step; standard graph-network training does not handle such changes well. r-adaptation, or moving-mesh adaptation, moves a fixed set of nodes and keeps both the node count and the graph topology. The classical methods of this family are described by W. Huang and Russell (2011). Earlier neural r-adaptation methods were supervised: they required optimal meshes from an expensive classical solver as training targets.

The mesh is guided by a monitor function. Hu et al. (2024) use the matrix-valued monitor $M = (1 + \|\nabla u\|_2/\alpha_M)\, I$, which is large where the solution varies sharply, that is, where its gradient is large. The identity matrix I makes the refinement the same in every direction, the constant 1 keeps M away from zero so that nodes do not collapse in smooth regions, and the constant α_M sets the balance between adaptivity and numerical stability. The mesh density is $\rho(x) = \sqrt{\det M(x)}$. This form follows from an upper bound on the interpolation error of finite elements.

The aim of the adaptation is equidistribution. A coordinate map produces an M-uniform mesh if every cell carries the same integral of ρ, namely σ/N, where σ is the integral of ρ over the domain Ω and N the number of cells. Cells therefore shrink where M is large.

Many maps satisfy this condition, so the one closest to the identity is chosen, which is an optimal transport problem. By Brenier's theorem its solution is

the gradient $\nabla\phi$ of a convex scalar potential ϕ. Substituting this map into the equidistribution condition gives the Monge–Ampère equation

$$\det H(\phi) = \frac{\sigma}{|\Omega|\,\rho(\nabla\phi)},$$

a fully non-linear second-order equation for ϕ, with the condition that the boundary of the domain maps onto itself. If ϕ is convex, the map $\nabla\phi$ has a non-singular Jacobian, so the mesh cannot fold. Convexity of a scalar potential is easier to encourage in training than a non-folding condition on a vector field, and for this reason the potential is learned instead of the map. It is written as $\phi(x) = \frac{1}{2}\|x\|^2 + \psi(x)$, so that $\nabla\phi = x + \nabla\psi$. The network predicts only ψ, whose gradient is the displacement of the nodes, and the map is close to the identity at initialisation.

The data-free mesh mover (DMM) has the form of a DeepONet. A branch network encodes the state u, a trunk network encodes a continuous query coordinate, and together they give the potential ϕ. The DMM is trained without example meshes, on the loss $L = L_{eq} + \beta L_{bound} + \gamma L_{convex}$, where L_{eq} is the squared residual of the Monge–Ampère equation at sampled interior points, L_{bound} enforces the boundary condition and L_{convex} penalises violations of convexity. Without the convexity term, a small equation residual can still come with a folded mesh. Training uses Adam, followed by BFGS on the last layer. The DMM is trained in advance and kept frozen while the solver is trained. This separates the geometric problem from the forecasting objective and lets the same mesh mover serve several experiments.

The solver combines two branches, $\mathcal{M}(u_t) = G_1(u_t) + \mathcal{I}(G_2, \nabla\phi, u_t)$. Here G_1 is a graph network on the uniform mesh, G_2 a graph network on the moved mesh, and $\mathcal{I}$ a learned interpolation that maps the result of G_2 back to the uniform mesh. The uniform branch covers the whole domain at uniform resolution, and the moved branch adds a correction where the gradients are steep. The interpolation network, ItpNet, computes its weights from the geometry of the moved mesh instead of using fixed finite-element basis functions. It is pretrained with its own optimiser before the two branches are trained together, because an untrained interpolation would pass meaningless values to the uniform branch.

Hu et al. (2024) report four ablations of the dual-branch solver: removing the uniform branch, replacing the moved branch by a second branch on the original mesh, removing the residual-cut network of their model, and replacing the DMM mesh by the uniform mesh. Each of them increases the error. They conclude that the branch on the original mesh, the residual cut and a well-defined moving mesh are all needed for accuracy. The residual cut, a convolutional network that corrects the interpolation error between the two meshes, was implemented for GA as well, but its correction accumulated over the autoregressive rollout and made the dual model diverge, so it was removed. The original method thus has two parts that a GA version has to provide: a mesh mover for the GA state and a dual-branch composition.

E.2 The Moving-Mesh Extension for GA

The adaptation keeps the fixed framework of §4.2 and adds the two parts: a mesh mover for the GA mask (§E.2.1) and a dual-branch composition with

a gate (§E.2.2). Two design choices make the experiment a measurement. A learnable scalar gate, initialised to zero, sets how much of the correction computed on the moved mesh enters the prediction, so its trajectory shows whether the correction is used. A parameter-matched bypass control runs the same machinery on the uniform grid, which separates the effect of the mesh geometry from the effect of the extra parameters. All reported five-fold runs of the extension use an earlier configuration of the k-nearest-neighbour graph network as their branches, which aggregates the messages of the neighbours by their mean and their maximum (75 339 parameters, against 67 147 for the k-NN arm of Table 5.1); their scores are therefore compared only with each other (§5.7). The mesh mover pulls nodes together where the lesion border lies, to raise the resolution there, and the correction branch has to exchange messages between exactly these close nodes. A dilated stencil, which skips the nearest nodes in order to reach further, would pass over the nodes the mesh mover has gathered and defeat its purpose. On fold 2, a dual model whose uniform branch uses the dilated stencil, while its correction branch keeps the k-nearest-neighbour graph, scores 0.031 below its own single-branch counterpart, within the single-fold noise level of 0.036; no five-fold run and no bypass control exist for this configuration (§E.4).

E.2.1 Data-Free Mesh Mover for GA

The mesh mover reads only channel 0, the binary GA mask, on the native 49×1024 grid. At the first step of a rollout this is the observed mask. At later steps it is the model's own predicted mask, denormalised and thresholded back to binary, so the mesh follows the predicted lesion as it would in use. Only the mesh mover reads the mask alone; both solver branches process all eleven channels.

The architecture follows the DeepONet form (Lu et al., 2021) as adapted by Hu et al. (2024). A convolutional encoder, the branch network of the DeepONet, reads the mask with three stride-2 convolutions ($1 \to 16 \to 32 \to 64$ channels), adaptive average pooling to 4×4 and a linear layer to a 512-dimensional latent vector. Because of the pooling, it accepts any grid size. The trunk network, a tanh multilayer perceptron with widths 2, 32 and 512, reads the continuous coordinate $\xi \in [0,1]^2$. A head maps the concatenated outputs of encoder and trunk to the scalar potential ψ through widths 1024, 512, 512 and 1, one layer deeper than the base head. The moved mesh is $x(\xi) = \xi + \nabla\psi(\xi)$, computed by automatic differentiation. The mesh mover has 1 394 273 parameters. They are frozen during solver training and excluded from the optimiser.

The monitor differs from the one of §E.1. It is a scalar,

$$M \;=\; 1 + \log\Big(1 + \frac{\|\nabla u\|}{\alpha_{\text{scale}}\,\bar{g} + \epsilon}\Big), \tag{E.1}$$

where u is the mask after a Gaussian blur, $\bar{g}$ the mean gradient magnitude over the grid, $\alpha_{\text{scale}} = 0.1$ and $\epsilon = 10^{-10}$. The gradient is taken in the unit-square computational coordinate, which gives the two axes their correct relative weight on the anisotropic grid. Two parts of this monitor were necessary. On the binary mask, the gradient is a step at the lesion edge: zero everywhere except for one jump. The monitor then consists of a single sharp ridge that the mesh cannot equidistribute, and training became unstable. A Gaussian blur ($\sigma = 1.0$ px),

applied to the monitor only, spreads the step over a few pixels and makes the gradient finite; without it, the mesh on the native grid barely moves (§E.3). The logarithm bounds how far the monitor at the lesion border rises above its value elsewhere, so that the equidistribution target remains attainable. The encoder always receives the raw binary mask.

The loss has three terms, weighted 1, 1000 and 1 as in the original (Hu et al., 2024). The first is the Monge–Ampère residual at sampled interior points, $\mathrm{MSE}\left(M(\xi + \nabla\psi)\,\det(I + \nabla^2\psi)/\bar{M},\ 1\right)$. It requires the monitor at the moved point times the change of cell area to be the same everywhere. The second, a soft boundary term, penalises the normal derivative of ψ on the four edges, so that boundary nodes stay on the boundary. The third penalises negative diagonal entries of the Hessian of ϕ, an early sign of folding. This term stays close to zero throughout training. Validity is enforced instead when a checkpoint is selected, by requiring zero folded cells. The last-layer BFGS refinement of Hu et al. (2024) was not used.

Each iteration samples 5 masks and 1000 points per mask, 90 % of them in proportion to the monitor and 10 % uniformly. The mesh mover was trained on 478 of the 553 masks of the cohort, the start masks of all windows of the 75 eyes (the last visit of an eye starts no window). It was trained with Adam, learning rate 2×10^{-4} and weight decay 10^{-5}, for 500 epochs, with the learning rate multiplied by 0.2 at epochs 333 and 467. The configuration was selected from a 55-run search on the native grid. Mesh quality (§E.3) improved with the number of epochs up to 500. The two settings with the largest effect were the extra head layer and the sharper blur of $\sigma = 1$ instead of 2. The sharper blur raised the edge gain by 0.24 with the deeper head, but by only 0.003 without it. The architecture of the encoder made no measurable difference across five variants; the pooling encoder is kept because it accepts any grid size. Table E.1 summarises the configuration.

Training on masks of every eye, including the validation eyes, does not give the forecast access to future observations. The mesh mover has no target: its loss measures only how evenly the mesh distributes the monitor of the mask it is given, a property of that mask alone. Its weights were also fitted to masks of later visits of the validation eyes, but only to mesh each mask, never to relate one visit to the next. At use, it receives the current mask only, and it is frozen while the solver is trained. Any advantage from this would favour the dual branch, which still shows no gain (§5.7). A held-out set of masks is not needed either, because the mesh mover is not a predictor whose accuracy on new data is in question. A classical mesh mover computes the mesh for each mask directly, by solving the Monge–Ampère equation for that mask, without any training data. The network takes this role for the masks of the cohort, and whether its meshes are valid and concentrate nodes at the lesion border is checked directly on these masks (§E.3). From the second rollout step on, it receives predicted masks, on which it was not trained and whose meshes were not scored separately. The same mesh mover is used for every fold.

E.2.2 Dual-Branch Composition

The dual-branch model predicts the next state by Equation (5.2): f_θ runs on the uniform grid and g_θ on the moved mesh. The uniform branch alone gives the

Table E.1: Configuration of the data-free mesh mover used by the dual-branch model (canonical checkpoint of the 55-run search on the native grid).

Component	Setting
Input	Channel 0 only (binary GA mask), native 49×1024 grid
Encoder (branch)	Convolutions $1 \to 16 \to 32 \to 64$, pooling to 4×4, linear to 512
Trunk	Tanh MLP, widths 2, 32, 512
Head	MLP, widths 1024, 512, 512, 1
Monitor	Scalar, Gaussian blur $\sigma = 1.0$ px, $\alpha_{\text{scale}} = 0.1$, log compression
Loss weights	1 (equidistribution), 1000 (boundary), 1 (convexity)
Sampling	5 masks $\times$ 1000 points, 90 % by monitor, 10 % uniform
Optimiser	Adam, learning rate 2×10^{-4}, weight decay 10^{-5}
Schedule	500 epochs, learning rate $\times 0.2$ at epochs 333 and 467
Training data	478 window-start masks, all 75 eyes
Parameters	1 394 273, frozen during solver training

full next state, $u_k + \Delta t_k f_\theta$. The correction branch gives an increment scaled by Δt_k, which the interpolation $\tilde{I}$ carries back to the uniform grid. Both branches are the graph network of §4.3.2, with two message-passing rounds of width 64.

The gate α is a learnable scalar initialised to zero. At the start, the dual model is therefore identical to the uniform branch alone, and the correction enters the prediction only if training increases $|\alpha|$. In the following, the gate is said to open when $|\alpha|$ grows well beyond zero and to stay shut when it remains close to zero. The gate also fixes the scale of the correction. The loss constrains only the sum of the two branches, whose outputs could otherwise grow large and cancel each other. The gate has its own optimiser group without weight decay. Unlike every operator of the main text, the decoder of the correction branch is not zero-initialised. If both the decoder and α started at zero, the gradients of both would be exactly zero, and the correction branch could never become active. The zero start of the dual model therefore comes from α alone.

The moved nodes lie off the lattice, where the dilated stencil of §4.3.3 is not defined. The correction branch therefore builds its own k-nearest-neighbour graph ($k = 12$) on the moved positions, searched in index space, that is, with the coordinates divided by the grid spacing. A search in physical coordinates would return only neighbours from the same row, because the rows lie about 21 times further apart than the columns (Figure 4.2a).

The interpolation is a learned network (ItpNet). For each query point, a tanh multilayer perceptron with widths 26, 128, 64 and 12 and a linear output turns the coordinates of its 12 nearest source points into 12 weights. Two independent copies are used. One interpolates all eleven channels from the uniform grid to the moved mesh, as the input of the correction branch, and the other carries the correction back. The weights of the return direction are renormalised to sum to one at each point, with a sign-preserving floor of 0.1 on the denominator, which bounds the gain at 10. The floor is active at about 12.6 % of the points. ItpNet is pretrained at epoch 0 for 25 passes with its own AdamW op-

timiser (Loshchilov and Hutter, 2019), on the round trip uniform $\to$ moved $\to$ uniform, which should return the input unchanged. The pretraining uses the channel-weighted mean squared error only, because the soft-Dice term would push the interpolation away from the identity it is meant to learn. During the pretraining, the gradients are clipped and the gate is disabled.

The mesh is recomputed from the current state at every step, because a mesh built at the start of a multi-step window would no longer match the lesion at the later steps. The dual-branch model is therefore one more operator that cannot predict several steps in one pass, the reason why the framework predicts one step at a time (§4.2).

Gradients are clipped to a norm of 1.0 separately in five groups: the uniform branch, the correction branch, the ItpNet weights, the gate α and the layer encoder; the mesh mover is frozen. In the dual runs, the gradients of ItpNet have a median of 10^3 to 10^4, against about 1 for the uniform branch. Under one joint clip they would dominate the joint norm and shrink the updates of the uniform branch accordingly.

The dual-branch model has $175\,663$ trainable parameters against $75\,339$ for the single-branch model in the configuration of the five-fold runs, about $100\,000$ more for the second network and ItpNet. With the dilated stencil the counts are $159\,279$ and $67\,147$. The frozen mesh mover is not counted, because it is not trained with the solver.

The bypass control loads the same mesh mover and runs the second network and the full ItpNet round trip, but places the correction branch on the uniform grid instead of the moved mesh. It has the same parameters as the dual model and differs from it only in the mesh geometry. The Runge–Kutta wrapper of §4.3.7, the dense operators and the finite-element networks exist only as single-branch models, so they are compared with the single-branch graph network and not with the dual branch. In total, the extension adds a frozen mesh mover, a second graph network, two interpolation networks and a gate to the single-branch model.

E.3 Moving Mesh Quality

The mesh mover was judged on its own terms, independently of the prediction, so that a defective mesh can be told apart from a correction branch that does not use a good one. Four metrics were used. All of them depend only on the mask and use the same reference for every run. The number of tangled cells counts the mesh cells with a non-positive signed area, that is, folds. The edge gain measures whether nodes move to the lesion border. The border is widened to a band of half-width 0.02 of the domain, normalised to a grid mean of 1, and averaged at the moved node positions. The unmoved mesh scores exactly 1, and a higher value means more nodes on the border. The reference coefficient of variation, `cov_ref`, is the standard deviation divided by the mean of the cell areas weighted by one fixed reference monitor, the equidistribution measure of W. Huang and Russell (2011); lower values mean a more even mesh. The determinant margin, `det_min`, is the minimum of $\det(\mathrm{I}+\nabla^2\psi)$ over random points. A positive value means that the map folds nowhere, not only at the grid nodes.

Table E.2: Quality of the meshes of the canonical mesh mover at epoch 500, scored on 16 masks spread over 13 eyes of the cohort. Edge gain as mean $\pm$ sd over three seeds; the other columns are three-seed means. All values are in-sample, because the mesh mover is trained on masks of every eye of the cohort.

Edge gain	Tangled cells	`cov_ref`	`det_min`
2.6307 ± 0.0607	0	0.311	0.115

Table E.2 gives the result. No mesh folds: the number of tangled cells is zero on every scored mask, at every seed and every scored epoch. The values are in-sample (§E.2.1).

On these masks, the configuration before the changes of head and blur reaches an edge gain of 2.24, the same mesh mover without blur 1.34, and the same without the logarithmic compression 0.94, below the unmoved mesh. The seed spread of the selected configuration, a standard deviation of about 0.06, is the largest in the search. The boundary weight of 1000 from the original was kept, although a weight of 100 scored higher on edge gain, by 0.105. Edge gain alone does not judge the mesh: with a weaker boundary term the boundary condition is met less well (Hu et al., 2024, Appendix C.3), so boundary nodes can be pulled inward, which edge gain does not penalise.

Figure 5.7 in the main text shows the moved mesh for three eyes with a small, a medium and a large lesion. In all three, nodes move from the surrounding retina towards the lesion, most strongly at its border, and no cell folds. The mesh mover thus produces valid meshes that concentrate nodes at the lesion border, and a defective mesh does not explain the result of §5.7.

E.4 Details of the Result

The main result, the comparison of the single-branch model, the dual-branch model and the bypass control, is given in §5.7 (Table 5.7). All three use the earlier k-NN configuration of §E.2, with 75 339 parameters for the single branch and 175 663 for the two two-branch models. With the statistics of §C.3, the dual branch is lower than the bypass control by 0.0001 on average ($z = 0.0$), higher on three of the five folds and on 33 of the 75 eyes, and lower than the single branch by 0.0013 ($z = -0.2$), higher on three folds and on 32 eyes.

The gate stays shut, with and without the moving mesh. In every fold of both arms, α moves from zero to exactly ± 0.0020 in the first step, one optimiser step from the zero start, as designed. It then peaks at 0.058 to 0.093 in the dual branch and at 0.031 to 0.101 in the bypass control, averages 0.009 to 0.024 and 0.005 to 0.023 in absolute value, and ends near zero in every fold. The range of the peaks in the bypass control encloses that of the dual branch, and the largest peak of either arm, 0.1005, belongs to a bypass fold.

Three further runs on fold 2 varied the design, and in none of them did the correction branch improve the forecast. Without the gate, with the decoder zero-initialised instead, the change-region Dice falls to 0.3148 against 0.4878 for the

gated dual model, a loss of 0.173, about five times the single-fold noise level of 0.036. Without weight decay on the correction branch and ItpNet, the Dice is 0.0092 higher than its counterpart, within the noise level. The gate peaks lower (0.058 against 0.067), and the parameters of the correction branch grow instead of shrinking, so weight decay does not keep the gate shut. On the dilated stencil, the dual model scores 0.5754 against 0.6060 for its own single-branch counterpart, a difference of -0.031, within the single-fold noise level, although higher on only 3 of the 16 eyes of the fold, at roughly nine times the cost. The gain of the stencil carries over to the dual model, but the correction branch still does not engage.

Before the runs, $\alpha \to 0$ had been fixed as a reportable outcome, to be read as evidence that mesh adaptation does not help GA forecasting. The bypass control, part of the same design, shows that this reading is not valid, because its gate behaves in the same way without any mesh. The experiments support a narrower statement. The correction branch did not improve the forecast in any tested variant. With the gate, the gate stayed shut on the moved mesh and on the uniform grid, and on fold 2 also without weight decay. Without the gate, the correction entered at full weight and lowered the change-region Dice on fold 2 by 0.173. The dual-branch model gave no measurable gain at about ten times the cost. Hu et al. (2024) report that their dual branch improves on the uniform grid on their benchmarks.

Why the experiment was not developed further, with reach as the main reason, is discussed in §6.1.2.

The result therefore concerns this method on this task. One dual-branch design, without the residual cut of the original and adapted to a slowly growing lesion on an anisotropic grid, need not be representative of adaptive meshes, and the experiment is not a general statement on how well adaptive meshes work, for GA or elsewhere.

E.5 Outlook

The monitor of the mesh mover is computed from the current state, whereas the change happens in the next one. A monitor built on a surrogate of the next state, the open problem named by Hu et al. (2024), would place the nodes where the lesion is about to grow. Whether it would also help the correction branch to engage is open, since the branch did not engage even on the uniform grid.